INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

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Abstract

Decentralized finance (DeFi) protocols seek on-chain wallet reputation to screen counterparties and set risk parameters. Adaptive Weighted PageRank (AWP) ranks wallets on token transfers, but a transfer records a payment, not trust. An ERC-20 allowance records an owner authorizing a spender to move its tokens. This thesis asks what a PageRank walk gains when it follows allowance edges together with transfers.

Its method, a coupled PageRank (C-PR), mixes the allowance and transfer rows of every address in one walk with a fixed weight. Its single-layer references are AWP and EndorseRank, PageRank on the latest non-zero allowances with AWP's edge weights. User wallets are used in too many ways to share one scale, so the study scores contracts. It computes all scores, raw degrees included, from the twelve months before each freeze, for $6{,}483$ contracts that at least three owners approved on one of five tokens on Arbitrum One between July 2025 and June 2026, and compares them with Kendall's τ and paired bootstrap intervals. It tests scores frozen on 31 March 2026 on $7{,}914$ spender contracts and registered a July–September 2026 window before extracting its logs. After the registered results were known, it revalued amounts in US dollars, kept five tokens and set the twelve-month score windows; its main analysis re-applies the rules fixed in advance to these revised scores, with the registered verdicts on the coupled walk.

Under these rules C-PR beats neither layer walked alone: it is no worse than the allowance layer on later approval pairs ($\Delta\tau = +0.031$ $[0.021, 0.042]$) but below the transfer layer on later senders (-0.030 $[-0.040, -0.020]$), and in the registered window it gains on approvals ($+0.170$ $[0.149, 0.189]$) but falls behind on senders by more than the margin (-0.025 $[-0.038, -0.015]$). In contrasts computed afterwards, C-PR ranks both labels above EndorseRank in both windows (approval pairs $+0.026$ $[0.016, 0.037]$ and $+0.017$ $[0.006, 0.029]$) and later approval pairs above AWP ($+0.133$ $[0.114, 0.153]$), mainly through the spenders that AWP scores zero, and it ranks

later senders below AWP (-0.073 $[-0.083, -0.063]$). In these contrasts no PageRank ranks later arrivals above the raw count of the same edge type: the approval count beats EndorseRank in both windows (-0.090 $[-0.100, -0.079]$ and -0.082 $[-0.090, -0.073]$), also with AWP's activity restarts (-0.060 $[-0.070, -0.052]$ and -0.046 $[-0.055, -0.038]$), and the transfer in-degree beats AWP (-0.038 $[-0.043, -0.032]$ and -0.041 $[-0.047, -0.034]$). On later liquidation avoidance of GMX V2 contract traders, built from neither edge type, the approval count led the transfer count in the registered window ($\Delta\tau = +0.141$, 97.5% interval $[0.026, 0.256]$, 70 traders) but not resolvably in the earlier window ($+0.095$, 97.5% interval $[-0.041, 0.263]$), whereas the registered analysis resolved it only in the earlier window. C-PR takes 1.36 times AWP's solve time; EndorseRank, on a graph with 2.21 times fewer edges, takes 3.21 times less.

The thesis contributes a walk that couples authorization and payment, its failed test under rules fixed in advance, the allowance as a reputation edge, a same-cohort and out-of-window evaluation of every score against the counts it smooths, and a closed form of such walks at depth one that reduces EndorseRank to an owner-normalized in-strength. About 25 colluding addresses can lift a contract past the cohort's 90th percentile on the allowance graph, and the same bound covers C-PR, so no score should decide rewards without Sybil controls; none measures creditworthiness or the safety of a contract's code.

Key terms: coupled PageRank; decentralized finance; on-chain reputation; ERC-20 allowance; EndorseRank; Adaptive Weighted PageRank; smart contract; temporal holdout; pre-registration; Arbitrum One; Kendall's tau

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Abbreviations and Notation

The abbreviations and mathematical notation used throughout this research are listed below. Each symbol is also defined where it first appears in the main chapters.

Abbreviations

Table I: Abbreviations used in this research.

Abbreviation	Meaning
AWP	Adaptive Weighted PageRank
C-PR	Coupled PageRank (per-node mixture of the allowance and transfer layers, weight λ)
DeFi	Decentralized Finance
DEX	Decentralized Exchange
EIP	Ethereum Improvement Proposal
EOA	Externally owned account (an address controlled by a private key; the wallet of a user)
ER	EndorseRank (in table headings)
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
FC	Financial Cryptography and Data Security
PR	PageRank
RQ	Research Question
SoK	Systematization of Knowledge
S-PR	Endorsement-Seeded PageRank (transfer layer walked with restarts drawn from the allowance walk)
USD	United States dollar, the unit of every amount in the revised analysis
USENIX	USENIX Security Symposium
W0	Holdout window: scores frozen on 31 March 2026, labels counted from April to June 2026
W1	Registered window: scores frozen on 30 June 2026, labels counted from July to September 2026
WETH, USDC, USDT0, ARB, WBTC	The five tokens of the revised analysis: wrapped ether, USD Coin, Tether USD on Arbitrum One, the Arbitrum governance token, wrapped bitcoin

Mathematical notation

Table II: Mathematical notation used in this research.

Symbol	Meaning
G_{approve}	Directed endorsement graph from latest ERC-20 allowances (owner $\rightarrow$ spender)
G_{transfer}	Directed transfer graph (sender $\rightarrow$ receiver; AWP input)
$\lvert V\rvert$	Number of nodes in a reputation graph
$\lvert E\rvert$	Number of directed edges in a reputation graph
w_{ij}	Weight of the directed edge from node i to node j
o_i	Out-strength of node i, $o_i = \sum_j w_{ij}$ (written $o(u)$, or $o_A(u)$ and $o_T(u)$ per layer, in Chapter 3)
P	Transition matrix, $P_{ij} = w_{ij}/o_i$ for $o_i > 0$; a dangling row is uniform in Equation (2.1) and zero in Equation (3.2), where its mass is redistributed separately
G_{PR}	Google matrix $dP^\top + (1-d)\,\mathbf{1}\mathbf{1}^\top/\lvert V\rvert$ (Equation (2.2))
$\mathbf{r}$, r_i	PageRank vector and its entry for node i; $\mathbf{r}^{(t)}$ is the iterate after t power-iteration steps
$\boldsymbol{\pi}$	Teleportation (restart) distribution: uniform for EndorseRank and C-PR, proportional to activeness X_u for AWP, and the normalized allowance-walk score for S-PR
d	PageRank damping factor (default 0.85); the teleport probability is $1-d$
ε	Convergence tolerance of power iteration (10^{-8}, on the $\lVert\cdot\rVert_1$ change between iterates)
$I_{\max}$	Iteration cap of power iteration (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 1$ and $\lambda = 0$: one layer walked alone)
$P^{(\lambda)}$	Coupled transition matrix of C-PR with allowance-layer weight λ (Equation (3.9))
$\alpha_A(u)$, $\alpha_T(u)$	Weights of the allowance row and the transfer row in the coupled transition row of node u
k	Logistic time-decay steepness
t_0	Logistic time-decay midpoint (days)
Δt	Age in days of a transfer or approval
$\sigma(\Delta t)$	Logistic time weight of a transfer (AWP) or latest approval (EndorseRank)
$V(z)$	Bounded value weight of AWP and EndorseRank on a US-dollar amount z, $2/(1+e^{-bz}) - 1$ with $b = \ln 3/m$, so that the median amount m ($m_T = 170$ dollars for transfers, $m_A = 100$ for allowances) gets $V = 1/2$
U	Cap on one allowance in the allowance layer of C-PR and in the in-approve value (19,000 dollars)
X_u	Activeness of address u, AWP's restart weight
n	Number of contracts in a correlation (matched cohort $n = 6{,}483$; spender cohorts $n = 7{,}914$ in W0 and $7{,}960$ in W1)
ρ	Spearman's rank correlation coefficient
τ	Kendall's rank correlation coefficient; every reported value is τ_b
T_{obs}	End of the score window of the same-window evaluation and score freeze of W1 (30 June 2026, 23:59:59 UTC)
t_1, t_2	Score freeze of W0 (31 March 2026, 23:59:59 UTC) and end of its label window (30 June 2026)
$\mathcal{W}$	Set of contracts in the evaluation cohort
$s(w)$	Reputation score of contract w under a given method
$p(w)$	Proxy value for contract w

1 Introduction and Research Problem

Decentralized Finance (DeFi) has transformed how capital is allocated on public blockchains: lending, borrowing, and leveraged trading can now run without traditional intermediaries (Schär, 2021). Most credit-like transactions are still heavily collateralized, though, because protocols cannot draw on off-chain identity frameworks or existing credit data (Harvey et al., 2021). Under pseudonymity, the central issue is how to allocate capital efficiently among unknown parties.

This thesis asks what ERC-20 allowance edges add to PageRank as a measure of on-chain wallet reputation when one walk follows them together with transfers. The allowance edges come from the `Approval` events that `approve` and `permit` emit. The method of the thesis, a coupled PageRank (C-PR), walks the allowance edges and the transfer edges of the same addresses in one chain and mixes the two at each address with a fixed weight λ. Two single-layer walks serve as its references: EndorseRank, PageRank on the allowance edges alone with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of T. Do et al. (2023), and AWP itself on the transfers. All of them are run on the same Arbitrum One contract accounts, beside the raw degrees that the walks smooth, and evaluated by the rank-correlation method used with AWP (T. Do et al., 2023), extended with paired contrasts, raw-degree baselines and temporal holdouts. In this thesis, an on-chain reputation metric is a value derived from on-chain activity and interactions without any off-chain identity information (H.-D. Do & Do, 2023).

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains make programmable settlement possible without a central authority (Nakamoto, 2008). A smart contract can encode the terms of a deal (Szabo, 1997), and a DeFi platform that uses such contracts clears trades on the blockchain instead of routing them through a custodian

(Buterin, 2014). Protocols then stack these contracts into decentralized markets for lending, automated market making and perpetual futures (Werner et al., 2022). On Arbitrum One and other L2 networks, fees are low enough that ordinary users transact often, and every transfer, approval and protocol event can be inspected wallet by wallet.

Even with this visibility, DeFi credit markets remain inherently over-collateralized. Most protocols make a borrower or trader post more collateral than the position is worth, because the counterparty is a pseudonymous account whose owner has not been verified (Gudgeon, Perez, et al., 2020; Harvey et al., 2021). Pseudonymity is a basic property of public ledgers. Anyone can create addresses at almost no cost, and one entity can own more than one wallet (Douceur, 2002). Conventional credit ratings depend on personal histories that translate poorly to this setting (Packin & Lev-Aretz, 2024), so protocols control risk through collateral reserves, liquidation mechanisms, and protocol rules instead of personalized reputation.

Collateralization protects protocols, but it uses capital inefficiently. Traders and borrowers have to tie up funds they could otherwise invest, and the platforms offer none of the under-collateralized products that traditional finance extends to people with a credit record. Researchers have therefore turned to on-chain signals as a substitute for off-chain credit histories: transfer-graph centrality (H.-D. Do & Do, 2023), network risk propagation (D. Lin et al., 2025), prospect theory and blockchain-based credit scoring (Hassija et al., 2020), and machine learning pipelines over lending protocols (Kotb, 2023). Each of these lines of work assumes that public ledger data contains information usable for risk assessment without deanonymizing users.

Graph ranking algorithms appear well suited to this problem. PageRank and its variants rank nodes by recursive endorsement, so that a node's score depends on the endorsements it receives from other nodes (Page et al., 1999). Earlier web trust algorithms such as EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) apply similar ideas to peer-to-peer trust and to web spam. H.-D. Do and Do (2023) use both PageRank and HodgeRank on a graph of transfers to judge how trustworthy Ethereum users are, and call the result a form of social credit. AWP (T. Do et al., 2023) is another transfer-graph ranking algorithm. It weights each transfer by a bounded function of its

value and a logistic function of its age, and it restarts the walk at addresses that sent recently; its authors validate it by showing that it correlates with inbound transfer count and value, taken as proxies for economic importance. These studies establish transfer-graph reputation as a credible baseline, but they leave out explicit spending authorization, a separate on-chain action recorded as ERC-20 allowances (Lundfall, 2020; Vogelsteller & Buterin, 2015).

1.2 Motivation for on-chain reputation

Four main factors drive the need for on-chain reputation research in leveraged DeFi.

- Capital efficiency: protocols are seeking ordinal ranking signals that could support tiered services (more frequent limit increases, lower fees, or higher priority in monitoring, for example), and perhaps even differentiated collateral requirements in future controlled experiments, while settlement stays on chain, where the contract enforces collateralization ratios and liquidation conditions (Schär, 2021).
- Auditability: off-chain credit scores are opaque, but graph-based scores computed from public event data can be recomputed and audited, which would help to ease some of the concerns about black-box decentralized credit scoring (Packin & Lev-Aretz, 2024).
- Operational cost: a score that is refreshed often, before each leverage adjustment for instance, should be cheap to recompute.
- Standardization: other markets use screening ratios to tell counterparties apart, but pseudonymous DeFi has no standard ordinal language for doing so at the wallet level. Chapter 5 discusses where the scores studied here could serve that role and where they cannot.

Transfer-based AWP is public and auditable, but its graph grows with every payment (T. Do et al., 2023). Allowance graphs are a structurally different alternative. Whenever a user approves an ERC-20 token, the contract records how much of that token a spender may transfer (Lundfall, 2020; Vogelsteller & Buterin, 2015), and each new approval supersedes the older ones, so the latest approval of each owner–spender pair and token stands for the history of that pair. Approvals are

also rarer than transfers. On the same contracts the endorsement graph G_{approve} therefore has fewer than half as many edges as the transfer graph G_{transfer}, and the PageRank solve, whose cost per iteration grows linearly with $|E|$, takes less time (Table 4.1). The same solver runs in both cases, so the speedup comes from the sparser graph and the iterations its walk needs there, and is not an algorithmic improvement; Claim C5 treats it as an operational property of the representation.

Efficiency aside, allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend funds. Treating allowances as endorsement edges carries PageRank's citation reading (Page et al., 1999) over to a trust relationship that protocols already depend on. Whether this produces different rankings, whether those rankings agree differently with authorization and transfer checks, and whether one walk over both kinds of edge ranks contracts better than a walk over either, is examined on a matched cohort of $n = 6{,}483$ contract accounts on Arbitrum One.[1]

1.3 Research problem

AWP infers trust from transfer history through weighted sums and logistic time-decay (T. Do et al., 2023). It (i) must aggregate the whole transfer history, (ii) models a high-value wallet as one that receives many inbound transfers, and (iii) assigns no role to spending authorization. EndorseRank, the allowance reference of this thesis, keeps AWP's edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token. The resulting graph is sparser, encodes a different signal and ranks a different side of each relationship. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals, such as routers and vaults, and gives every owner that receives none the same value. Each of the two walks therefore follows one kind of act: AWP the payments that reach an address, EndorseRank the spending rights it is granted.

[1] Matched means that every method scores the same contracts: those approved by at least three distinct owners on WETH, USDC, USDT0, ARB or WBTC between 1 July 2025 and 30 June 2026, a list drawn from the approval logs alone (Section 3.2).

Because both edge types live on the same addresses, a single walk can follow both, mixing them node by node with a fixed weight λ. This walk, the Coupled PageRank (C-PR), is the method of the thesis; at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The mixture is worth testing because the two edge types record different acts on the same contracts: an operator that already runs a transfer walk could add allowance edges to it instead of running a second score. The main research question is: what does a single PageRank walk over both ERC-20 allowance and transfer edges add to ranking contract accounts on Arbitrum One, compared with walks over either edge type and the raw degrees they smooth? Research questions RQ1–RQ5 (Section 1.6) break it down. The analysis plan fixed one decision rule, and its object is C-PR (Section 3.7).

Agreement with statistics of a score's own edges, or prediction of more of the same edges, does not yet show that a score measures reputation in a substantive sense. A comparison registered with the second holdout window, which adds no research question, therefore also uses an outcome built from neither edge type and observed after the freeze: the later liquidation avoidance of the GMX V2 accounts that are contracts, in small samples (Section 3.7.3); it is applied to the revised scores as a re-analysis.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

The main terms are defined as follows:

- *DeFi (Decentralized Finance):* Financial services such as lending and borrowing, implemented as smart contracts on public blockchains and generally run without a central operator in charge of custody or settlement (Schär, 2021).
- *Smart contract:* Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary (Szabo, 1997).
- *Wallet / Address:* A blockchain account that holds funds and interacts with smart contracts, identified by its address (Buterin, 2014). It is either an externally owned account (EOA), a user's wallet controlled by a private key, or a contract account governed by its code

Table 1.1: The scores evaluated in this thesis (matched contract cohort, $n = 6{,}483$).

Score	Input graph	Role
C-PR	Both layers, US-dollar amounts with each allowance capped, mixed per node with weight λ	The method of this thesis, object of rules A and B. Its layers walked alone, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), are weighted by amounts, so they are not EndorseRank and AWP. Its ablation S-PR restarts a transfer walk from the allowance walk
EndorseRank	Latest ERC-20 allowances of five tokens in US dollars (G_{approve}), weighted as in AWP	Allowance delegation; the single-layer reference on allowances. Its variant with restarts at approving owners is reported beside it
AWP	Transfers of five tokens in US dollars (G_{transfer})	Transfer flow; the single-layer reference on transfers, the published transfer-graph method of T. Do et al. (2023) (Section 3.6.1)
Raw degrees	G_{approve} and G_{transfer}, counted per node	In-approve degree and transfer in-degree, the local counts that the PageRanks smooth: proxies in the same window and baselines out of window

(Antonopoulos & Wood, 2018). The wallets ranked here are contract accounts; EOAs enter only as owners, senders and recipients. No off-chain identity is assumed or recovered.

- *On-chain reputation score:* A number computed from on-chain data that expresses a wallet's relative position for financial activity under pseudonymity (H.-D. Do & Do, 2023). Here the scored addresses are contracts, and the only off-chain inputs are the five-token list and daily US-dollar prices. Scores are ordinal and meant for ranking; their absolute scale has no meaning for credit limits.
- *PageRank:* A graph ranking algorithm that spreads influence through a directed graph with a damping factor (Page et al., 1999). Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders (T. Do et al., 2023); here values are in US dollars, and the value weight is one half at the median amount. EndorseRank applies AWP's edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR):* The method of this thesis: one random walk over both the al-

lowance and the transfer layer, which at each wallet mixes the two rows with weights λ and $1 - \lambda$ over the layers that wallet occupies (Section 3.6.3). Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank (Gyöngyi et al., 2004) restarts from a seed set.

- *ERC-20 allowance / Approve / Permit:* Contract methods that record how many of the owner's tokens a spender may spend on the owner's behalf (Lundfall, 2020; Vogelsteller & Buterin, 2015). `Approve` is an ordinary on-chain call; EIP-2612 `permit` makes the same state change through a signature. For each owner–spender pair and token, EndorseRank uses the latest non-zero allowance.
- *Domain alignment:* The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938) and read as evidence on construct validity (Cronbach & Meehl, 1955); with proxies built from a score's own edges it is only an intended-construct check.

1.4 Temporal holdout

The contemporaneous allowance Kendall τ (Table 4.3) is an intended-construct check: PageRank on G_{approve} should agree with spender-side approval degree, because it smooths that degree globally. To test whether the score conveys information beyond the same-window graph, the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC) and compared with two labels that only become available in April–June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). In this first window, W0, the holdout cohort ($n = 7{,}914$) consists of every contract spender that holds at least one positive latest allowance of one of the five tokens, set in the twelve months before t_1, at t_1 in the extracted approval data. Each PageRank is also compared with the raw degree it smooths at t_1.

A second window, W1, was registered at 16:49 UTC on 9 October 2026, after the W0 results were known and before the study extracted any log dated after 30 June 2026. Its scores are frozen on

30 June 2026 and its labels counted from July to September 2026. It repeats the two spender labels on the $n = 7{,}960$ contract spenders of that freeze and adds one for the GMX V2 traders that are contracts ($n = 70$): the share of their closes that did not end in liquidation (Section 3.7.2).
The analysis plan of 9 October 2026 and the registration of W1 define the registered analysis, with amounts in raw base units of every token. After its results were known, the revised analysis was defined, with amounts in US dollars on five widely traded tokens and scores from the twelve months before each freeze (Section 3.5.1). The revised analysis is the main one; its verdicts under the rules of the registered analysis are a re-analysis, not registered tests, and the registered analysis reached the same verdicts except on R1 (Appendix 8.6).

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with the measurement used to operationalize it and with the criterion that Chapter 5 applies when judging whether the objective was met.

O1. Establish whether EndorseRank and AWP, the two single-layer walks, order the contracts of the same cohort differently, and assess how closely each score agrees with each construct in the same window (RQ1). Measurement: Kendall τ between the two score vectors ($n = 6{,}483$) with a 95% bootstrap interval, and family-level Kendall τ with 95% intervals for every score and construct. Criteria: the interval of the agreement between the two walks contains neither 0 (unrelated orderings) nor 1 (identical orderings); a score agrees with a construct when its interval on that family excludes 0, and agreement with the construct of the score's own edges is read as an intended-construct check, because those proxies come from the same events as the graph.

O2. Measure what each score costs to compute on the same contracts, C-PR on the union of both graphs included, and how runtime responds to the number of sampled contracts (RQ2). Measurement: solve time (mean and standard deviation of five timed runs after one warm-

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1, H2 / RQ1	Do EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, order the same contracts differently, and does each agree most with the checks of its own edge type?	§4.3 and §4.4; Tables 4.3 and 4.4; Claim C4
H3 / RQ2	What does each walk cost to compute at equal n, and how much does walking both edge sets add?	Tables 4.1 and 4.2; Claim C5
RQ3	Do t_1 scores predict t_2 new approvals and new senders, and does any walk, the coupled one included, do better than the raw degree of the same edge type?	§4.6 and §4.7.4; Tables 4.9, 4.10, 4.11, 4.12, 4.15, 4.13 and 4.18; Claims C2 and C3
RQ4	Does C-PR rank contracts better than either of its layers walked alone, in window and out of window (rule A)?	§4.7.1 and §4.7.2; Tables 4.3, 4.14 and 4.10; Claim C1
RQ5	Does adding the allowance layer improve a transfer walk on new data (rule B)?	§4.7.3; Tables 4.15 and 4.16; Claim C1
C1	Under rules A and B, applied to the revised scores, C-PR beats neither layer walked alone and does not improve on its transfer layer within the margin (the gain on new approval pairs holds, non-inferiority fails on new senders and narrowly on the liquidation-free close rate); against AWP it fails clearly; in S-PR allowance edges help mainly when they carry the walk	Tables 4.14, 4.10 and 4.16
C2	Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows and new approval pairs above AWP, largely through the spenders AWP scores zero; it ranks new senders below AWP and the transfer in-degree, and new approval pairs below the in-approve degree and, in W1, below EndorseRank with activity restarts (contrasts computed after the labels were known)	Table 4.18; §4.7.4
C3	Out of window neither single-layer walk beats the raw degree it smooths: AWP ranks later new senders below the transfer in-degree and EndorseRank later new approval pairs below the in-approve degree, in both windows and with either restart rule (contrasts computed after the labels were known)	Tables 4.9, 4.10, 4.11 and 4.13
C4	Same-window agreement of each single-layer walk with the checks of its own edge type, partly built in, and different orderings of the cohort by the two walks	Tables 4.3 and 4.4; Figure 4.3
C5	Runtime and graph size of each score: speed follows graph size and the iterations each walk needs, not the algorithm; C-PR and S-PR are slower than AWP	Table 4.1
C6	On later liquidation avoidance of GMX V2 contract traders, a count-level lead of al-	Tables 4.19 and 4.20

up run), memory, iterations, $|V|$ and $|E|$, all under one protocol. Criterion: the time ratio is reported only together with the ratio of $|E|$, so that any difference can be traced to the size of the graph.

O3. Test whether scores frozen at t_1 predict new approvals and new transfer senders at t_2 among the cohort's spenders, and whether any PageRank, the coupled one included, does better than the raw degree of the same edge type (RQ3). Measurement: Kendall τ with 95% intervals between every score computed at the freeze, the two raw degrees included, and both labels, in W0 (freeze 31 March 2026, labels April–June 2026, $n = 7{,}914$ contract spenders) and W1 (freeze 30 June 2026, labels July–September 2026, $n = 7{,}960$), with paired contrasts of each single-layer PageRank against its own degree and of C-PR against both degrees. Criterion: a score predicts a label when its interval excludes 0, and a PageRank improves on a degree only when the interval of the paired difference lies above 0. Neither the degree contrasts nor this criterion was fixed before the W0 labels were known.

O4. Test whether one walk across both edge types beats each of its layers walked alone (RQ4). Measurement: C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and its ablation S-PR, compared through paired intervals on $\Delta\tau$ with the two layers walked alone, on the holdout and same-window labels. Criterion: rule A, fixed in the analysis plan of 9 October 2026 before any log of the study was extracted, under which C-PR does no worse than the allowance walk on new approvals and better than the transfer walk on new senders in W0, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on Sybil stability. The rule is applied to the revised scores as a re-analysis.

O5. Test whether adding the allowance layer to a walk over the transfers improves it on new data (RQ5). Measurement: paired contrasts of C-PR with its transfer layer walked alone in W1, on new approval pairs, new senders and the traders' liquidation-free close rate. Criterion: rule B, registered on 9 October 2026 after the W0 results were known and before the study extracted the W1 logs, under which C-PR beats the transfer walk on new approval pairs in

W1 and is not worse by more than 0.01 on new senders or on the traders' liquidation-free close rate. The rule is applied to the revised scores as a re-analysis. A registered sensitivity analysis repeats it with AWP in place of the transfer walk; it does not change the rule. The contrasts of C-PR with EndorseRank, AWP and the other scores are reported beside both rules and decide neither.

1.6 Research questions

Three hypotheses go with the first two research questions. H1: on the same set of contracts, EndorseRank and AWP produce different orderings. H2: each score agrees with the checks built from the edges it walks on. H3: at the same value of n, a score's computation time grows with the size of its graph, so EndorseRank, on the sparsest graph, is the fastest to compute. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ3–RQ5, the last two answered by rules fixed before the study extracted the relevant data and re-applied to the revised scores (Section 3.7).

RQ1. On the same set of Arbitrum contracts, do EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, produce different orderings, and how closely does each score agree with current allowance checks and with current transfer checks (Spearman's ρ and Kendall's τ)?

RQ2. What does each score cost to compute on the same set of contracts, as measured by time, memory, and iterations, and how much does walking both edge sets add?

RQ3. Given scores computed at time t_1, do the scores predict t_2 new owner–spender approvals to inbound spenders and new transfer senders, and does any PageRank, the coupled one included, predict them better than the raw degree of the same edge type at t_1?

RQ4. Does a coupled PageRank running on both the allowance and the transfer layer rank contracts better than a walk over either layer alone, in sample as well as out of sample?

RQ5. Does adding the allowance layer to a walk over the transfers, such as AWP, improve it out of

sample?

RQ3 uses later labels of the same two constructs because no default or repayment label exists for the contracts ranked here (Section 2.8). RQ1 and RQ2 establish the two single-layer walks and what they cost; RQ4 and RQ5 move from comparing the two edge types to combining them. The rule for RQ5, rule B, came after the W0 labels had been seen, so it is judged only on W1 (Sections 3.7 and 3.7.2).

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank (Page et al., 1999) ranks nodes by recursively propagated endorsement at a per-iteration cost linear in the number of edges (Langville & Meyer, 2006), and EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) adapt it to peer-to-peer trust and web spam with normalization and seed sets. In on-chain reputation, H.-D. Do and Do (2023) apply PageRank and HodgeRank to Ethereum transactions, and AWP (T. Do et al., 2023) weights transaction edges by value and age and checks its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. D. Lin et al. (2025) propagate account risk through the transaction network, and surveys of blockchain trust and reputation systems (Bellini et al., 2020) and of reputation systems built on smart contracts (Almasoud et al., 2020) describe designs that rest on transfers or explicit ratings. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person (Human Passport, n.d.). None of this work takes the ERC-20 allowance (Vogelsteller & Buterin, 2015) as a ranking edge; address clustering uses approvals only to link addresses of one entity (Victor, 2020). In DeFi risk and measurement, systematizations classify risk by protocol layer (Werner et al., 2022) and by the attacks seen in practice (Zhou et al., 2023), and wash trading (Cong et al., 2023; Victor & Weintraud, 2021), extractable value (Qin et al., 2022), scam tokens (Xia et al., 2021) and airdrop farming (Luo et al., 2025) show that much transfer activity is artificial. In measurement theory, construct validity (Cronbach & Meehl, 1955; Messick,

1995) separates what a score measures from external criteria, and the bootstrap (DiCiccio & Efron, 1996; Efron, 1979) gives distribution-free intervals for rank statistics. The gap (Section 2.21) is a walk that couples allowance and transfer edges, evaluated on one cohort beside the walks over each edge type and the raw degrees, with confidence intervals, paired differences, a fixed timing protocol and temporal holdouts.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 `Approval` and `Transfer` logs on Arbitrum One from 1 October 2023 to 30 September 2026 are extracted from Google BigQuery in two scans, the second after W1 was registered. The main analysis keeps the events of five tokens in the twelve months before each freeze around the cohort of that freeze (the matched cohort, $n = 6{,}483$, at T_{obs}), valued in US dollars; these choices were made after the registered results were known (Section 3.5.1). One power-iteration PageRank implementation scores the coupled walk over both layers, the latest-allowance graph and the transfer graph. Each score is compared with Spearman's ρ and Kendall's τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approval pairs and new transfer senders on the spender contracts of W0 and W1, with the raw degrees at each freeze as baselines. Except in the value-slope and sample-definition checks, every coefficient has a 95% interval from 400 paired contract resamples (seed 42), and method comparisons are paired. A comparison registered with W1 sets allowance-side against transfer-side scores on the later liquidation avoidance of GMX V2 contract traders in both windows (2,000 paired resamples; Bonferroni 97.5% intervals for its two decision contrasts). The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.13), and the study uses only public data (Section 3.16).

1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation and DeFi risk. EndorseRank and AWP both apply weighted PageRank (Page et al., 1999) to a single edge type, each with its own restart distribution; the coupled walk C-PR changes the transition rule itself. What is new is therefore a transition rule over two on-chain relations, its test under rules fixed in advance, an edge type, an evaluation design, and an analytic account of what such a walk computes on this sample.

- Coupled walk over authorization and payment (C-PR): one random walk over a US-dollar allowance layer, with each allowance capped, and a US-dollar transfer layer, which at each address mixes the two transition rows with a fixed weight λ over the layers that address occupies; S-PR is its seeded ablation. No reputation method reviewed couples the two relations (Section 2.21).
- Its test under fixed rules: rule A, fixed with λ in the analysis plan, asked C-PR in its primary part to do no worse than its allowance layer on new approval pairs and better than its transfer layer on new senders, and rule B, registered with W1, asked it to improve on its transfer layer; only these contrasts change the coupling and nothing else. On the revised scores, in a re-analysis, both fail as in the registered analysis (Appendix 8.6), though the gain over the transfer layer on new approval pairs recurs in W1 (Claim C1). Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows, but it ranks later senders below AWP and later approval pairs below the in-approve degree, and its lead over AWP on later approvals comes largely from contracts that AWP cannot order (Claim C2). The ablation shows that the allowance signal survives when allowance edges carry the walk and is mostly lost when they only decide where it restarts.
- Latest-allowance edge construction (EndorseRank), the allowance reference: an endorsement graph in which an edge $u \to v$ stands for the latest in-window ERC-20 allowance, set by `approve` or `permit`, that the spender v holds from the owner u, valued in US dollars and weighted as AWP weights a transfer. The edge records a delegation of spending rights, not a history

of payments. Approvals have been used to cluster the addresses of one entity (Victor, 2020), but no reputation method reviewed ranks on them (Section 2.21).

- Same-cohort evaluation with out-of-window tests and degree baselines: every score is computed on one set of contracts and checked against identical transfer, allowance and Sybil-stability proxies under the protocol used with AWP (T. Do et al., 2023), with 95% bootstrap intervals for the coefficients and paired intervals for the contrasts (Efron, 1979); scores frozen at t_1 are tested against the new approvals and new transfer senders that arrive after t_1, beside the raw degrees that PageRank smooths, which separates in-sample agreement from prediction and the propagated score from its local count (Claims C2 and C3). In contrasts computed after the labels were known, no PageRank beat its degree out of window: AWP ranked later senders below the transfer in-degree, EndorseRank, with either restart rule, ranked later approval pairs below the in-approve degree, and C-PR ranked neither label above the degree of the same edge type. A comparison registered with W1 adds an outcome built from neither edge type: on later liquidation avoidance the approval count led the transfer count in the registered window, though not resolvably in W0, and the like-for-like PageRank pair is not resolved, in small trader samples (Claim C6).
- Closed form on a depth-one graph: with uniform restarts and uniformly redistributed dangling mass, a spender whose in-neighbors have no in-edges scores $c\,(1 + d\,\delta(v))$, where c is the common score of nodes without in-edges, d the damping factor and $\delta(v)$ the sum of the transition probabilities into v; in EndorseRank this is the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves. The proposition needs only the uniform restarts, so it describes C-PR's walk as well. On EndorseRank's graph it holds to within 1% for every depth-one spender but for only a minority of the deeper ones, so multi-hop propagation is present (Section 5.3.2). Besides the time and token weights, that normalization and this propagation separate EndorseRank from the raw approval count, but over all spenders they did not make it rank later approvals above that count, and deeper spenders were not tested apart.

1.10 Scope and limitations

The study extracts Arbitrum One from 2023-10-01 to 2026-06-30, W1 adds July to September 2026, and the main analysis scores the twelve months before each freeze. Its main sample is the matched cohort of $n = 6{,}483$ contracts, and its holdout samples the contract spenders of each freeze ($n = 7{,}914$ in W0, $n = 7{,}960$ in W1). The scored wallets are contract accounts, because user wallets serve too many different purposes for their activity to be put in one order (Section 3.2); account type is read from the code at a pinned block, not at the time of each event. Only events of the five tokens with a cohort contract at one end enter the graphs, so each graph is the ego network of the cohort, and a multi-hop path is seen only where each of its edges touches a cohort contract (Section 5.3.2). Few GMX V2 contract accounts in the freeze-date graph closed at least three positions in these windows ($n = 87$ in W0, $n = 70$ in W1), so every trader result has wide intervals (Section 4.8). Runtime is also measured on subsamples of the cohort. No claim is made that the results generalize beyond Arbitrum One or the months studied. Amounts in US dollars and the twelve-month score windows were chosen after the registered results were known (Section 3.5.1); EndorseRank and AWP pass them through a bounded function that gives the median amount half weight, while the layers of C-PR use the amounts themselves, with each allowance capped (Chapter 3). Adversarial manipulation of allowances or transfers is a real threat, because addresses are cheap to create (Douceur, 2002): Section 5.6 shows that a small farm of Sybil addresses can lift a contract's score on the allowance graph above most of the cohort cheaply, while reaching the top is expensive, that the bound behind this holds for C-PR as well, and that the Sybil-adjusted proxies describe some manipulation patterns without protecting the score against them.

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the cohorts, the tokens, prices and the revision of the amounts, the graphs and scores, the construct

checks, the temporal holdouts, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the benchmark (Claim C5), the same-window alignment of every score and the divergence of the two single-layer walks (C4), robustness checks, the temporal holdouts of the single-layer walks (C3), the coupled walk under its rules and beside every other score (C1 and C2) and the comparison on liquidation avoidance (C6); Appendix 8.5 collects the direct comparisons between EndorseRank and AWP, and Appendix 8.6 reports the registered analysis. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.

2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that transfer-graph reputation scores give no role to spending authorization, an explicit act of trust that every ERC-20 token records, and that their graphs grow with every payment. The objectives O1–O5 and research questions RQ1–RQ5 follow from it. To study a PageRank walk that follows allowances and transfers together, and to compare it with the walks over each, EndorseRank and Adaptive Weighted PageRank (AWP), we run each method's graph through PageRank and check the resulting rankings with rank correlations. The gap that remains is taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it (Nakamoto, 2008). Once a majority of the participating nodes has accepted a block, rewriting it is costly. That cost allows parties to settle on the ledger without each of them needing a known legal identity (*trust-minimized* settlement).

Bitcoin built a peer-to-peer electronic cash network on this kind of ledger (Nakamoto, 2008). Ethereum keeps the append-only ledger but lets an account hold a program alongside its balance. A transaction can then call one of the program's functions and change the contract's state (Buterin, 2014), and these functions run on the Ethereum Virtual Machine (EVM) (Wood, 2025). Programs stored on the ledger in this way are called smart contracts (Szabo, 1997); any honest node that replays a function call gets the same output (Wood, 2025). As a result, collateralization ratios, liquidation conditions and fee schedules can be enforced by the contract instead of by a trusted third party (Schär, 2021).

Collectively, these contracts make up Decentralized Finance (DeFi). Between them they do the

work of exchanges, lending platforms, derivatives markets and asset managers; Werner et al. (2022) define the protocol classes, and Schär (2021) describes how protocols are layered on one another. Because one protocol can interact with another, new financial instruments can be composed from existing ones (Kitzler et al., 2023). The same property lets a failure spread across several protocols. Successful attacks have included flash-loan exploits (Qin et al., 2021) and price-oracle manipulation (Zhou et al., 2023). Transaction histories on these chains are public, yet the addresses in them are pseudonymous, which makes the identity checks of conventional finance hard to carry out (D. Lin et al., 2025).

2.1.1 Pseudonymity and the credit problem

Conventional credit scoring ties a credit history to a legal entity. On a public blockchain the basic unit of analysis is instead an address (a wallet), which may stand for a legal entity, a trust, an algorithmic agent or a smart contract. One entity can own several addresses (Douceur, 2002), and statistical clustering can recover ownership only in part (Victor, 2020). On-chain lending therefore requires borrowers to post collateral worth more than the amount they borrow; if the collateral value falls below a set threshold, a liquidator may repay part of the debt and receive the corresponding collateral at a discount (Bartoletti et al., 2021). The arrangement is secure, but it ties up capital. An on-chain credit score tries to quantify from on-chain behavior alone how likely an actor is to repay. Recent proposals draw on prospect theory (Hassija et al., 2020), on machine learning applied to lending data (Kotb, 2023; Palaiokrassas et al., 2024), and on models of risk spreading through account networks (D. Lin et al., 2025).

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications have been deployed on the Ethereum mainnet, historically the main venue for DeFi innovation and liquidity, but while Ethereum ran on proof of work, it could not raise its block size or shorten its block interval far without weakening its security (Gervais et al., 2016), so growing use of the mainnet raised gas costs and constrained throughput. Layer-2 (L2) proto-

cols try to keep Ethereum's security guarantees while making transactions cheaper. Some execute transactions off-chain, while others batch them and post a succinct summary of the execution to the Ethereum mainnet (Gudgeon, Moreno-Sanchez, et al., 2020). Thibault et al. (2022) survey rollups and compare their implementations. Optimistic L2s such as Arbitrum treat each submitted batch of transactions as valid unless evidence against it is presented within a dispute period (Kalodner et al., 2018).

Arbitrum One is an optimistic L2 of this kind and can execute EVM-based smart contracts. It has become a popular platform for DeFi applications, among them GMX, an exchange for perpetual swaps (GMX, n.d.). We use Arbitrum One because the public BigQuery dataset includes both its ERC-20 `Approval` and `Transfer` events, and because its fees are low enough that the windows studied contain dense activity. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further validation. The framework can be applied to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of `transfer`, `approve`, `transferFrom`, `allowance` and the associated events (Vogelsteller & Buterin, 2015). After the owner calls `approve(spender,amount)`, the token contract records that the spender may move up to `amount` out of the owner's balance. The spender can then call `transferFrom` to carry out a token transfer. With EIP-2612 `permit`, an allowance can be updated through an off-chain signature that settles in a single transaction (Lundfall, 2020). The semantics of authorization are unchanged.

Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and involves no actual transfer. Token owners can grant allowances to routers, vaults or trading contracts that they trust to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force, and `transferFrom` reduces

it; some tokens record the remainder in a new `Approval` event. EndorseRank keeps, for each owner, spender and token, only the most recent approval, so a newer allowance replaces the previous one outright, and the surviving approval loses weight as it ages even while it stays in force (Section 2.7).

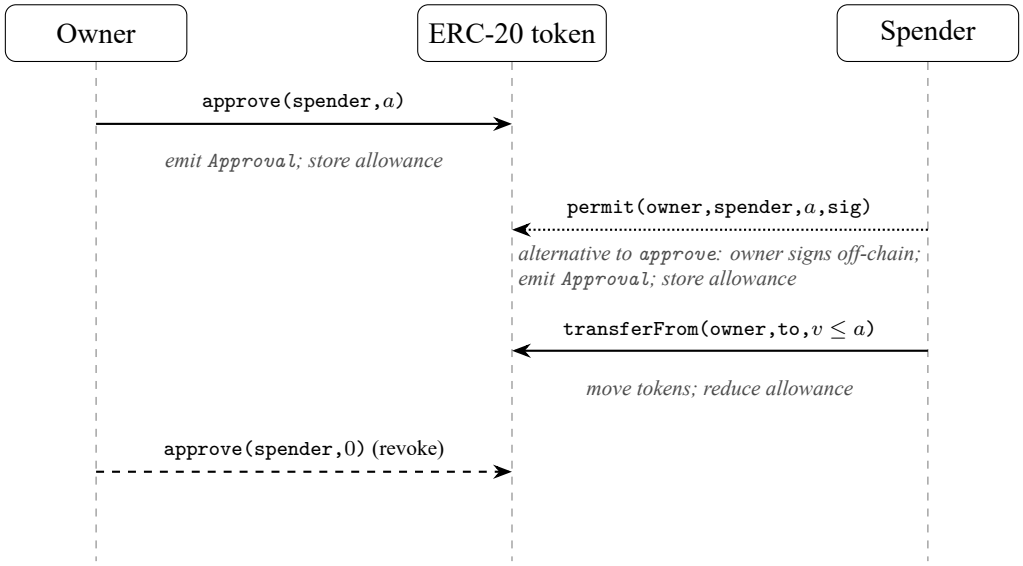


Figure 2.1: ERC-20 allowance lifecycle: `approve`, or its signature-based alternative `permit` (dotted), an optional `transferFrom`, and revoke. EndorseRank keeps the most recent `Approval` event of each owner, spender and token in the score window and weights it by its age and US-dollar amount; a later approval replaces an earlier one.

In the available data, allowances are sparse compared with transfers. A single approval can cover many transfers, and many wallets have never granted an approval to any external account. The sparsity lowers the cost per iteration, and it also marks a semantic difference, since EndorseRank quantifies authorization to spend whereas a transfer graph records payment flows.

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges (Page et al., 1999). Two recent studies apply this approach to blockchain transaction graphs. H.-D. Do and Do (2023) run PageRank and HodgeRank on Ethereum transactions as a measure of social credit, meant for people who lack a credit history at financial institutions. T. Do et al. (2023)

propose Adaptive Weighted PageRank (AWP), the transfer-graph method that this thesis evaluates beside EndorseRank (Section 2.18). D. Lin et al. (2025) treat the risk rating of Ethereum accounts as a node-ranking problem and propagate risk through the transaction network, so that accounts without labels are rated as well, and Wu et al. (2022) detect phishing addresses from a network embedding of the transaction graph that takes the amounts and times of transactions into account. Bellini et al. (2020) survey blockchain-based trust and reputation management, and Almasoud et al. (2020) review reputation systems built on smart contracts, in which users rate how trustworthy the providers of online services or products are. Data mining on cryptocurrency transactions likewise works from transaction histories, for address linking and for the study of user behavior (Liu et al., 2021). Packin and Lev-Aretz (2024) warn that decentralized credit scores can turn biased decisions into automated processes that outsiders cannot inspect.

Outside the research literature, trust in deployed DeFi and web3 systems is usually established another way. An address builds standing through activity that anyone can inspect on the ledger, and tooling bound to the address summarizes it. One such tool, Human Passport (formerly Gitcoin Passport), collects verifiable credentials, called stamps, for checks such as government identity, biometrics, vouching and earlier web2 and web3 activity. Each stamp carries a weight that the documentation describes as the strength of its human signal, and an address passes when the weighted sum reaches a threshold (Human Passport, n.d.). According to the same documentation, Passport also classifies EVM addresses on several networks, Arbitrum among them, as human or Sybil from their on-chain activity (Human Passport, n.d.). Its purpose is Sybil resistance, so it combines on-chain with off-chain evidence and works as a gate and not as a graded ordering. A review of proof-of-personhood protocols describes how they resist Sybil attacks with subjective inputs such as vouching and voting in place of attestations from central issuers (Siddarth et al., 2020), and a recent review of Sybil attacks on blockchains traces airdrop farming on layer-2 networks such as Arbitrum and assesses proof-of-personhood systems among the defenses (Bordeianu & Popescu, 2026). Such tools answer whether an address is controlled by a distinct person. EndorseRank answers a different question, which addresses other addresses authorize to spend their tokens, and it

orders them from on-chain records alone (Table 2.1).

Graph-based methods are attractive because they (i) use only public information, (ii) produce continuous scores that suit ranking, and (iii) admit complexity analysis by number of edges. They are, however, vulnerable to Sybil attacks (Douceur, 2002) and wash trading, and they suffer from the mismatch between graph centrality and actual default risk; PageRank scores are not credit scores in the regulatory sense (Section 2.8).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the Web graph, PageRank is on average proportional to in-degree and can be approximated from it (Fortunato et al., 2008). New links concentrate on nodes that already have many, as preferential attachment assumes (Barabási & Albert, 1999) and measured network growth shows (Newman, 2001), so link prediction, on Ethereum transaction networks too (D. Lin et al., 2022), uses degree-based predictors as standard baselines (Liben-Nowell & Kleinberg, 2007).

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j. That weight stands for influence from i to j, in either the endorsement or the transfer sense introduced above. Let W be the corresponding weighted adjacency matrix, and let $o_i = \sum_j w_{ij}$ be the out-strength of node i. The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \tag{2.1}$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d\, P^{\top} + \frac{1-d}{|V|}\, \mathbf{1}\mathbf{1}^{\top}, \tag{2.2}$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ (Langville & Meyer, 2006; Page et al., 1999). Element by element, with the dangling rows of P written as a separate middle term, this becomes

$$r_j = d \sum_{i:(i\to j)\in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:\, o_i=0} r_i + \frac{1-d}{|V|}. \tag{2.3}$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d, choosing it in proportion to its weight, or teleports to a uniformly chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique (Langville & Meyer, 2006). Appendix 8.3.1 gives the argument.

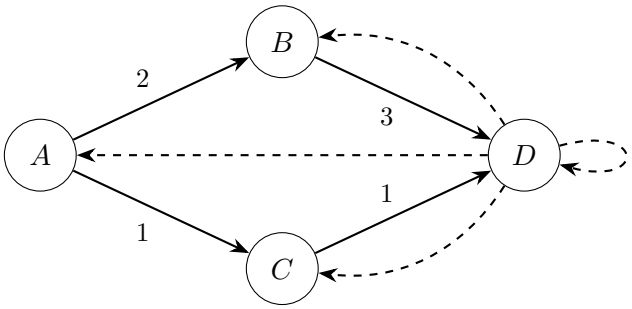


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.13; edge labels are the allowance weights. Solid arrows are graph transitions: with total probability d the surfer follows one of them, chosen in proportion to its weight. Dashed arrows show uniform teleportation, which reaches every node (the current one included) with probability $(1 - d)/|V|$; the dashed fan is drawn only from D, although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node, each with probability $1/|V| = 0.25$. That rule, and not damping, keeps the mass that reaches D in the chain.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\mathrm{PR}}\mathbf{r}^{(t)}, \tag{2.4}$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap $I_{\max}$ is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations (Leskovec et al., 2020). A method that generates fewer edges, for instance by using latest-allowance snapshots instead of full transfer histories, therefore lowers the cost per iteration and sometimes the memory required.

By default this work sets damping to the conventional $d = 0.85$ (Page et al., 1999) and uses two study-specific solver settings, a convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of $I_{\max} = 300$ iterations.[1] Damping is varied over $d \in \{0.75, 0.85, 0.95\}$ in the robustness section.

2.6 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. Kleinberg's HITS algorithm assigns each page two scores, one as a hub and one as an authority (Kleinberg, 1999). EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node cannot raise its own score at will (Kamvar et al., 2003). To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward (Gyöngyi et al., 2004). Reputation is relational in all of these methods. Where trust flows only from a vetted seed set, as in TrustRank, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker. Gyöngyi and Garcia-Molina (2005) derive the link farms and farm alliances that most raise a target's PageRank, and Cheng and Friedman (2005) show that symmetric reputation functions such as PageRank can be manipulated with Sybils, whereas some asymmet-

[1] The tolerance and the iteration cap were chosen to keep run-times reproducible; they are not put forward as part of the classical PageRank formulation.

ric functions based on flows or paths from a trusted node resist them. The Sybil cost model of Section 5.6 prices such a farm on an allowance graph.

On-chain, the Sybil attack remains a central concern (Bordeianu & Popescu, 2026). New wallets cost little to create, whereas wallets that receive allowances or transfers from distinct counterparties cost more. The Sybil-adjusted stability proxies used in this work (inbound counterparty ratio, tenure, active months) partly capture that diversity of interaction, but they do not prevent a malicious actor from manipulating them.

Machine-learning credit models fitted to DeFi lending data (Kotb, 2023) learn from labeled credit outcomes, and an ensemble of fitted models is harder to audit than a fixed rule. EndorseRank and AWP keep to fixed propagation rules that anyone can recompute from the public logs.

2.7 ERC-20 allowances as endorsement

Through the ERC-20 `approve` function (Vogelsteller & Buterin, 2015) and EIP-2612 `permit` (Lundfall, 2020), users state explicitly how much a spender may move on their behalf. EndorseRank reads the latest allowance from an owner to a spender as one endorsement. On each of the five tokens of the study (Section 3.5.1), the latest in-window approval of the pair contributes its time weight times the bounded value weight of its US-dollar amount, as AWP weights a transfer; a pair whose latest value is zero (a revocation or, for tokens whose `transferFrom` emits `Approval`, an allowance spent in full) is dropped, and the terms are summed over tokens (Equation (3.7) in Section 3.6.2). PageRank on the resulting graph G_{approve}, which is sparser than the transfer graph over the same contracts and window, yields the EndorseRank scores $\mathbf{s}_{\text{ER}}$.

Treating allowances as endorsements is a deliberate choice, and an imprecise one. Users approve routers because it is convenient, not necessarily to make a social statement about whether the router is trustworthy. An allowance manager such as Uniswap's Permit2, which owners approve once so that applications can spend through it, collects the endorsements meant for those applications (Section 3.6.2). Contract addresses and EOAs coexist on the owner side as well. Even so, an

allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading (Cong et al., 2023), airdrop farming (Luo et al., 2025) or routing without any intent to endorse.

Approvals have been read as a relation between addresses before: Victor's self-authorization heuristic uses ERC-20 approvals to cluster addresses that it attributes to one entity (Victor, 2020). That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-endorsement, the edge that the Sybil cost model of Section 5.6 prices at gas alone.

2.8 What this research does not claim

Trading outcomes enter this thesis only out of window: in an exploratory run on the GMX V2 contract traders of W0 (Section 4.6.1), as the liquidation-free close rate of the registered window W1 (Section 3.7.2), and in a comparison of the two edge types (Section 4.8). That comparison needs an outcome built from neither edge type. Construct validity expects a score to agree with checks of the construct it was built to measure (Cronbach & Meehl, 1955; Messick, 1995), so each edge type's lead on its own checks is partly built in; the GMX V2 trader outcomes give neither score that head start, and for counterparty-risk reputation the relevant one is forced liquidation. No score in this thesis is offered as a predictor of trading profit, loss avoidance or liquidation, or as a measure of creditworthiness or default risk. No score is a security audit of a contract's code either: a contract can contain malicious functions whatever its standing among owners (Z. Lin et al., 2024), and the scores read neither code nor upgrade permissions.

2.9 Rank correlation theory

Reputation scores and the proxies of the construct checks are ordered quantities, and neither has a natural boundary for classification. Following the AWP validation design (T. Do et al., 2023) and classical psychometrics (Cronbach & Meehl, 1955), this research measures two kinds of rank agree-

ment, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every coefficient of the main analysis (Efron, 1979).

Spearman's rank correlation (Spearman, 1904) is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} (R_i - S_i)^2}{n(n^2 - 1)}. \tag{2.5}$$

This no-ties form does not hold when ties occur. Spearman's ρ is then the Pearson correlation of average ranks (mid-ranks), in which tied items share the mean of the positions they occupy, and every reported ρ is computed this way. In both forms ρ measures monotonic association.

Kendall's τ (Kendall, 1938) rests on counts of concordant and discordant pairs. Of the $n_0 = n(n-1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k-th group of tied values in the first ranking, and define n_2 in the same way for the second ranking. The τ-b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \tag{2.6}$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b, computed on the raw scores and proxies. The difference matters for heavily tied scores, such as the raw degrees, which are integer counts; EndorseRank on the matched cohort leaves few ties (Section 4.4). For agreement about pairwise orderings, Kendall's τ is usually the easier coefficient to interpret, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

2.10 Theoretical framework

This study first compares the two single-layer walks, EndorseRank and AWP, on reputation structure (H1), construct alignment (H2) and computational efficiency (H3), and sets every score against the raw degrees out of window (RQ3); it then tests the walk that couples their layers (RQ4 and RQ5). Three bodies of theory frame the first part: hyperlink–citation propagation, construct (domain) alignment, and computational complexity. Two further arguments bear on the second part.

- Hyperlink–citation semantics: In the original PageRank algorithm (Page et al., 1999), an inbound hyperlink counts as a citation. H.-D. Do and Do (2023) apply this reasoning to Ethereum transfers, and AWP (T. Do et al., 2023) weights each transfer edge by the age and the value of its transfers, through a logistic decay in age and a bounded transform of value (Section 3.6.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges. An allowance is an explicit, on-chain delegation of spending authority that remains in force until it is spent, revoked or replaced (Vogelsteller & Buterin, 2015), whereas a transfer may reflect trading or routing without any intent to trust. Graph-based risk models propagate risk through the transaction network to rate accounts that carry no label (D. Lin et al., 2025); whether a directed graph of authorizations adds to such propagation over transfers is the question taken up here.
- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track (Cronbach & Meehl, 1955). Messick (1995) widens the idea into one unified judgment about what a score means and what using it implies. Graph coherence is checked with rank correlations (Section 2.9) against contemporaneous transfer and allowance statistics. The out-of-window checks are temporal holdouts: new approvals and new transfer senders for spenders, and, in the registered July–September window W1, the liquidation-free close rate for traders.
- Computational complexity: Each PageRank iteration costs $O(|E|)$ (Section 2.5.2), so on the sparser allowance graph an iteration needs less time and memory; this argument underlies

H3. The number of iterations is a separate factor: the damping factor bounds its rate of convergence (Langville & Meyer, 2006), but the count also depends on the graph and the restart vector. Both methods use the same solver, so a difference in runtime comes from the graphs and the restart vectors and does not count as an algorithmic contribution (Appendix 8.3.3).

- Local counts as a baseline: Where contracts that already have many counterparties gain the most new ones (Section 2.4), the raw degree is a strong predictor of future arrivals. A PageRank differs from it in three ways: the endorsers' own scores propagate along the edges, the edges are weighted by age and amount, and the walk restarts by a rule of its own; a contrast with the degree tests the three together. RQ3 therefore sets each PageRank against the raw degree of its own edge type at the freeze date and does not credit a difference to propagation alone.
- Layer coupling: Multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) treat several relations as layers over one node set. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ4 and RQ5 therefore ask two things: whether the mixture beats both layers, and whether it improves on its transfer layer walked alone. Both layers weight edges by US-dollar amounts without AWP's bounded value weight and are walked with uniform restarts, so neither walk is AWP or EndorseRank, and C-PR is compared with AWP separately (Section 4.7.4). Coverage matters there: a transfer walk leaves all nodes without transfer edges in one tie, and any walk over the allowance edges, EndorseRank included, orders them, so coupling can add coverage over a transfer walk even if it adds nothing where both layers reach.

The conceptual model in Figure 2.3 links the scores of the thesis to H1–H3 and RQ1–RQ5. The two single-layer walks feed H1–H3, RQ1 and RQ2, and the holdout against the raw degrees (RQ3); the dashed box on the right is the method of the thesis, the coupled walk C-PR, which takes input

from the construct checks and the holdout and asks whether the two constructs combine (RQ4 and RQ5).

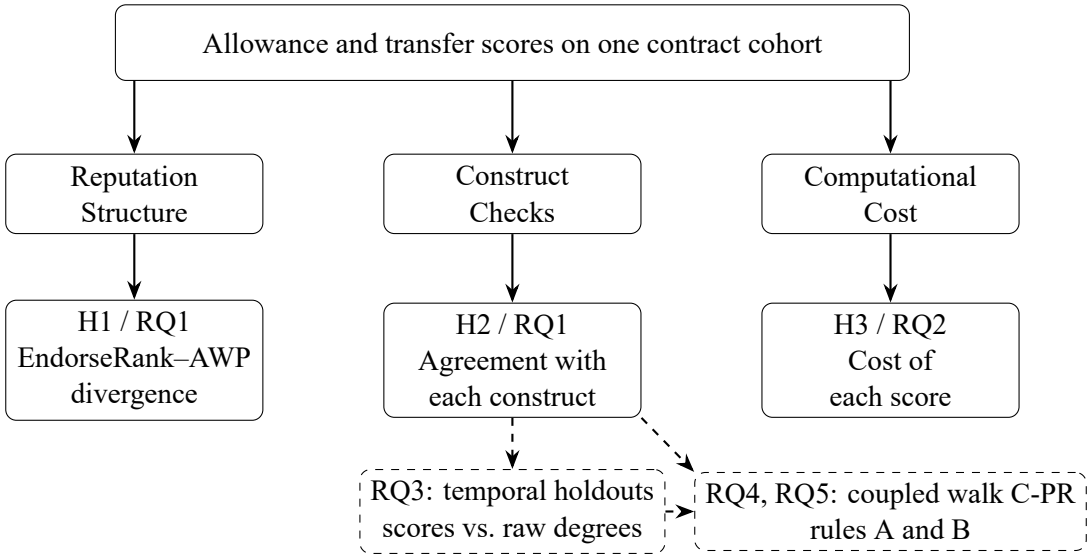


Figure 2.3: Conceptual model linking the allowance and transfer scores of one contract cohort to reputation structure and construct checks (H1, H2/RQ1), computational cost (H3/RQ2) and the temporal holdout, in which each PageRank is set against the raw degree it smooths (RQ3). The dashed box on the right is the method of the thesis, the coupled walk C-PR, tested under rules A and B (RQ4, RQ5).

Only EndorseRank and AWP enter H1. H2 and H3 concern every score. RQ3 is a time-split test on the spender cohort, with the raw degrees at the freeze as its baselines, and does not rank trading outcomes; RQ4 and RQ5 test C-PR under rules A and B.

2.11 Terminology

The formal constructs are defined in Chapter 1. The terms below are used when results are interpreted:

- Single-layer methods: EndorseRank (allowance graph) and AWP (transfer graph), the two PageRank scores that walk on one edge type. The coupled walks of Section 3.6.3 and the raw degrees are evaluated beside them.

- Primary graph construct: the graph that a method is *designed* to rank, for example G_{approve} for EndorseRank, as distinct from the checks applied after scoring.
- Contemporaneous check: a local transfer or allowance statistic taken from the same window as the score (Cronbach & Meehl, 1955).
- Temporal holdout: in the first window, W0, scores are frozen at t_1 (31 March 2026), and new owner–spender approval pairs and new transfer senders are counted up to t_2 (30 June 2026); a registered second window, W1, freezes scores on 30 June 2026 and counts labels from July to September 2026 (Section 3.7.2).
- Matched contract cohort, spender holdout cohorts of W0 and W1, trader cohorts and subsamples of the matched cohort: the address sets of Section 4.1.1.

2.12 Domain alignment metrics

Construct alignment is measured by Spearman's ρ and Kendall's τ between scores and check rankings (Section 2.9), on contemporaneous transfer and allowance checks and on the holdout labels; Chapter 3 gives the operational definitions.

2.13 Worked example: PageRank on a four-node endorsement graph

Take four addresses $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \to B\ (2), \quad A \to C\ (1), \quad B \to D\ (3), \quad C \to D\ (1),$$

with damping $d = 0.85$. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \to D$ and $C \to D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906,\ 0.2323,\ 0.1615,\ 0.5156)$ for (A, B, C, D).

The mass already tilts toward D, since both B and C endorse D, while A splits its mass between B and C. The iteration converges to the stationary vector $\mathbf{r} = (0.1375,\ 0.2154,\ 0.1765,\ 0.4706)$, so D comes first, B and C follow, and A is last. No address endorses A. Apart from teleportation, its whole score comes from the dangling rule, which spreads D's mass evenly over the four nodes ($d\, r_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is precisely the citation logic of PageRank (Langville & Meyer, 2006; Page et al., 1999).

Suppose the same addresses were linked by transfers instead of allowances. The transfers would in general join other pairs, or the same pairs with other per-pair totals. That gives another transition matrix P, and possibly a different ranking of the same nodes. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree. They also restart differently, so H1 compares edges and restarts together; Section 4.5.4 walks the allowance graph with AWP's restarts.

2.14 DeFi risk taxonomy and where reputation fits

Werner et al. (2022) separate the technical security of DeFi protocols from their economic security, and Zhou et al. (2023) place the attacks and accidents seen in practice in one reference frame. Bugs in contract code are smart-contract risks, one of the several levels at which DeFi vulnerabilities arise (Li et al., 2022). A stale price, or a feed under someone's control, is an oracle risk (Angeris & Chitra, 2020). Market risks (Perez et al., 2021) cover volatility and liquidation cascades, including stress in stablecoin and funding markets (Klages-Mundt et al., 2020), and the market crash of March 2020 showed how these two layers can interact (Gudgeon, Perez, et al., 2020). Token-weighted voting belongs to governance risk: in the on-chain governance of Compound, Uniswap and ENS, a small number of addresses hold most of the voting power (Fritsch et al., 2024). We study a narrower layer, *counterparty* or *behavioral* risk: whether an address's past behavior makes it a safe counterparty, as a trader that handles leveraged exposure responsibly or as a contract that many owners trust to spend their tokens.

On-chain reputation scores cover only part of the behavioral risk layer and are no substitute for collateral, auditing, or oracle design. Their purpose is to compress a system's relational structure (who authorizes whom, who pays whom) into an ordinal measure that others can audit and reproduce from public data. Within behavioral risk, EndorseRank looks at authorization and AWP at payment flows. Neither tries to predict exploits or governance attacks.

Finally, fake activity makes behavioral risk harder to measure. Wash trading inflates volume on decentralized exchanges (Victor & Weintraud, 2021) and, on a larger scale, on unregulated centralized exchanges (Cong et al., 2023). Front-running and related forms of extractable value create transfers that follow ordering strategies instead of relationships (Daian et al., 2020); Qin et al. (2022) estimate how much of that value is captured on Ethereum. Scam tokens on Uniswap (Xia et al., 2021) likewise add transactions without forming sustained relationships. The transfer graph is particularly sensitive to fake volume, while the allowance graph is vulnerable to approval farming. Time-based splits and Sybil detection solve the problem only in part.

2.15 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce (Resnick & Zeckhauser, 2002), in peer-to-peer networks (Kamvar et al., 2003), and in work on web spam (Gyöngyi et al., 2004). Four insights from this work apply here: (i) reputation is a property of relations; (ii) when identities are easy to create, Sybil attacks become possible (Douceur, 2002); (iii) seeding and normalization limit the effect of inflation; and (iv) evaluation needs external benchmarks, because internal coherence alone is not enough (Cronbach & Meehl, 1955). EndorseRank takes up (i) directly and treats (ii) as a threat. Of (iii) it takes up normalization, since each owner's endorsement is divided over the spenders it approves, but not seeding, since it restarts uniformly (Section 2.6). On (iv) it goes part of the way: the contemporaneous checks, measured by Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938), test coherence, while the temporal holdouts test prediction out of window, on the same constructs for spenders and on liquidation for traders.

Personalized and topic-sensitive PageRank (Haveliwala, 2003; Jeh & Widom, 2003) bias the teleportation vector toward a chosen group of nodes, the mechanism that TrustRank uses to turn a small seed set into a global ranking (Gyöngyi et al., 2004); AWP's restarts in proportion to sending activity are another instance (T. Do et al., 2023). Multiplex PageRank lets a node's centrality on one layer shape the walk on another (Halu et al., 2013), and the multilayer framework of De Domenico et al. (2015) lets one walker move within and between the layers of an interconnected network. Both treat layers as different relations over one node set, which matches allowances and transfers among the same addresses. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR borrows the personalized-teleportation idea, with the allowance score as the teleportation vector (Section 3.6.3).

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores (Antonopoulos & Wood, 2018). This thesis scores contract accounts only (Section 3.2). Every state-changing transaction consumes gas. Logs come from a contract's `LOG` instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must prefer logs to off-chain databases, because logs are a permanent part of the blockchain history that any full node or index provider can recompute independently (Wood, 2025).

Transfers recur routinely, while an approval is granted once so that later transfers can go through, so G_{transfer} is expected to be the denser graph, and low L2 fees may widen the gap; Chapter 4 reports the resulting graph sizes.

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the approved account is compromised, every one of the owner's tokens can be stolen until the owner resets the approval. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or revoked (Vogelsteller & Buterin, 2015). Transaction-based phishing exploits this: fake websites lead users to sign transactions that let scammers take their assets (He et al., 2023). EndorseRank interprets a token approval as endorsing one spender's capability to move those tokens. An account could receive many transfers and never be granted permission to move funds on anyone else's behalf.

Wang et al. (2022) find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry. Here they are close to half of the positive latest allowances (Appendix 8.4.2 gives the counts) and appear as amounts of $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units. EndorseRank's value weight gives them the full weight of one, which a finite allowance approaches only at a few times the median amount, while in the allowance layer of C-PR, which counts every allowance up to a cap in US dollars (Section 3.6.3), an unlimited approval weighs no more than a large finite one. Just over half of the owners hold at least one.

2.18 AWP in depth: what the transfer-graph baseline measures

Adaptive Weighted PageRank (AWP), proposed by T. Do et al. (2023), ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. Ranking accuracy is then validated against inbound degree and inbound value, taken as proxies for the economic attention a wallet attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; Section 3.6.1 gives the values

used here. The approach is entirely on-chain and rank-based and needs no off-chain identity. EndorseRank keeps AWP's weight functions under PageRank and its rank-correlation check, and changes what the edges mean and where the walk restarts: EndorseRank restarts uniformly, AWP at active senders (Section 3.6.2). In Chapter 4 we test whether allowance edges yield a score that is construct-distinct, allowance-aligned, and cheaper, whether scores taken at t_1 can predict new approvals by t_2, and whether one walk over allowance and transfer edges ranks contracts better than a walk over either.

2.19 Comparison of graph reputation methods

Table 2.1 compares EndorseRank with the other approaches covered in this chapter by edge semantics, temporal model and validation design.

As Table 2.1 indicates, the main contribution of the thesis lies on the operator side: it couples the two relations into one walk, C-PR, with a seeded ablation to measure how much this coupling contributes, and tests the coupling out of window and against both constructs, under rules fixed before this study extracted the data; on the revised scores of the main analysis they are applied as a re-analysis (Section 3.5.1 and Appendix 8.6). The coupling rests on an edge that the reviewed methods do not use. EndorseRank's contribution is the use of the latest allowance as the edge of a reputation ranking, evaluated out of window; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.7).

2.20 Summary of the reviewed literature

Table 2.2 collects the sources underlying this study, with the strengths this thesis draws on and the limitations it tries to address.

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank (web) (Page et al., 1999)	Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust (Kamvar et al., 2003)	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank (Gyöngyi et al., 2004)	Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank on Ethereum (H.-D. Do & Do, 2023)	Token transfers	Static window	Not reproduced in this thesis
AWP (T. Do et al., 2023)	Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp (D. Lin et al., 2025)	Transactions, with a de-anonymization score	Propagation of risk	Labeled illicit accounts
Multiplex / multilayer PageRank (De Domenico et al., 2015; Halu et al., 2013)	Several relations over one node set	Static layers	Versatility of top-ranked nodes
Human Passport (n.d.)	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold; known human and Sybil addresses
EndorseRank (this study)	Latest allowances of five tokens in US dollars, AWP's weight functions	Logistic decay of each latest approval; uniform restarts	Allowance checks; t_2 new approvals
C-PR / S-PR (this study)	Allowances mixed with, or seeding, transfers; US-dollar amounts, allowances capped	Allowance snapshot and logistic decay of transfers	Holdout labels; rules fixed in advance, re-applied to the revised scores

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1999); Langville and Meyer (2006)	Eigenvector ranking of a hyperlink graph, solved by power iteration	Interpretable propagation; $O(\|E\|)$ per iteration; damping yields a unique Perron vector	Edges can only mean hyperlinks; no notion of time	Same solver and complexity argument, used unchanged for both methods compared
Kamvar et al. (2003); Gyöngyi et al. (2004)	Aggregation of trust in P2P networks; propagation from seeds to demote spam	Normalization and seed sets curb inflation; the adversary is modeled explicitly	Depends on explicit ratings or curated seeds; not applied on-chain	Informs the Sybil-cost discussion and the uniform-teleportation baseline
H.-D. Do and Do (2023)	PageRank and HodgeRank applied to Ethereum transaction graphs as a measure of social credit	Treats transfer connectivity as relevant to credit; data are fully public	Transfers conflate payment with trust; allowance semantics absent	Earlier transfer-graph work, kept separate from AWP in this thesis
T. Do et al. (2023), AWP	Transaction PageRank with bounded value weights, logistic recency and restarts weighted by sending activity; rank correlation with in-degree/in-value as validation	Validates on ranks, so no default labels are needed; recency modeled explicitly	Proxies are computed from the same transfer data that forms the graph (mechanical alignment); dense graph; no parameter values reported; contracts without transfer edges all tie at zero	Baseline, re-implemented from the published formulas with k, t_0 and b set here (Section 3.6.1); EndorseRank takes the same weight functions with its own slope

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
D. Lin et al. (2025), RiskProp	Network propagation of account risk, paired with a de-anonymization score	Rates every account, including those that clustering and classification leave unassessed; checked against labeled accounts	Checked against illicit-account labels; aimed at fraud, not trust or authorization	Shows that propagation over transactions adds to per-account methods; not reproduced here, since this study uses no fraud labels
Wu et al. (2022)	Phishing detection through network embedding of Ethereum transaction graphs	Embedding that uses the amounts and times of transactions; more effective than earlier embedding methods	Needs labeled phishing addresses; its labels mark phishing, not creditworthiness	Backs the premise that graph position is informative; not a competing method
Hassija et al. (2020); Kotb (2023)	Blockchain-data credit scoring using prospect theory and machine learning	Aimed directly at lending decisions	Need labels or off-chain features; only partly interpretable	A contrasting class; EndorseRank remains within auditable graph propagation
Bellini et al. (2020); Almasoud et al. (2020)	Surveys of blockchain-based trust and reputation systems	Taxonomies of the design space; a review of thirty studies of reputation systems built on smart contracts	Reputation taken from users' ratings of services; authorization edges not considered; nothing measured on L2	None of the surveyed designs uses latest-allowance edges

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Victor (2020); Victor and Weintraud (2021); Cong et al. (2023)	Heuristics for clustering addresses, one of which links addresses through ERC-20 approvals; detection of wash trading on DEXs and exchanges	Measure the share of on-chain activity that is artificial or multi-address	No reputation score proposed	Closest prior use of approvals, as a clustering link and not a ranking edge; show why transfer-based signals are open to inflation
Cronbach and Meehl (1955); Messick (1995)	Psychometric construct validity	Distinguish what a score is designed to measure from external criteria	Not specific to blockchains	Basis for treating the contemporaneous checks as construct checks and the temporal holdouts as out-of-window evidence
Human Passport (n.d.); Siddarth et al. (2020); Bordeianu and Popescu (2026)	Weighted credentials about an address holder, on-chain and off-chain, against a passing threshold; reviews of proof-of-personhood defenses	In deployed use; adds off-chain evidence against Sybil attacks	A gate and not a graded ordering; relations between addresses play no part; Sybil strategies keep changing	Shows how trust is established in practice, and what EndorseRank does not offer: no personhood guarantee
Efron (1979); DiCiccio and Efron (1996)	Bootstrap resampling and bootstrap-based confidence intervals	Uncertainty for rank statistics with no distributional assumption; paired contrasts	Units must be exchangeable	Source of the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Five gaps emerge from Table 2.2 and Section 2.15. First, every on-chain reputation method discussed here (PageRank/HodgeRank (H.-D. Do & Do, 2023), AWP (T. Do et al., 2023), RiskProp (D. Lin et al., 2025), and the designs in the surveys (Bellini et al., 2020)) builds its graph from transfers or explicit ratings, and the attestation tooling deployed in DeFi scores the holder of an address without any graph of relations between addresses (Human Passport, n.d.). None uses the ERC-20 allowance, the on-chain record of a delegation of spending power that every ERC-20 token carries. Approvals have served address clustering and security analysis (Sections 2.7 and 2.17), but in the work reviewed here they have not served as the edges of a reputation ranking. It remains unknown whether a latest-allowance endorsement graph yields a distinct and meaningful ordering (O1 and RQ1). A second gap lies in validation. The transfer-graph baseline checks its scores against in-degree and in-value computed from the same transfer data used to build the graph, so some agreement is unavoidable; no report applies this procedure to an allowance construct and its transfer counterpart with confidence intervals and paired difference tests on the same set of addresses (O1 and RQ1). Third, the transfer-graph studies reviewed here give no runtimes with edge counts, memory use and iteration counts under a fixed protocol, which makes it hard to attribute any efficiency claim to sparsity as opposed to implementation (O2 and RQ2). The fourth gap is methodological and has to do with time. Graph statistics from the same window are often presented as validation, although such scores cannot be shown to predict future behavior; the out-of-window check used here is instead a temporal holdout with a future-tense label for each construct and raw degree at the time of the freeze as the baseline (O3 and RQ3). Finally, multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) have been applied to social networks and to other networks with several kinds of relation, but, in the work reviewed here, not to two on-chain relations over one set of addresses. No study reports whether combining an authorization layer with a payment layer beats both layers, or improves on the payment layer alone, under rules fixed before the outcome is seen (O4 and RQ4, O5 and RQ5). The method of the thesis, C-PR,

addresses this fifth gap; the first four give it its reference walks and its evaluation.

2.22 Open problems motivating this research

The literature leaves several problems in on-chain reputation unresolved, and these motivate the present study:

1. Edge semantics: When identities are pseudonymous (D. Lin et al., 2024), which on-chain relations capture trust or authorization best (Packin & Lev-Aretz, 2024)?
2. Validation without defaults: On what basis can methods be compared when loan-default labels are scarce (H.-D. Do & Do, 2023)?
3. Computational cost: As graphs grow, can reputation scores still be refreshed at interactive latencies (Langville & Meyer, 2006)?
4. Time-split checks: What keeps an evaluation from mistaking same-window graph statistics for out-of-window evidence (Cronbach & Meehl, 1955)?
5. Adversarial robustness: How much does it cost to forge reputation when identities are cheap (Douceur, 2002)?

The thesis takes on (1)–(4) through allowance edges and a walk that couples them with transfers, contemporaneous checks, runtime benchmarks, and a temporal holdout. Problem (5) is analyzed at the model level in Chapter 5; a full empirical adversarial evaluation is left for future work. Chapter 3 turns this gap statement into an operational design.

3 Research Methodology

Chapter 2 ended with five gaps. This chapter gives each of them a measurable procedure and sets out how the coupled walk C-PR, the method of the thesis, and its two single-layer references, EndorseRank and Adaptive Weighted PageRank (AWP), are evaluated as reputation scores for contract accounts on Arbitrum One. EndorseRank runs on the ERC-20 allowance graph and AWP on the transfer graph (T. Do et al., 2023); C-PR walks both in one chain (Section 3.6.3). The two single-layer walks weight their edges as AWP's authors specify, and every walk is solved with weighted PageRank (Page et al., 1999); AWP restarts the walk in proportion to sending activity, and EndorseRank and C-PR restart it uniformly.

Methodologically, the study follows construct (domain) alignment. Reputation scores are continuous rankings, and the checks rest on observable contract-level statistics instead of binary default labels; each ranking method is one indicator of a reputation construct that cannot be observed directly, in the sense of construct validity (Cronbach & Meehl, 1955). The claim is comparative, every score set against the checks of each construct and against the raw degrees it smooths; it is not a prediction of trading profit or credit default.

Four principles guide each operational choice.

- Same-seed comparability: every score is computed on an identical set of contracts, so no difference between scores can come from sampling.
- Fixing choices in advance: the analysis plan and the versioned configuration were committed to the public repository at 13:41 UTC on 9 October 2026 (commit `43d8045`), before the study extracted any log.[1] They fix the windows, the cohort and account-type rules, every score with its parameters, the values of λ, the proxies, the W0 labels, the bootstrap, the benchmark protocol, the robustness checks and rule A, the plan's one decision rule, whose object is C-PR

[1] All times are UTC; the commits record UTC+9.

(RQ4; Section 3.7). Scan A, the cohort, the ego extraction and the same-window and W0 results followed. At 16:49 UTC the same day the registration of W1 was committed (commit `0d3a057`): rule B with its margin, F4–F6, two sensitivity analyses and the comparison on outcomes built from neither edge type, with the wording of its conclusions (Sections 3.7.2 and 3.7.3). Scan B, the registered evaluation, the supplementary contrasts, the robustness checks, the Sybil model and the benchmark came next. After these results were known, the revised analysis was defined (Section 3.5.1): amounts in US dollars, five tokens, and scores computed from the twelve months before each freeze. The revised analysis, which Chapter 4 reports, applies the same rules to these scores; every other contrast, the degree contrasts among them, was computed after its data were known and is reported as such, with its interval.

- Construction and timing: each cohort's edge lists are built once, before any timer starts, and the benchmarks time the solver call.
- Scope: H1 and the divergence half of C4 compare the orderings of EndorseRank and AWP; the other hypotheses and Claims C3, C5 and the agreement half of C4 concern every score, C1 and C2 the coupled walk and its ablation, and C6 the approval and transfer counts and their PageRanks on trader outcomes. RQ3–RQ5 draw on the spender cohorts of W0 and W1, and rule B adds a trader cohort.

3.1 Notation

The notation used in this chapter and in Chapters 4 and 5 is collected in Table 3.1 below, and a condensed list appears in Table II. Vectors are set in bold and sets in calligraphic type.

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched contract cohort ($\lvert\mathcal{W}\rvert = n = 6{,}483$)
$G = (V, E)$	Graph passed to the PageRank solver: V holds the end points of the retained edges E
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph

Table 3.1: (continued)

Symbol	Meaning
$a^\star(u, v, \text{token})$	Latest in-window allowance of owner u to spender v on one token (US dollars)
$\Delta t^\star(u, v, \text{token})$	Age of the approval that set $a^\star$
$w_A(u, v)$	EndorseRank edge weight, owner $u \to$ spender v: $\sigma(\Delta t^\star)V(a^\star)$ summed over tokens
$w_T(u, v)$	AWP edge weight, sender $u \to$ receiver v: $\sigma(\Delta t_e)V(x_e)$ summed over transfers
x_e	Amount of transfer e (US dollars)
$V(z)$, b	Bounded value transform of AWP, $V(z) = 2/(1+e^{-bz}) - 1$, with $b = \ln 3/m$ (b_T for transfers, b_A for allowances)
m_T, m_A	Median transfer (170) and allowance (100) amounts in US dollars in the W0 score window
U	Cap on one allowance in C-PR and in the in-approve value (19,000 dollars)
X_u, X_u^A	Activeness of address u in the transfer and the allowance graph, the restart weights of AWP and of EndorseRank with activity restarts
$\tilde{w}_A(u, v)$, $\tilde{w}_T(u, v)$	Weights of the two layers of C-PR, in US dollars
$o(u)$; $o_A(u)$, $o_T(u)$	Out-strength $\sum_v w(u, v)$ of node u; the same in the allowance and transfer layers
P	Transition matrix, $P_{u,v} = w(u, v)/o(u)$; rows of dangling nodes are zero
$\mathbf{1}[\cdot]$	Indicator: 1 if the condition holds, 0 otherwise
$\mathbf{1}_{\text{dang}}$	Indicator vector of the dangling nodes, entries $\mathbf{1}[o(u) = 0]$
$\mathbf{r}$, $\mathbf{r}^{(t)}$	PageRank vector and its iterate after t steps
$\boldsymbol{\pi}$	Restart (teleportation) distribution, also used for dangling mass: uniform for EndorseRank and C-PR, proportional to X_u for AWP and to X_u^A for EndorseRank with activity restarts, and for S-PR the allowance layer's score restricted to V_T and renormalized
d	PageRank damping factor (default 0.85)
ε	PageRank convergence tolerance (10^{-8})
$I_{\max}$	Maximum PageRank iterations (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 0.5$ primary; 0.25, 0.75 sensitivity; at $\lambda = 1$ and $\lambda = 0$ one layer is walked alone)
$\alpha_A(u)$, $\alpha_T(u)$	Layer weights in node u's C-PR row; $D_\lambda(u)$ is their common denominator
$P^{(\lambda)}$	Coupled transition matrix of C-PR
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_q$	Proxy value vector for proxy q
$\sigma(\Delta t)$	Logistic time-decay on the age of an approval or transfer
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before the decay anchor (T_{obs}; t_1 in W0)
T_{obs}	Observation end timestamp and W1 freeze (2026-06-30 23:59:59 UTC)
t_1, t_2	W0 freeze (2026-03-31 23:59:59 UTC) and end of its label window (2026-06-30 23:59:59 UTC)
ρ	Spearman's rank correlation

Table 3.1: (continued)

Symbol	Meaning
R_i, S_i	Mid-ranks of contract i's score and proxy value (tied values share their average position)
τ, τ_b	Kendall's rank correlation; every reported value is the tie-corrected τ_b
n_c, n_d	Numbers of concordant and discordant contract pairs
n_0; n_1, n_2	All pairs, $n(n-1)/2$; pairs tied in the first and in the second ranking

Except where stated otherwise, PageRank means the weighted variant of Page et al. (1999), whose update at iteration t is

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\left(\mathbf{r}^{(t)\top}\mathbf{1}_{\text{dang}}\right)\boldsymbol{\pi}, \tag{3.1}$$

where P is the transition matrix of Equation (3.2) below, $\boldsymbol{\pi}$ is the restart distribution over the nodes, uniform ($\pi_u = 1/|V|$) unless a method specifies another (AWP in Section 3.6.1, EndorseRank with activity restarts in Section 3.6.2, S-PR in Section 3.6.3), and $\mathbf{1}_{\text{dang}}$ is the indicator vector of the dangling nodes, with entries $\mathbf{1}[o(u) = 0]$. The third term therefore collects the mass held by nodes without out-edges. With $w(u,v)$ denoting the weight of the directed edge from u to v and $o(u) = \sum_v w(u,v)$ the out-strength, the transition matrix is

$$P_{u,v} = \begin{cases} w(u,v)/o(u) & \text{if } o(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

The rows of dangling nodes are zero, so P is substochastic; the mass lost at those nodes returns through the third term of Equation (3.1) and is spread according to $\boldsymbol{\pi}$. Filling each zero row with $1/|V|$ gives the row-stochastic matrix of Equation (2.1), and with uniform $\boldsymbol{\pi}$ Equation (3.1) is then the Google-matrix form of Equation (2.2). The iteration starts from $\mathbf{r}^{(0)} = \mathbf{1}/|V|$ and stops once $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or after $I_{\max}$ iterations. V is the set of end points of the retained edges, and the final vector is normalized to sum to one over V. Contracts are also listed in descending order

of score, ties broken by address, but these ordinal ranks serve only for listing; the correlations of Section 3.10 are computed on the raw scores, where tied scores stay tied.

The damping factor d interpolates between pure propagation ($d \to 1$) and uniform scores ($d \to 0$). Its default of 0.85 is the classical web setting (Page et al., 1999), and the robustness sweep varies it with edge weights held fixed (Section 3.12). The strict $\varepsilon = 10^{-8}$ and $I_{\max} = 300$ keep the measured runtimes tied to the structure of the graph and not to early termination.

3.2 Data sources and sampling

This subsection fixes the chain, time interval, public tables, events and contracts of the main sample; no later step brings in events from outside it. Figure 3.1 places every kind of on-chain record used in the thesis on one calendar, with the months each analysis scores, the months it holds out as labels, and what each record is used for.

- Primary chain: All events come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum (Thibault et al., 2022) whose logs are in the public BigQuery export. One chain avoids cross-chain aliases, where one user's addresses on several chains would appear as distinct nodes, and every address in the sample faces the same fees and latencies.
- Extraction and score windows: Scan A extracts the events from 2023-10-01 00:00:00 UTC to 2026-06-30 23:59:59 UTC (the queries take 2026-07-01 as an exclusive end), and scan B those of the label window W1, July to September 2026, so the data cover three years. Every score of the main analysis is computed from the twelve months before its freeze: from 2025-07-01 00:00:00 UTC to T_{obs} = 2026-06-30 23:59:59 UTC in the same-window evaluation and in W1, and from 2025-04-01 00:00:00 UTC to t_1 = 2026-03-31 23:59:59 UTC in W0 (Section 3.7). A window of one year bounds the age of every event that enters a score, a proxy, a baseline or a label history, gives the two freezes windows of the same length, and lets the tenure and active-month proxies range up to 365 days and 12 months. Both graphs use exactly the same dates, and the time decay is anchored at the freeze, so an event on the

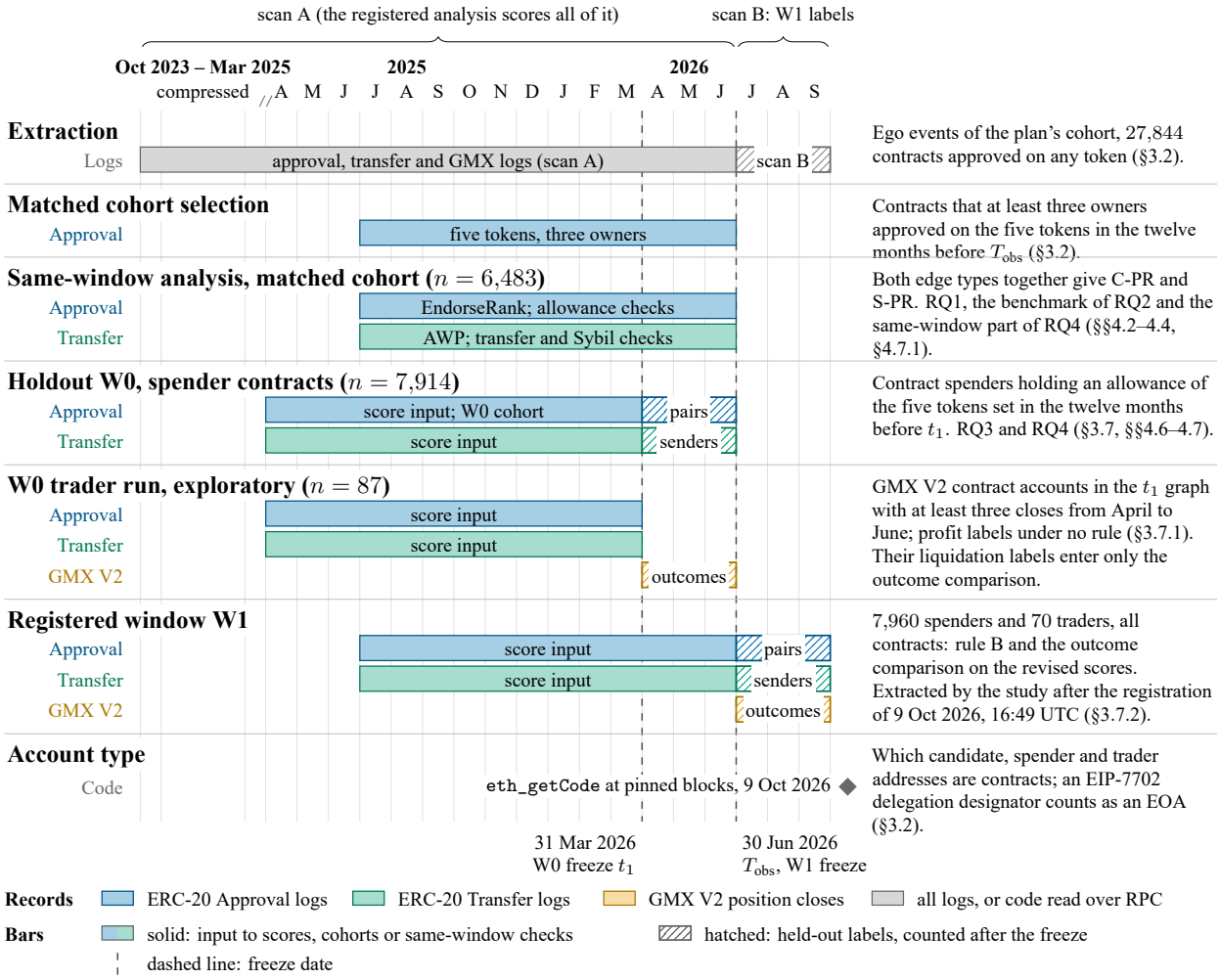


Figure 3.1: Blockchain data used in this thesis, all from Arbitrum One. Each lane is one analysis, and each bar is one kind of on-chain record placed on the months it covers; the 18 months before April 2025 are drawn at a compressed scale. Solid bars are inputs to scores, to cohorts or to same-window checks. Hatched bars are labels counted after a freeze date and kept out of the scores; the study extracted the July–September logs only after the registration of W1 was committed at 16:49 UTC on 9 October 2026. The text in a bar says what the records compute, and the right-hand column names the cohort, the research questions and the sections that report the results. The approval logs of five tokens in the twelve months before each freeze select the cohort of that freeze, and every approval and transfer query of the main analysis is then filtered to it, to those tokens and to those months; the registered analysis scores every event of scan A.

final day has $0 \leq \Delta t < 1$ and one on the first day is 365 days old. The registered analysis (Appendix 8.6) scores every event of scan A from 2023-10-01.

- BigQuery source: Logs are read from the public BigQuery table

  `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

  One record is one event emitted by a contract, with its block timestamp, address, topics and data. Scans A and B copy the events the study needs into its own Google Cloud project, and later queries read those copies (Section 3.3). Anyone with a Google Cloud account can rerun the same queries against the same public table; Appendix 8.2 shows the query patterns.

- Reputation events: EndorseRank uses ERC-20 `Approval` logs (Vogelsteller & Buterin, 2015), which also cover approvals made through EIP-2612 `permit` (Lundfall, 2020), and AWP uses `Transfer` logs. Both are kept only for logs with exactly three topics and for the five tokens of Section 3.5.1. GMX V2 position closes supply the trader labels.

- Scored addresses: Only contract accounts are scored. An externally owned account (EOA), the wallet of an individual user, can serve for trading, holding, payments, bridging, airdrop farming or an automated bot, and users mix these uses in proportions that differ from one to the next, so EOAs have no common pattern of activity on which they could be put in one order. A contract's role is fixed by its code (a router, a vault, a pool), and the owners that approve it grant the same kind of permission for that role, so the approvals and transfers that different contracts receive can be compared. EOAs therefore enter the graphs only as owners, senders and recipients. That comparability holds within a role, however. The cohort mixes roles and can include contracts that individuals use as wallets, and no analysis is stratified by role, so differences between roles enter every coefficient. An address counts as a contract when `eth_getCode` at block $513{,}216{,}506$ (9 October 2026) returns code other than an EIP-7702 delegation designator, which an EOA can set while keeping its key (Buterin et al., 2024); addresses outside the cohort candidates were read at later blocks within three hours (Appendix 8.1.3).

- Matched cohort: Alignment is applied to the $n = 6{,}483$ contracts that received a non-zero `Ap`

`proval` of one of the five tokens from at least three distinct owners inside the score window of T_{obs}, the part of the plan's cohort ($27{,}844$ contracts approved on any token from 2023-10-01) that meets the rule on the five tokens within that window. W0 draws a cohort of its own by the same rule inside its own score window, so that no event after t_1 chooses the contracts around which its data are drawn (Section 3.7). The list is fixed from the approvals and the code check alone, but it is not independent of the allowance graph: every cohort contract held allowances at some time in the window. A cohort contract need not appear in either graph; the method concerned then scores it 0 (Section 3.6), and Section 4.1.1 counts these contracts.

- Ego network: Only events in which a cohort contract is owner or spender, or sender or recipient of a non-zero transfer, are used. The extraction keeps those of the plan's cohort, and each freeze keeps those of its own cohort in its score window and its label window. This bounds the extracted volume and makes each graph a one-hop ego network around the cohort, since an event between two addresses outside it is never used. An address outside the cohort, such as a holdout spender that fewer than three owners approved or almost any GMX trader, is therefore seen only through its events with cohort contracts: its degrees, activeness and scores, and a spender's labels, count only those. Endorsement passes over more than one hop only along edges that each touch a cohort contract; Section 5.3.2 examines how far the allowance walk does so.
- Exclusions: Mempool transactions, off-chain order books, centralized exchange accounts and bridges lie outside the data, and no attempt is made to merge addresses controlled by one operator into one entity. Tokens other than the five are left out, and native ether enters only as WETH.

3.3 Two-phase collection strategy

Data collection and evaluation run as two separate phases, so the tables, benchmarks and robustness runs can be rebuilt as often as needed without cloud costs, by us or by anyone who has run the

extraction.

- Phase 1 (reputation streams): Scan A copies the approval, transfer and GMX events from 1 October 2023 to 30 June 2026 into the study's project, and scan B those of July to September 2026 once the registration of W1 is committed (Section 3.7.2). One query lists the spenders approved by at least three owners, the code check fixes the plan's cohort, and a further query extracts its ego events. From these the main analysis keeps the five tokens, draws the cohort of each freeze from the approvals of its score window and keeps the events of that cohort. At each anchor date, T_{obs} and t_1, the approvals of the anchor's score window are reduced to the most recent allowance per (token, owner, spender) triple in (block number, log index) order and kept when positive, the transfers of the window are aggregated per sender–recipient pair, and amounts are valued in US dollars (Section 3.5.1); the proxies and holdout labels are computed alongside. These tables are the only inputs to graph construction.
- Phase 2 (local evaluation): From the downloaded Phase 1 tables a local machine decodes the GMX closes and computes the scores, the alignment statistics, the holdouts, the benchmarks and the robustness runs, without querying BigQuery.
- Cost guardrails: Every query is first run as a dry run, and a query whose estimate would take the spending past the study's budget of USD 50 is aborted unless the operator confirms it with an extra flag.[2] Every job is written to a cost ledger with its billed bytes and cost (Appendix 8.1), so the cost of acquisition can be audited.

3.4 Pipeline architecture

Figure 3.2 traces the whole pipeline, from public logs to the research tables. All remote steps write tables in the study's BigQuery project, and the downloaded files are never changed.
Unit tests run the scoring and statistics code on synthetic fixtures.

[2]This budget is enforced by the study's own pipeline; BigQuery does not impose it.

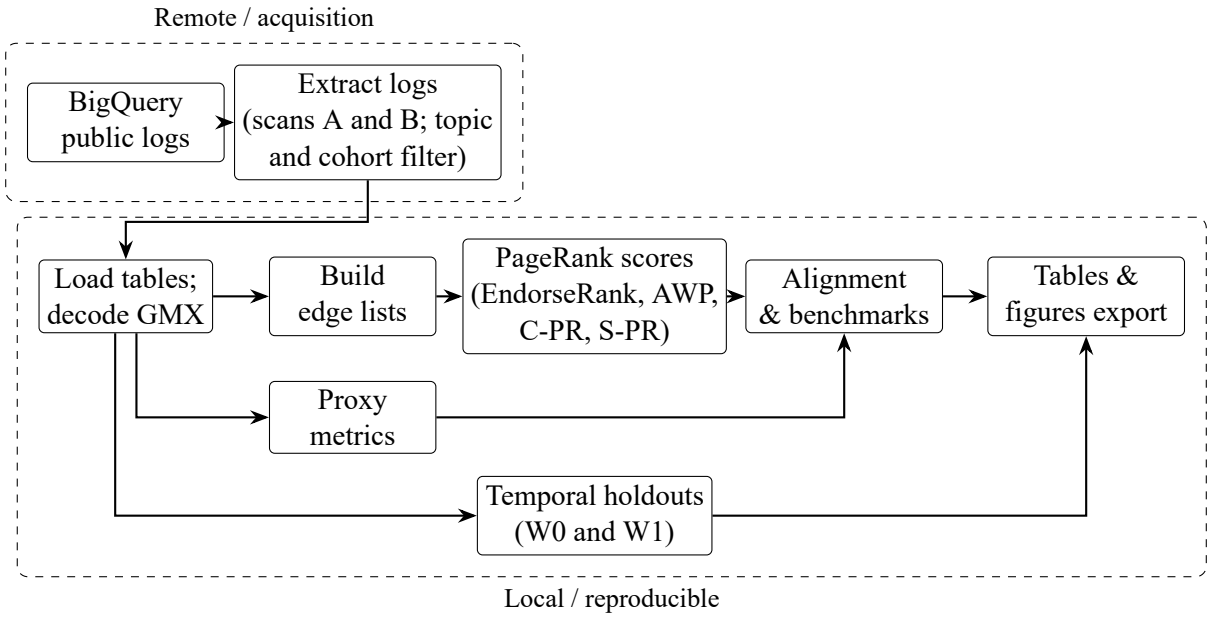


Figure 3.2: Pipeline overview. Approval, transfer and GMX logs are copied remotely in two scans and aggregated into pair tables for the cohort. Locally, the GMX closes are decoded and the pair tables turned into edge lists, from which EndorseRank, AWP, C-PR and S-PR are computed; the proxy metrics come from the same events on a parallel branch. The alignment and benchmark calculations and the temporal holdouts follow, and the result tables and figures are exported last.

3.5 Event decoding

A log emitted by a smart contract has topics and a data payload but no field names. Decoding turns each row into a typed record for graph construction and the checks, and an error here would corrupt both graphs silently. The topic hashes and field mappings are listed below.

- Log anatomy: Each log records the emitting contract, a sequence of topics and an ABI-encoded data blob. The first topic is the keccak-256 hash of the event signature; indexed parameters follow as 32-byte words (addresses left-padded), and non-indexed parameters go into the data field in ABI order (Vogelsteller & Buterin, 2015). ERC-721 tokens emit `Transfer` and `Approval` under the same signatures but put the token ID in a fourth topic, so the condition of exactly three topics keeps them out. SQL filters by topic and decodes the ERC-20 fields, and Python decodes the GMX events.
- ERC-20 `Approval`: Its topic hash is

  `0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925`,

  the keccak-256 hash of `Approval(address,address,uint256)`. Owner and spender occupy Topics 1 and 2, and the allowance is decoded from the data field in the token's base units and valued in US dollars (Section 3.5.1). An allowance of zero is kept, since it resets the allowance in force. A `permit` emits the same `Approval` event, so an allowance counts whether it came from an on-chain `approve()` call or from an off-chain signature (Lundfall, 2020).
- ERC-20 `Transfer`: Its topic hash is

  `0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef`,

  the keccak-256 hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed, and the amount is valued in US dollars at the price of its day. Zero-value transfers and transfers made before their token's first price are dropped. Mint and burn logs, which

use the zero address, are kept whenever a cohort contract is involved. Self-transfers stay in the events table but are left out of the inbound-counterparty computations (Section 3.9), since they reveal no counterparty.

- GMX V2 position closes: A close is one `PositionDecrease` log of the GMX V2 event emitter, so partial closes and liquidations count. Python decodes its account, base profit or loss and order type; every close of a contract account decoded without error.
- Address normalization and units: Addresses are lower-cased before the join. Amounts are the `uint256` value as a double-precision float, divided by 10^q for a token with q decimals and multiplied by its US-dollar price, so an unlimited allowance, $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units, becomes a very large dollar amount (Sections 3.6.2 and 3.6.3 give its effect).

3.5.1 Tokens, prices and the revision of the amounts

The analysis plan fixed amounts in raw base units, every ERC-20 token and $b = 1$ in Equation (3.4). That setting has three defects. First, $V(z)$ is within 10^{-4} of 1 for any amount of 10 base units or more, which almost every transfer and allowance reaches, so the amount played no part in EndorseRank or AWP. Second, the layers of C-PR added raw base units across tokens with different decimals, and an unlimited allowance took almost the whole row of every owner that held one. Third, any token contract can emit `Approval` and `Transfer` logs naming an owner or sender that took no part, so the log of an arbitrary token is no evidence of an endorsement or a payment.

The revised analysis was therefore defined after the registered results were known. Its amounts are in US dollars, it keeps five tokens, and it scores each contract on the twelve months before each freeze (Section 3.2). A token qualifies when Chainlink's directory of data feeds on Arbitrum One lists a market-price feed for it, or for the asset it represents, with a low or medium market-risk category (Chainlink, n.d.), and DefiLlama has a daily price for it (DefiLlama, n.d.). Of the 133 tokens the directory maps, 64 qualified and 46 of these are priced; the analysis keeps the five with the most rows in the score window of T_{obs}: WETH, USDC, USDT0, WBTC and ARB. A transfer is valued at the DefiLlama daily close of its UTC day, and a latest allowance at the price

of the anchor's day. The slope of V is $b = \ln 3/m$, so that an amount of m dollars weighs one half; m is the median US-dollar amount in the W0 score window, so that no event after t_1 enters it: $m_T = 170$ dollars for transfers and $m_A = 100$ dollars for the finite latest allowances at t_1. In the allowance layer of C-PR an allowance counts at most $U = 19{,}000$ dollars, the 99th percentile of single transfers in the same window. The 99th percentile of the finite allowances themselves is about 10^{63} dollars, so a cap taken from them would leave unlimited allowances in charge of the rows. The cohorts are drawn on the five tokens from the ego events already extracted, without a new scan.

Rules A and B and the comparison on outcomes built from neither edge type are applied to the revised scores with their thresholds and wording; since these scores were defined after the registered results were known, the verdicts are a re-analysis and not registered tests. Appendix 8.6 sets the registered analysis beside the revised one, and Appendix 8.4.2 gives the tokens, the rows left out and the price checks.

3.6 Graph construction

EndorseRank and AWP weight their edges the same way and run the same PageRank solver, Equation (3.1); they differ in the events that make the edges and in where the walk restarts. Each method keeps the edges with at least one endpoint in the seed set $\mathcal{W}$, which brings in their one-hop neighbors as nodes, and reports a score for every evaluated contract; a contract missing from the subgraph scores 0.

Edges in EndorseRank stand for endorsements and edges in AWP for transfers (Section 2.7). Both methods weight an event by its age, so that recent events count for more, and pass its US-dollar amount through a bounded function, so that a larger amount counts for more but one event counts at most once. AWP is described first, since EndorseRank takes its weights from it, and C-PR, which walks both edge types, comes last.

3.6.1 AWP transfer graph

AWP is the method of T. Do et al. (2023), presented at IEEE RIVF 2023. A transfer e worth x_e US dollars, made Δt_e days before the score window closes at its anchor (T_{obs}, or t_1 in W0), enters the edge from its sender to its receiver with the weight $\sigma(\Delta t_e)\, V(x_e)$. The time weight is the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp\big(k(\Delta t - t_0)\big)}, \qquad k = 0.01, \quad t_0 = 180 \text{ days}, \tag{3.3}$$

and the value weight is

$$V(z) = \frac{2}{1 + e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \tag{3.4}$$

which is 0 at $z = 0$, $1/2$ at the median amount $z = m$ and rises towards 1 as the amount grows. For transfers $b = b_T = \ln 3/m_T$ (Section 3.5.1). The edge weights are sums over transfers:

$$w_T(u, v) = \sum_{e:\, u \to v} \sigma(\Delta t_e)\, V(x_e). \tag{3.5}$$

The sum runs over every transfer from u to v in the window, whatever the token. A self-transfer ($u = v$) is kept and becomes a self-loop on u. The walk restarts at an address in proportion to its activeness,

$$X_u = \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e), \tag{3.6}$$

which adds up, over the addresses that u has sent to, the time weight of its latest transfer to each. The restart distribution of Equation (3.1) is $\pi_u = X_u / \sum_{v \in V_T} X_v$, and it also takes the mass of the dangling nodes. An address that never sends to another has $X_u = 0$ and receives no restart mass, so its score comes from its in-edges alone. The damping factor is $d = 0.85$.

The paper scores Ethereum transactions and gives no parameter values; its time weight is the logistic function of Equation (3.3) written in calendar time instead of age, and k, t_0 and b are this study's choices. A transfer of the median amount weighs one half and one of twice the median 0.8, so an edge weight grows with the amounts transferred as well as with their number. Within a score

window the decay discounts older events strongly (Figure 3.3): the weight falls from $\sigma(0) = 0.858$ through $\sigma(182) = 0.495$ to $\sigma(365) = 0.136$ for an event from the first day of the window.

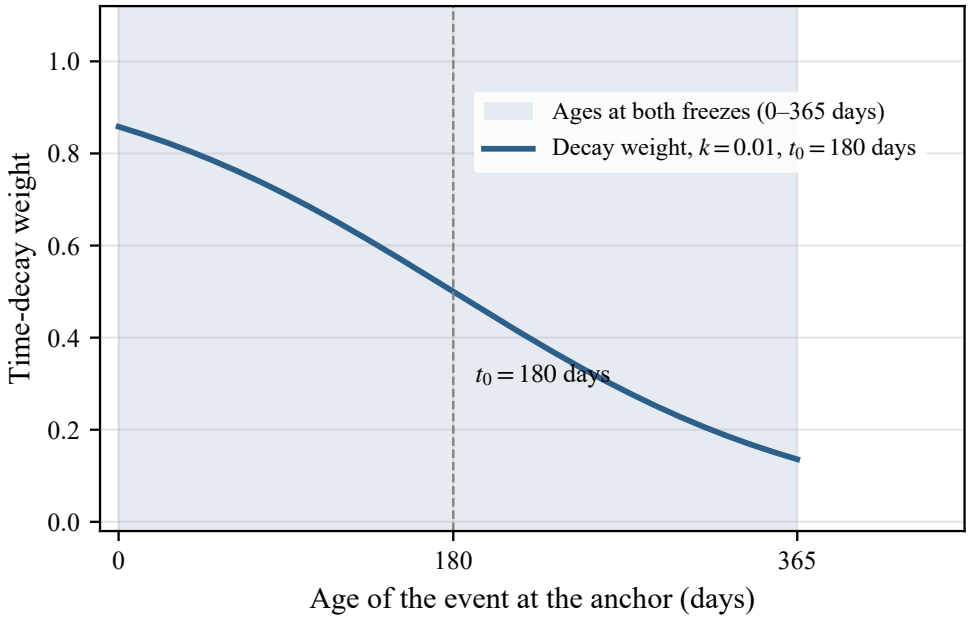


Figure 3.3: Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). The horizontal axis shows the age of an event in days before the anchor, $T_{\text{obs}} = 30$ June 2026 or, in W0, $t_1 = 31$ March 2026, and the vertical axis shows the time weight of a transfer or an approval. The shaded band marks the ages that occur at either freeze (0 to 365 days), over which the weight falls from 0.86 to 0.14.

Algorithm 1 states the construction and scoring steps formally.

3.6.2 EndorseRank allowance graph

EndorseRank puts AWP's edge weights on the allowance graph. Let $\mathcal{A}$ denote the set of latest allowances after preprocessing. For each (token, owner–spender) triple only the most recent `Approval` event inside the score window is kept, in (block number, log index) order, and a zero allowance deletes the edge. The surviving allowance $a^{\star}(u, v, \text{token})$, in US dollars, was set $\Delta t^{\star}(u, v, \text{token})$

Algorithm 1 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$ in US dollars, seed contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_T, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$
$E \leftarrow \emptyset$; $X_u \leftarrow 0$ for every address u
for each transfer $e = (u, v, x_e, t_e)$ in $\mathcal{T}$ **do**
    $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days
    accumulate $\sigma(\Delta t)\, V(x_e)$ on edge $(u \to v)$ ▷ Equations (3.3)–(3.5); $u = v$ gives a self-loop
end for
for each sender u **do**
    $X_u \leftarrow \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e)$ ▷ Equation (3.6)
end for
remove non-positive edges; retain edges incident to $\mathcal{W}$
$\pi_u \leftarrow X_u / \sum_v X_v$ for the nodes u of the retained edges
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{AWP}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

days before T_{obs}, and it enters the edge as one event of that age and amount:

$$w_A(u, v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)\, V\big(a^\star(u, v, \text{token})\big), \tag{3.7}$$

with σ, V, k and t_0 as for AWP and $b = b_A = \ln 3 / m_A$. A directed edge $u \to v$ means that owner u endorses spender v. An allowance of the median amount enters with half its time weight and a large or unlimited one with close to its full time weight, so $w_A(u, v)$ grows with the amounts that u has approved to v, but no single allowance counts for more than one. The damping factor is $d = 0.85$. The walk restarts uniformly. AWP's restart rule, carried over to allowances, restarts the walk at owner u in proportion to $X_u^A = \sum_{v \neq u} \max_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)$, summed over the spenders v of u's positive latest allowances, a weight that grows with the number of spenders u approved recently. That rewards granting endorsements, while EndorseRank is meant to rank the contracts that receive them. The analysis plan fixed uniform restarts and named this variant, EndorseRank with activity restarts, as one to report next to EndorseRank (Table 4.8).

The weight keeps the amount and the time of the latest approval; it is not the allowance in force at

the end of the window. Within a pair, on each token, the newest approval replaces the older ones (Vogelsteller & Buterin, 2015). An allowance last set before the score window began is not an edge, even if it is still in force, so the allowance graph holds the authorizations that owners granted or renewed in the year before the freeze. An owner who approves an allowance-manager contract such as Uniswap's Permit2 appears only with an edge to that contract, not to the application that later spends through it. Some tokens also emit `Approval` from inside `transferFrom` with the reduced allowance. For them the latest record can be the remainder after the spender's last use, dated at that spend, so the time weight partly records use, an allowance spent in full ends at zero and drops out, and an authorization granted before the score window and used in a label window can appear as a new approval pair. For the other tokens a reduced allowance keeps its approved value. How often the latest record is such a remainder was not measured.

The steps for building the graph and running PageRank are listed in Algorithm 2.

Algorithm 2 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$ in US dollars, seed contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_A, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$
$E \leftarrow \emptyset$
for each latest allowance $(u, v, \text{token}, a^\star, t^\star)$ in $\mathcal{A}$ **do**
    $\Delta t^\star \leftarrow (T_{\text{obs}} - t^\star)$ in days
    accumulate $\sigma(\Delta t^\star)\, V(a^\star)$ on edge $(u \to v)$ ▷ Equation (3.7)
end for
remove non-positive edges; retain edges incident to $\mathcal{W}$
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{ER}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

Approvals are rarer than transfers, so on the matched cohort the approval subgraph is sparser (Chapter 4); since one PageRank iteration costs $O(|E|)$, this sparsity is the structural basis of H3.

For both methods, edge lists are stored as (from, to, weight) triples; duplicate edges are merged by summing their weights, and non-positive weights are dropped before the solve.

3.6.3 Coupled operator over both layers (C-PR) and the seeded ablation (S-PR)

EndorseRank and AWP each walk on a single edge type, so each ranks a contract by one kind of act: the spending rights it is granted or the payments it receives. C-PR, the method of the thesis, walks both edge types of the same addresses in one chain. RQ4 asks whether it ranks contracts better than a walk over either layer alone, and RQ5 whether it improves on a walk over the transfers alone. The analysis plan fixed the coupled operator on two layers that weight each event by its amount, which the revision values in US dollars (Section 3.5.1):

$$\tilde{w}_A(u,v) = \sum_{\text{token}} \min\big(a^{\star}(u,v,\text{token}), U\big), \qquad \tilde{w}_T(u,v) = \sum_{e:\,u\to v} x_e\,\sigma(\Delta t_e). \tag{3.8}$$

The allowance layer has the edges of EndorseRank and the transfer layer those of AWP, apart from the pairs whose value weight is zero in floating point (Section 4.1.1), which only the layers of C-PR keep; otherwise only the weights differ. Amounts enter with no bounded transform, an allowance counts at most U, and the allowance layer carries no time decay. Both rules for C-PR were fixed on these layers in raw base units. The two layers form one graph on the union node set $V = V_A \cup V_T$, with transition matrices P_A and P_T from Equation (3.2). For node u the coupled row is a convex mixture of the two layer rows, renormalized over the layers in which u has out-edges:

$$\begin{aligned} P^{(\lambda)}_{u,\cdot} &= \alpha_A(u)\,P_{A,u,\cdot} + \alpha_T(u)\,P_{T,u,\cdot}, \\ \alpha_A(u) &= \frac{\lambda\,\mathbf{1}[o_A(u)>0]}{D_\lambda(u)}, \qquad \alpha_T(u) = \frac{(1-\lambda)\,\mathbf{1}[o_T(u)>0]}{D_\lambda(u)}, \end{aligned} \tag{3.9}$$

where $D_\lambda(u) = \lambda\,\mathbf{1}[o_A(u)>0] + (1-\lambda)\,\mathbf{1}[o_T(u)>0]$, and $o_A(u)$ and $o_T(u)$ are the out-strengths of u in the two layers. A node with out-edges in only one layer follows that layer with weight one. A layer of weight 0 is dropped with its edges and nodes, and a node with $D_\lambda(u) = 0$ is dangling, its mass handled by the third term of Equation (3.1). Teleportation is uniform, $d = 0.85$, and $\mathbf{s}_{\text{CPR}}$ is the PageRank vector of Equation (3.1) with $P = P^{(\lambda)}$, under the solver settings of the single-layer methods. Figure 3.4 shows one step of the walk, and Algorithm 3 lists the construction and scoring

steps.

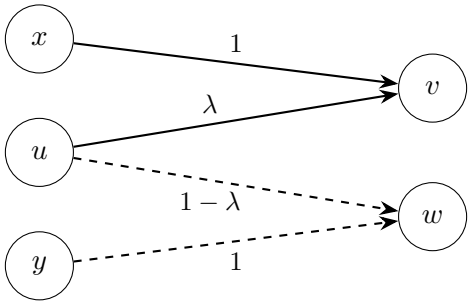


Figure 3.4: One step of the coupled walk C-PR (Equation (3.9)); solid arrows are allowance edges and dashed arrows transfer edges, labelled with their transition probabilities before damping. The address u has out-edges in both layers and mixes them with weights λ and $1-\lambda$; x and y have out-edges in one layer only and follow it whatever λ is. With probability $1-d$ the walk restarts at a uniformly chosen address.

Algorithm 3 C-PR graph construction and scoring

Require: Latest allowances $\mathcal{A}$ and transfers $\mathcal{T}$ in US dollars, seed contracts $\mathcal{W}$, observation end T_{obs}, cap U, weight λ, parameters $k, t_0, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{CPR}}$ on $\mathcal{W}$
allowance layer: accumulate $\min(a^\star, U)$ on $(u \to v)$ for each latest allowance in $\mathcal{A}$ ▷ Equation (3.8)
transfer layer: accumulate $x_e\,\sigma(T_{\text{obs}} - t_e)$ on $(u \to v)$ for each transfer in $\mathcal{T}$
remove non-positive edges; retain edges incident to $\mathcal{W}$; drop a layer of weight 0
$V \leftarrow V_A \cup V_T$; $P_A, P_T \leftarrow$ row-normalized layers ▷ Equation (3.2)
for each node $u \in V$ with $D_\lambda(u) > 0$ **do** ▷ the others are dangling
    $P^{(\lambda)}_{u,\cdot} \leftarrow \alpha_A(u)\,P_{A,u,\cdot} + \alpha_T(u)\,P_{T,u,\cdot}$ ▷ Equation (3.9)
end for
$\mathbf{s} \leftarrow \text{WeightedPageRank}(P^{(\lambda)}, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equation (3.1)
return $\mathbf{s}_{\text{CPR}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

At $\lambda = 1$ the operator walks the allowance layer alone and at $\lambda = 0$ the transfer layer alone; we write these walks C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). They are the comparators of both rules for C-PR (RQ4 and RQ5; Sections 3.7 and 3.7.2). Unit tests check on a fixture that each end point gives the scores of its layer walked alone with uniform restarts. The family is not continuous at the end points. For $0 < \lambda < 1$ the node set is $V_A \cup V_T$ and a node with out-edges in only one layer follows it whatever λ is, whereas at $\lambda = 1$ a node whose out-edges are all transfers is dangling or absent; as λ

approaches 1, C-PR therefore tends to a walk that still follows the transfer edges of such nodes, and the same holds at the other end. The values $\lambda = 0.5$ (primary) and $\lambda \in \{0.25, 0.75\}$ (sensitivity) were fixed in the plan's configuration.

Uncapped amounts would make the allowance layer very uneven: close to half of the positive latest allowances at T_{obs} are unlimited (Appendix 8.4.2), each would take almost the whole row of its owner, and the finite allowances of the owners who hold both kinds would have almost no effect. With the cap an unlimited allowance weighs as much as a finite allowance of U dollars or more, and the smaller allowances keep a share of the row in proportion to their amounts. EndorseRank and AWP bound every amount through V instead.

In the S-PR ablation the transfer layer is walked, but the uniform teleportation vector $\boldsymbol{\pi}$ of Equation (3.1) is replaced by the score of C-PR ($\lambda = 1$), restricted to V_T and renormalized over it. Let $\mathbf{r}_A$ be the PageRank vector of the allowance layer walked alone on V_A, extended by $r_{A,u} = 0$ for $u \notin V_A$. Then

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}} \qquad (u \in V_T),$$

so a transfer-graph node without an allowance score receives zero restart mass. If no node of V_T lies in V_A, the denominator is zero and $\boldsymbol{\pi}$ falls back to the uniform distribution on V_T. The same vector redistributes dangling mass. The result is the TrustRank construction (Gyöngyi et al., 2004), but with allowance-derived seeds instead of hand-labeled ones. Allowances decide where the walk restarts; transfers decide how it spreads. A comparison of S-PR with C-PR separates what the allowance edge contributes as a propagation medium from what it contributes as a seed. Like AWP and C-PR ($\lambda = 0$), however, S-PR scores zero every contract without a transfer edge, about one spender in six at each freeze (Section 4.6). Allowance edges order these spenders, so any contrast of a transfer-only walk with a score that orders them, F1 of rule B included, mixes ranking with coverage.

The edge set determines what the coupled operator costs. Because C-PR walks on $E_A \cup E_T$, one of its iterations costs as much as an iteration on each layer, and its runtime is expected to be at or

above AWP's. The speed result of Claim C5 is therefore EndorseRank's alone. S-PR needs one solve on each layer. Both operators are timed under the benchmark protocol of Section 3.11.

3.7 Temporal holdout protocol

In the holdout W0, scores use the events of the W0 score window, from 1 April 2025 up to $t_1 =$ 31 March 2026, 23:59:59 UTC. Labels are counted on events from 1 April 2026, 00:00:00 UTC, up to $t_2 =$ 30 June 2026, 23:59:59 UTC. The cohort consists of the $n = 7{,}914$ contract spenders that hold at least one positive latest allowance of one of the five tokens, set inside the window, at t_1. These data are the ego network of the W0 cohort, which the rule of Section 3.2 draws from the approvals of the same window, so no event after t_1 chooses the contracts around which the W0 data are drawn. Every score in the holdout table (EndorseRank, AWP, C-PR at the three values of λ and at its two end points, S-PR and two raw-degree baselines) is computed from these events alone. The time decay is anchored at t_1 as well: Δt in Equation (3.3) is measured back from t_1, so ages run up to 365 days and the weights from 0.858 down to 0.136.

We count two labels per spender, one for each construct, so that in the future window each single-layer method is tested on its own construct and the hybrids on both:

1. New approval pairs: the count of owner–spender pairs, attributed to the spender, that receive a positive approval in the label window but were absent from the latest-positive snapshot at t_1. This label measures the endorsement construct.
2. New transfer senders: the count of distinct addresses that send the spender a transfer in the label window without having sent it one between the start of the score window (1 April 2025) and t_1, self-transfers excluded. It measures the flow construct and is the transfer-side counterpart of the first label.

For baselines we take each spender's in-approve degree (the number of distinct owners with a positive allowance) and its transfer in-degree (the number of distinct senders other than the spender itself), both at t_1. Each single-layer PageRank smooths one of these local statistics. Because their

rows appear in the table, the increment of each PageRank over its own raw degree can be read off directly. That increment answers RQ3; the PageRank coefficient on its own does not. C-PR smooths both statistics, so it is read against both.

Uncertainty comes from the paired bootstrap over contracts of Section 3.10 ($B = 400$, one shared index matrix), so every difference between two coefficients on the same label is a paired difference. The contrasts fixed in the plan are the two of rule A below. Every other contrast in this window, each PageRank against its own degree and those of S-PR among them, was computed after the labels were known; the contrasts of C-PR with every other score are in Table 4.18, and the differences between EndorseRank and AWP in Appendix 8.5.

Rule A, fixed in the analysis plan, judges C-PR against its own layers walked alone and has two parts. The primary part uses W0: C-PR at $\lambda = 0.5$ should do no worse than C-PR ($\lambda = 1$) on new approval pairs (the interval of $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=1}$ contains zero or lies above it) and should be better than C-PR ($\lambda = 0$) on new transfer senders (the interval of $\tau_{\text{CPR}} - \tau_{\lambda=0}$ lies above zero). The secondary part uses the same window and the matched cohort. Its condition is that C-PR's point estimate lies within the 95% interval of the better of the two layers on the transfer family and on the allowance family, and that C-PR exceeds both layers on the Sybil-stability family. The paired contrasts of Table 4.14 are reported beside this rule as a stricter check; they do not decide it. If its primary part fails, RQ4, whether C-PR beats a walk on either layer alone, receives a negative answer, and the result is reported with its intervals. RQ5, whether C-PR improves on a walk over the transfers alone, is not covered by rule A; rule B tests it in W1 (Section 3.7.2). Once labels have been observed, λ is not tuned. When rule A was fixed, the study had extracted no log; its verdict on the revised scores is a re-analysis (Section 3.5.1).

The pipeline also counts, over the label window, revocations, the pairs held at t_1 that receive a zero approval (for some tokens an allowance spent in full, Section 3.6.2), and an approval-then-outbound drain heuristic. Neither is offered as supporting evidence (Section 4.6).

3.7.1 Trader outcomes in W0

The spender cohort says little about contracts that trade. An exploratory run applied the freeze t_1 to the 87 GMX V2 accounts that are contracts, appear in the freeze-date graph and close at least three positions from April to June 2026, with success labels from that window: the number of profitable closes, the realized gain (the sum of the positive base profits of the closes, losses not subtracted) and the shares of closes that were profitable and that were not losses; a liquidated close counts as not profitable and, since the pipeline does not count it as a loss, as not a loss. Every score of the spender holdout was set against these labels under no rule, with 400 paired resamples, and Chapter 4 reports the run as exploratory. It has no liquidation label, and with so few contract traders every interval is wide.

3.7.2 Registered window W1: July to September 2026

The W0 holdout also showed something that no rule had asked about. Against its transfer layer walked alone, C-PR ranked the spenders that later gained new approval pairs higher, with an interval above zero, while it was worse on new transfer senders (in the registered analysis; Appendix 8.6). We noticed this only after the labels were in, so it counts as a hypothesis and not as a finding, and it is tested again on a window that the study had not extracted when the test was fixed. The registration document, written after the W0 results were known, and the configuration, with the status of the window set to registered, were committed to the public repository at 16:49 UTC on 9 October 2026 (commit `0d3a057`), and scan B, the only query that reads logs dated after 30 June 2026, ran after that commit. The extraction script refuses to run unless both files are committed without local changes and the status is registered, and it records the commit and the hashes of both files in an extraction manifest (Appendix 8.1).

The method is unchanged. C-PR is used at $\lambda = 0.5$, as fixed in the plan, together with the two sensitivity values and S-PR. The solver settings, the SQL, the extraction filter of the matched cohort, the GMX decoder and the paired bootstrap (400 resamples, seed 42) are those used for W0. Scores

are frozen at T_{obs}, 30 June 2026 at 23:59:59 UTC, so they are the scores of the same-window evaluation: in the registration, which belongs to the registered analysis, they use every extracted event from 1 October 2023 onwards, and in the revised analysis the events of the score window of T_{obs} (Section 3.2). Labels are counted from 1 July to 30 September 2026.

Two cohorts are evaluated, and Section 4.1.1 gives their sizes. The spender cohort is defined as in W0: every contract spender holding at least one positive latest allowance at the freeze. It holds about five in six of the W0 spenders, so W1 is a later label window on largely the same contracts. The trader cohort is the GMX V2 accounts that are contracts, are nodes of the freeze-date graph in either layer and close at least three positions in the label window. The spenders keep their two labels. The traders get the liquidation-free close rate, one minus the share of a contract's July–September closes that ended in liquidation. It is the nearest thing to a default that these data contain, it is built from neither graph, and because it is a rate it does not reward a contract merely for trading more. When it was chosen, no score of the study had been set against a liquidation label; the study knew only, from the decoding step, how common liquidations are among the closes of contract accounts (Section 4.1.1).

Rule B compares C-PR with its transfer layer walked alone, C-PR ($\lambda = 0$), through the paired difference $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=0}$:

- F1, spenders, new approval pairs: the 95% interval of $\Delta\tau$ lies above zero;
- F2, spenders, new transfer senders: the lower end of the interval lies above $-\delta$;
- F3, traders, liquidation-free close rate: the lower end of the interval lies above $-\delta$.

C-PR counts as improving on a walk over the transfers alone only if all three hold, so no correction for multiple testing is needed. The margin $\delta = 0.01$ is the half-width of the paired interval for the F2 contrast in W0 in the registered analysis, rounded up to two decimals, as the registration fixed it (Appendix 8.6). In the registered analysis the spender cohorts could resolve differences smaller than the margin, so it was wider than those data required; on the smaller cohorts of the revised analysis the paired interval of F2 is about as wide as the margin (half-width 0.012 in W1), and on

the traders the lower end can clear $-\delta$ only if the two walks rank them almost alike, so F3 had little power to pass. One weakness of F3 was written down with the rule. If neither score predicts liquidation, F3 can pass with both coefficients near zero, so the two coefficients are reported next to the verdict. Three further results are reported in a fixed order and do not enter the decision: C-PR against C-PR ($\lambda = 0$) on the liquidation-free close rate as a superiority test (F4), C-PR against C-PR ($\lambda = 1$) on the same label (F5), and the primary part of rule A, rerun on the new window (F6).

Two sensitivity analyses were registered in the same commit, before the study extracted any July–September log. Both are reported next to the decision, and neither enters it. The first repeats F1–F4 with AWP (Section 3.6.1) in place of C-PR ($\lambda = 0$). The second repeats F1–F6 with the cohort contracts that a graph leaves out kept as isolated nodes instead of scored zero (Section 4.4). Only the two layers walked alone change under the second analysis, since every spender is in the allowance graph and every trader in the union graph that C-PR walks. Other contrasts in W1, each PageRank against its own degree and C-PR against every other score among them, were computed after the labels were known and are reported apart from the decision.

3.7.3 Outcomes built from neither edge type

A comparison of the two edge types on the GMX trader labels, which are built from neither, was registered in the same commit as rule B. As in a preregistration (Nosek et al., 2018), its contrasts, rules and wording were written down before the study extracted the July–September logs and before any W0 liquidation label was computed; on the revised scores they are applied as a re-analysis. It reuses the trader cohorts and label code of Sections 3.7.1 and 3.7.2 in two windows: W1 (primary), frozen on 30 June 2026 with labels from July to September (70 traders), and W0 (secondary), frozen on 31 March 2026 with labels from April to June (87 traders).

Four pairs set allowance against transfer: in-approve degree and transfer in-degree; EndorseRank with AWP's activity restarts and AWP, the like-for-like PageRank pair; EndorseRank and AWP; and the amount-weighted layers C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). C-PR ($\lambda = 0$) minus AWP

isolates AWP's weights and restarts. All six trader labels are reported: the liquidation-free close rate (primary), a no-liquidation flag, profitable closes, realized gain, and the profitable and non-loss shares of closes. Two contrasts on the liquidation-free close rate in W1 decide: P, in-approve degree minus transfer in-degree, and Q, EndorseRank with activity restarts minus AWP, each with a Bonferroni 97.5% interval from $B = 2{,}000$ paired resamples (seed 42). Because the chance of at least one liquidation rises with the number of closes, τ_b is also computed within strata of 3–4, 5–9, 10–24 and 25 or more closes in the label window and weighted by each stratum's contract pairs. Recipients (traders with an in-approve degree above zero at the freeze) are compared with the other traders on every label, and an influence check drops the 10 recipients of highest in-approve degree. Every W1 recipient and 38 of the 39 W0 recipients were approved by one and the same cohort contract, so the recipients form one cluster. The two trader cohorts share 56 contracts, all W1 recipients among them, so W0 is not an independent second sample.

R1, a count-level lead on liquidation avoidance, needs the 97.5% interval of P and the 95% interval of the stratified P above zero in W1, and a positive P in W0. R2, a PageRank-level lead, needs the 97.5% interval of Q above zero in W1 and a positive Q in W0. R3 applies if P on the profitable or the non-loss share in W1 is below zero with a 95% interval excluding zero; the allowance signal is then read as specific to liquidation avoidance. The wording of the conclusion for each outcome was fixed in the same commit.

3.8 Construct-check design

Every score enters the same-window evaluation: EndorseRank, AWP, C-PR at its five values of λ and S-PR. For each score M and check q the pipeline reports Spearman's ρ and Kendall's τ between $\mathbf{s}_M$ and $\mathbf{p}_q$ with 95% bootstrap intervals (Section 3.10), each check oriented so that a higher value means higher expected reputation, as in the AWP evaluation protocol (T. Do et al., 2023).

The same-window evaluation is organized around two contemporaneous families:

1. Transfer (the proxies used to evaluate AWP (T. Do et al., 2023)): the contract's inbound

transfer activity.

2. Allowance: inbound approval activity, where the contract is the spender.

Both families are graph-adjacent: their inputs are the event classes that make up G_{transfer} and G_{approve}, although each check is a local statistic and not a global score. Alignment of AWP with the transfer checks, or of EndorseRank with the allowance checks, is an intended-construct check: it shows coherence within a construct and is not evidence of independent predictive power. A third family, the Sybil-adjusted activity summaries of Section 3.9.3, enters the secondary part of rule A, but no hypothesis rests on it.

The temporal holdout of Section 3.7 provides the out-of-window check. Packin and Lev-Aretz (2024) warn that decentralized credit scores can become opaque; publishing explicit formulas (Section 3.9) and keeping same-window checks apart from the time split reduce that opacity.

3.9 Operational definitions of proxies

In what follows, $\mathcal{W}$ denotes the matched cohort. The notation is that of the check-computation module in the open evaluation pipeline.

3.9.1 Transfer family

Write $\mathcal{T}_{\text{in}}(w)$ for the set of transfers received by contract w in the window and u_e for the sender of transfer e. Self-transfers belong to $\mathcal{T}_{\text{in}}(w)$, so a contract that sends tokens to itself counts itself as one sender, and amounts x_e are summed in US dollars across the five tokens. Define

$$\text{in_degree}(w) = \big|\{\, u_e : e \in \mathcal{T}_{\text{in}}(w) \,\}\big|, \tag{3.10}$$

$$\text{in_value}(w) = \sum_{e \in \mathcal{T}_{\text{in}}(w)} x_e. \tag{3.11}$$

Both are baselines defined in the AWP paper (T. Do et al., 2023), on the premise that economically important addresses receive transfers from more distinct counterparties and in greater total value.

Unlike PageRank they are local and ignore the senders' reputations, so comparing AWP with them shows how far propagation, recency weighting and restarts at active senders move its ordering away from the raw inflow.

3.9.2 Allowance family

On the allowance side, w is the spender (owner $\rightarrow$ w). With $a^\star$ the latest allowances of Equation (3.7), define

$$\text{in_approve_degree}(w) = \big|\{\, u : a^\star(u, w, \cdot) > 0 \,\}\big|, \tag{3.12}$$

$$\text{in_approve_value}(w) = \sum_{u,\text{token}} \min\big(a^\star(u, w, \text{token}), U\big). \tag{3.13}$$

Like the transfer baselines, they count and sum incoming approvals without following multi-hop endorsement chains; the in-approve value uses the cap U of C-PR.

3.9.3 Sybil-adjusted stability family

The inbound counters leave out self-transfers ($u = w$). Let $N_{\text{tx}}(w)$ be the number of transfers that w receives from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Tenure and active months use a second set of rows. Every transfer that involves a cohort contract is attributed to its sender if the sender is in the cohort, and otherwise to its recipient; a transfer between two cohort contracts thus counts only for the sender. We take $t_{\text{min}}(w)$ and $t_{\text{max}}(w)$ to be the earliest and latest timestamps among the rows attributed to w. $N_{\text{mon}}(w)$ is the larger of two counts of distinct calendar months: the months in which w received a transfer from another address, and the months

of the rows attributed to w. Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \tag{3.14}$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad \text{(seconds to days)}, \tag{3.15}$$

$$\text{active_months}(w) = N_{\text{mon}}(w). \tag{3.16}$$

The counterparty ratio penalizes contracts that receive many transfers from few addresses, a simple wash pattern. It does not penalize the usual Sybil pattern, in which many fresh addresses send one transfer each; a contract fed that way scores close to the maximum of one. Tenure and active months track how persistent on-chain activity is over the score window. These proxies are therefore weak Sybil checks, and colluding adversaries can satisfy all three.

3.10 Statistical tests

Neither the reputation scores nor the proxies are binary default indicators; both are continuous and are handled as ranks. Following the AWP evaluation protocol (T. Do et al., 2023) and the approach to construct validity of Cronbach and Meehl (1955), concordance is measured with rank correlations, which are invariant under any monotonic re-scaling of either variable.

- Spearman's ρ: the Pearson correlation of the mid-ranks R_i and S_i of the raw score and the raw proxy value of contract i, tied values sharing the average of their positions (Section 2.9). It measures monotonic agreement: if higher reputation consistently goes with higher proxy values, ρ is positive whatever the nonlinearity in the raw scores.
- Kendall's τ: every reported value is the tie-corrected τ_b of Equation (2.6), computed on the raw scores and proxies. Without ties, $(1+\tau)/2$ is the probability that a randomly chosen pair is ordered the same way by both rankings. With ties that reading belongs to $\tau_a = (n_c - n_d)/n_0$ and not to τ_b. Ties are rare among the PageRank scores of the matched cohort (Section 4.4) but common in the raw degrees and in the labels. When the two sides of a paired contrast tie

very different shares of pairs, as a raw count and a PageRank do, τ_b gives the more tied side the smaller denominator, so the contrast can favour it under τ_b and not under τ_a; the degree contrasts and P, computed with τ_b alone, are exposed to this.

- Why both: the AWP protocol (T. Do et al., 2023) reports both, and reporting both keeps the conclusions from hinging on one coefficient. The score distributions are heavy-tailed, with most contracts near zero and a few hubs holding much of the mass, which is why rank coefficients are used instead of a Pearson correlation on raw scores, which the hubs would dominate.
- Family aggregation: A family is summarized by the plain average over its proxies. The transfer-family mean τ, for example, is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. Conclusions are drawn from the confidence intervals, and no absolute τ cutoff is used as a decision criterion.
- Uncertainty: Sampling uncertainty is estimated with a paired bootstrap over contracts (Efron, 1979). The evaluated cohort is resampled with replacement $B = 400$ times (seed 42, fixed in the configuration), and one index matrix is reused for every method and proxy, so the resampled statistics are paired across methods. For each proxy τ, family mean and inter-method difference, the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval (DiCiccio & Efron, 1996). Because τ_b is a smooth function of U-statistics, the percentile interval is first-order accurate at the sizes of the matched and spender cohorts if contracts are independent; contracts that share owners or belong to one protocol are not, so the intervals may be too narrow. The small number of resamples keeps the $O(B\,n \log n)$ cost of the tie-corrected τ tractable across all methods and proxies, at the price of a Monte Carlo error in each interval end of about 0.13 times the bootstrap standard error. Only the analysis of Section 3.7.3 departs from this, with $B = 2{,}000$.
- Difference tests: Every comparison is a paired contrast on the same resamples. Resample by resample we compute $\Delta\tau = \tau_A - \tau_B$ and report the 95% percentile interval of its bootstrap distribution; a contrast counts as a difference when its interval excludes zero. The intervals

of the more than a hundred contrasts computed after the labels were known are not adjusted for multiplicity, and a few would exclude zero by chance alone; a post hoc contrast resolved in one window only is therefore read as a hypothesis, and one resolved in the same direction in both windows as less exposed to chance. Because resampling is paired, the contrast keeps the covariance of the two estimates, $\mathrm{Var}(\hat{\tau}_A - \hat{\tau}_B) = \mathrm{Var}\,\hat{\tau}_A + \mathrm{Var}\,\hat{\tau}_B - 2\,\mathrm{Cov}(\hat{\tau}_A, \hat{\tau}_B)$, and with positive covariance its interval is narrower than the two separate intervals suggest. The contrasts that decide a rule were fixed before the study extracted the data they are judged on: those of rule A (Section 3.7) before any log, and those of rule B and Section 3.7.3 before the W1 logs; the revised analysis applies them after the registered results were known. C-PR against its transfer layer walked alone on new approval pairs in W0 was not among them; it is marked as exploratory wherever it appears. In the same window, eight contrasts set C-PR and S-PR against whichever layer, walked alone, leads each family (the transfer walk on transfer, the allowance walk on allowance, both on Sybil stability), and six set C-PR against AWP on transfer and allowance, against EndorseRank on allowance and transfer and against both on Sybil stability; none of these fourteen was fixed in the plan, and two of them, C-PR against AWP on allowance and against EndorseRank on transfer, were computed afterwards from the saved scores. Out of window, the contrasts of C-PR with every other score and with the two raw degrees (Table 4.18) were also computed after the labels were known. Five same-window contrasts that involve EndorseRank and AWP, none fixed in advance either, are reported in Appendix 8.5.

- Directionality and imputation: Each proxy is oriented so that a higher value means a better reputation. After re-alignment to the evaluated cohort, a contract absent from a given proxy source is imputed as zero. With sparse transactions this is a conservative choice, and both scoring approaches apply it in the same way. After this step every evaluated contract has a value in every score and proxy vector, so each correlation uses the whole cohort.
- Non-inferiority: Two of the three conditions of rule B ask only that C-PR not fall behind its transfer layer walked alone by more than a margin fixed in advance, $\delta = 0.01$. Such a

contrast passes when the lower end of its 95% interval lies above $-\delta$. The test is weaker than a difference test and says nothing about superiority on that label. It is used where the claim under test is that C-PR gives up nothing that matters.

- What is excluded: The study attempts no causal inference, trains no supervised models on separate training and testing sets, and reports no p-values; it asks whether the interval of a contrast excludes the null value.

3.11 Computational benchmark (Claim C5)

Claim C5 concerns the computational cost of each score. The benchmark times the solver call that turns an edge list into scores. It leaves out the one-time cost of building the edge lists, mostly loading and preprocessing, which a scoring service that refreshes its scores periodically would amortize.

- Timed component: The edge lists and AWP's restart weights are built before the timer starts. The timed call indexes the nodes, assembles the sparse weighted adjacency matrix, row-normalizes it, forms the restart vector and runs power iteration with $d = 0.85$, $\varepsilon = 10^{-8}$ and $I_{\max} = 300$. Matrix assembly and each iteration take time that grows with $|E|$ and $|V|$, and a deployment would repeat both at every update. The same graphs give the alignment scores.
- Measurement protocol: Every configuration is measured in one session. An untimed warm-up run, on which alone peak memory is tracked, is followed by five timed runs, from which we report the mean and standard deviation of the run times and the iteration count, with the nodes and edges of the input graph. The same graph under the same settings occurs in the main benchmark, in the $d = 0.85$ row of the damping sweep and in the scaling stage with $n = 6{,}483$, which is the full cohort graph (Section 3.12.1); those tables reuse the one measurement by design, and they are not independent measurements of one quantity. Every runtime table lists edge counts, since edges drive the cost of an algorithm with $O(|E|)$ work per itera-

tion (Langville & Meyer, 2006), and the runtimes are written to the machine-readable output (Section 3.13).

- Cohorts: The main benchmark runs on the graphs of the matched contract cohort ($n = 6{,}483$). For the growth of runtime with graph size we draw deterministic subsamples of the cohort (Section 3.12).
- Fairness: Every algorithm runs with the same solver, tolerance, iteration cap, damping (outside the damping sweep) and seed set, so differences in cost come from the graphs and from AWP's restart vector, which can change the number of iterations, and not from the implementation. Hardware and software are recorded in the benchmark summary, and the cost of two scores is compared through time ratios, each reported beside the corresponding edge ratio.

3.12 Scaling and robustness extensions

We also examine how runtime grows with the size of the graph, and whether the family-level pattern survives changes to the damping, the slope of the value weight and the sample. These analyses qualify Claim C5 and test the reach of the family-level conclusions; neither supplies an alternative primary sample. Unless stated otherwise, alignment claims in Chapter 4 refer to the matched cohort.

3.12.1 Cohort subsamples for runtime scaling

The scaling stages are drawn from the matched cohort by deterministic SHA256 subsampling with seed string `contract-benchmark-v1`: contracts are sorted by

$$\texttt{SHA256}(\textit{seed}:\textit{address}) \text{ with } \textit{seed} = \texttt{contract-benchmark-v1},$$

that is, by the hash of one string made of the seed, a colon and the lower-cased address, and the subsample is the first n contracts in that order. The seed string is a constant fixed in our configuration. We report four stages, $n = 1{,}000$, $2{,}000$ and $5{,}000$ and the full cohort, $n = 6{,}483$. Since the hashing is deterministic, the subsamples cannot drift between library versions the way pseudorandom draws do, and each stage contains the one before it.

Two features of this design limit what the experiment can show. At every stage the graph keeps the edges of the cohort graph at T_{obs} that have at least one end point among the sampled contracts, and the solver's node set consists of the end points of those edges. The edge count $|E|$ thus grows with the stage but not in proportion to n, since a contract with many counterparties brings many edges. The other is that no stage goes beyond the full cohort graph, so the experiment shows how runtime grows with $|E|$ up to that size and says nothing about larger graphs. Every scaling table reports per-stage $|V|$ and $|E|$ for both methods, and the scaling statement in Chapter 4 goes no further than those columns allow.

3.12.2 Damping sensitivity

On the matched cohort we vary the damping over $d \in \{0.75, 0.85, 0.95\}$, with the graphs and proxies held fixed. A smaller d raises teleportation; a larger one gives multi-hop paths more weight and may slow convergence. The sweep asks whether alignment on the allowance and transfer families depends on the teleportation probability.

3.12.3 Value-slope sensitivity

For this check we recompute EndorseRank and AWP at T_{obs} with the slope b of V multiplied and divided by ten, which moves the amount of half weight to a tenth and to ten times the median. It replaces the plan's twenty-token check, which loses its point with five tokens (Section 3.5.1). The check asks whether the family-level results change when the amount weight rises more or less steeply.

3.12.4 Sample-definition sensitivity

At each scaling stage of Section 3.12.1 the family means of EndorseRank and AWP are recomputed on the stage's contracts, scored on the stage's graph. The stages differ from the cohort in size and in their graphs: a contract's score changes because the edges of the contracts left out of the stage are missing. The per-stage coefficients are therefore a sensitivity check on the sample definition

and not evidence of stability, and no intervals were computed for them; the question is how far each coefficient moves when the sample changes. This analysis was not in the plan.

3.13 Reproducibility

No table in Chapter 4 was transcribed by hand. Once the Phase 1 tables are on disk, the evaluation drivers recompute every check, every alignment statistic with its interval, every holdout result and the benchmark, and write them to machine-readable summaries, from which a separate stage exports the tables of Chapter 4 and Appendix 8.1. One versioned configuration file holds the windows, cohort threshold, event-topic IDs, damping, decay parameters, values of λ, bootstrap count and seed, benchmark seed and cost budget, and the revised analysis adds the token list, daily prices and constants of Section 3.5.1 and the start of each score window.

The plan, the registration of W1 and the documents of the revised analysis are in the public repository at https://github.com/abitocodes/phd_works: the plan and the registration in commits `43d8045` and `0d3a057`, and the revised analysis in `docs/revision_price_weighting.md` and `docs/revision_one_year_window.md`. The reproducible elements are these: deterministic subsampling of the cohort by SHA256 hashes; a fixed PageRank tolerance and iteration cap; one bootstrap seed and one resample-index matrix for each cohort; the proxy formulas of Section 3.9; the benchmarking procedure of Section 3.11, with the time of every run kept in the output; the coupling weight λ and the holdout logic of Section 3.7, both fixed in the configuration before the study extracted any log; the account types, read at a pinned block; and the extraction manifest of scan B. The machine-readable summaries behind all the tables, with the configuration, the plan and the registration, are committed to the Git repository under `experiments/data/3-published-results-for-thesis/` (the revised analysis in `revised-usd-12m/`), in the same folder as the source code. Appendix 8.1 lists the GitHub repository URL, the exact commit hash that generated the tables in this thesis, the configuration file and the summary files.

3.14 Sampling considerations and missingness

The matched cohort is a constructed sample, not a random draw from all Arbitrum addresses: eligibility needs non-zero approvals of one of the five tokens from at least three distinct owners in the twelve months before the freeze, and transfer activity plays no part in it. Same-window findings therefore apply to contracts on Arbitrum One that at least three owners approved on the five tokens in these months, and holdout findings to the contract spenders in the cohort's approval data. EOAs and contracts that fewer than three owners approved on these tokens are excluded by design (Section 3.2).

Data can be missing in a few ways. An amount that cannot be parsed is read as zero, which drops a transfer and resets an allowance. EndorseRank does not see approvals that emit no standard ERC-20 event, and AWP does not see transfers made off-chain or inside custodial accounts. Both are thus biased toward on-chain, standard-token behavior, which is acceptable for an on-chain reputation study but matters once results are extrapolated to all contracts of a protocol.

3.15 Implementation notes

The PageRank code is written in Python, with a SciPy sparse transition operator and NumPy power iteration. Appendix 8.1.1 names the machine on which the benchmark ran. Randomness enters only through the bootstrap, whose seed is fixed; given the downloaded tables and the configuration, evaluation is deterministic. Small floating-point differences between platforms are to be expected, but they would be negligible against the three-decimal-place precision of the τ metric.

3.16 Ethics

Only publicly available blockchain data are used in this research, and it involves no human subjects. The study was reviewed by the institutional ethics committee (Unisa SoC/CAES). Appendix 8.7 contains the ethics clearance certificate (reference [Unisa SoC/CAES ethics clearance ref: TBD],

dated [TBD]).

- Privacy and anonymity: An address on Arbitrum One is a pseudonym and does not by itself reveal the real-world identity behind it (Packin & Lev-Aretz, 2024). Pseudonymity is not the same as anonymity, however, and in theory a determined adversary could tie an address to a person through exchange deposits, online profiles or heuristics. In this work we deliberately make no such links. The scored addresses are contracts, and no result is reported for any EOA. Our data collection is limited to on-chain `Approval`, `Transfer` and GMX V2 position events, and we scraped no off-chain personally identifiable information (such as IP addresses, exchange KYC records, social media handles or deanonymization graphs). We did not cluster addresses into real-world entities, publish leaderboards of addresses likely to be personally identifiable, or try to join an address's transaction history with other databases to infer identity. The published results are aggregate measures only (correlation coefficients, runtimes, family means) and contain no individualized rankings of named participants.
- Data minimization: Scan A keeps every ERC-20 `Approval` and `Transfer` event from 1 October 2023 to 30 June 2026 in the study's cloud project for the cohort rule; every later query and downloaded table is restricted to the cohort's ego network and to the fields that graph construction and the checks require, namely the relevant addresses, token amounts and timestamps. The extracted tables are kept so that the research pipeline can be audited. They will not be released for the purpose of deanonymizing users.
- Data usage and compliance: Every query ran against public data hosted in the `bigquery-public-data.goog_blockchain_arbitrum_one_us` dataset and complied with the Google Cloud terms of service and rate limits. Dry runs and a study budget prevented unnecessary scans. Account types and token data were read with read-only RPC calls, and daily prices came from the public DefiLlama API. We sent no transactions and accessed no privileged or non-public indexing endpoints, and no smart contract was invoked in any way that might interfere with the normal operation of the network.

- Fairness and transparency: Decentralized credit scores raise concerns about fairness as well as opacity (Packin & Lev-Aretz, 2024). New contracts, rarely used contracts and contracts that receive no public approvals may all be undervalued by graph-based reputation systems. Both EndorseRank and AWP carry structural biases of this kind. In the allowance graph, for instance, an address that has received no incoming spend authority is at a disadvantage relative to addresses that have; transfer graphs, in turn, favor addresses embedded in value-flow networks. Chapter 5 discusses the limitations, among them cold-start effects and the Sybil susceptibility of approval graphs.
- Responsible disclosure and impact: EndorseRank may inform risk assessment in DeFi, but it is not financial advice, credit advice or a consumer reporting product. Lending, trading and compliance decisions still need human oversight. Chapter 5 acknowledges and discusses the potential for adversarial manipulation, for example Sybil attacks on G_{approve} or wash transfers on G_{transfer}. Any future operational deployment would require additional governance, appeal mechanisms and fairness monitoring, all of which lie beyond the scope of this research.
- Summary of methodological commitments: Before results are interpreted, the design has already fixed the chain, windows, cohort and account-type rules, graph formulas, proxy definitions, bootstrap and timing protocols, and statistical tests; Section 3.5.1 describes the one later change. The primary claims concern all the scores together, on the matched contract cohort and the spender holdouts, with each PageRank set against the raw degree it smooths.

Chapter 4 reports what this protocol yields for Arbitrum One.

4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one score window per freeze, one matched cohort of contracts for every same-window comparison, spender and trader cohorts for the temporal holdouts W0 and W1, and subsamples of the matched cohort for runtime against size. It reports the two single-layer walks first, their cost, their agreement in the window, their divergence, robustness checks and their holdouts (Sections 4.2 to 4.6), and then the coupled walk C-PR, the method of the thesis, under its rules and beside every other score (Section 4.7); a comparison on outcomes built from neither edge type follows (Section 4.8). It reports the revised analysis of Section 3.5.1, with amounts in US dollars, five tokens and scores from the twelve months before each freeze; this analysis was defined after the registered results were known, so no result in this chapter is a registered result. Rules A and B and R1–R3 are applied to the revised scores, with their thresholds and wording, as a re-analysis; the registered analysis reached the same verdicts except on R1, which it did not meet (Appendix 8.6, Table 8.12). Every table is generated from the evaluation summaries of Section 3.13, each Kendall τ comes with its confidence interval except in the value-slope and sample-definition checks, and each PageRank is read against the raw degree it smooths.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because one of its margin-trading protocols, GMX V2, records the position closes behind the trader labels. Each score window covers the twelve months before its freeze, 1 July 2025 to 30 June 2026 for the same-window evaluation and W1 and 1 April 2025 to 31 March 2026 for W0. The time decay is

anchored at the freeze (Figure 3.3), so the most recent six months of a window carry most of the edge weight in AWP and EndorseRank.

4.1.1 Cohorts

Four sets of contracts are used. Every address in them holds contract code; externally owned accounts (EOAs) enter the graphs as owners, senders and recipients and are never scored (Section 3.2).

- Matched cohort ($n = 6{,}483$). The contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners in the score window of T_{obs} (Section 3.2). They are part of the analysis plan's cohort of 27,844 contracts, which applied the same rule to every token from 1 October 2023 and left out as EOAs 1,121 of its 29,022 candidates without code and 57 with an EIP-7702 delegation designator. Membership does not depend on either graph: no allowance edge touches 191 cohort contracts (by T_{obs} every allowance set in the window by them or to them was zero), no transfer edge touches $1{,}192$, and 30 are touched by neither; a contract absent from a graph scores zero under that method. Every same-window alignment, difference test and benchmark is computed on this cohort. The W0 data are drawn around a cohort of their own, the $6{,}343$ contracts that meet the same rule in the W0 score window, $5{,}383$ of which are also in the matched cohort.
- Spender cohorts of W0 and W1 ($n = 7{,}914$ and $7{,}960$). In W0 the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC), and the cohort is every spender that holds at least one positive latest allowance of one of the five tokens, set inside the W0 score window, at t_1 and holds contract code: 7,914 of 7,962 such spenders. In W1 the scores are frozen at T_{obs}, and the same rule gives 7,960 contract spenders. The spender cohorts are not subsets of the matched cohort, since a spender needs only one owner with a positive allowance. The scores of these contracts are tested against owner–spender approval pairs and transfer senders first seen in the three months after each freeze (Section 3.7).
- Trader cohorts ($n = 87$ and 70). The GMX V2 accounts that hold contract code, are nodes of the freeze-date graph and close at least three positions in the label window: 87 in W0, 39 of

which had received an approval in the score window, and 70 in W1, 37 of which had. Contract accounts closed 133,685 GMX V2 positions from October 2023 to June 2026, but only 9,809 in the last six months of that span and 6,805 from July to September 2026 (Table 8.5 in Appendix 8.4.3). Every trader result therefore rests on few contracts and has wide intervals.

- Subsamples of the matched cohort ($n = 1{,}000$, $2{,}000$ and $5{,}000$). Deterministic SHA256 subsamples (seed `contract-benchmark-v1`), followed by the full cohort, used only for runtime against size and for the sensitivity to sample definition (Section 3.12.1). Each stage keeps the extracted edges incident to its sampled contracts, so its edge count grows with n.

At T_{obs} the allowance graph has 2,514,846 nodes and 3,808,949 edges, and the transfer graph 3,603,300 nodes and 8,426,124 edges. The solver graphs of Table 4.1 leave out 864 and 1,573 addresses whose pairs all carry a value weight of zero in floating point. On the same contracts the allowance graph has 45.2% as many edges as the transfer graph, because an owner–spender pair is an allowance edge only while its latest allowance of one of the five tokens, set in the window, is positive, whereas every pair of addresses that exchanged a non-zero transfer of these tokens in the window is a transfer edge. The sampling is the same for both, so the differences between the scores trace back to their edges and restarts.

The five tokens hold most of the extracted rows: of the ego-network events of the plan's cohort in the score window of T_{obs}, the revised analysis keeps 85,986,562 approvals (63.3%) and 442,722,039 transfers (76.2%), and every kept transfer has a price. Table 8.4 in Appendix 8.4.2 gives these figures per token.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two edge lists node by node (Equation (3.9)). Each evaluation run writes its scores, per-proxy statistics and bootstrap intervals to one summary file, and the benchmark its timings to another. The tables of this chapter come from runs made on 28 October 2026 from 09:16 UTC. At $d = 0.85$ every

solve converged well inside the iteration cap. Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C5)

Claim C5 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the full graphs of the matched cohort, the mean solve time and its standard deviation over five timed runs after one untimed warm-up, the peak memory of the warm-up solve, the number of iterations to convergence and the nodes and edges of the solver graph. Each graph and setting was timed once, so the same measurement reappears as the full-cohort row of Table 4.2 and as the $d = 0.85$ row of Table 4.5; Table 8.2 in Appendix 8.1.4 lists every timed run.

Table 4.1: Computational benchmark on the full graphs of the matched cohort ($n = 6{,}483$ contracts; AWP/EndorseRank runtime ratio 3.2× at an edge ratio of 2.2×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalisation, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.2 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	2.657	0.109	325.3	57	2,513,982	3,808,949
AWP	8.530	0.618	663.7	94	3,601,727	8,426,124
C-PR ($\lambda = 0.5$)	11.563	0.214	719.1	82	4,111,490	10,562,390
S-PR	10.883	0.330	509.4	151	3,603,300	8,432,635

EndorseRank solves the allowance graph in 2.66 s and AWP the transfer graph in 8.53 s, so AWP takes 3.21 times as long, at an edge ratio of 2.21. Each power iteration costs a sparse product over the $|E|$ edges and several vector operations over the $|V|$ nodes (Appendix 8.3.3). Divided by the iterations, assembly included, AWP's solve takes 0.091 s and EndorseRank's 0.047 s, 1.95 times as long, between the node ratio of 1.43 and the edge ratio, and AWP needs 1.65 times as many

iterations (94 against 57); together the two factors give the time ratio. On the same graphs that ratio is 3.00, 3.21 and 2.56 at $d = 0.75$, 0.85 and 0.95 (Section 4.5.1), so it does not follow the edge ratio. Both scores use the same solver, so the difference in speed comes from the size of each graph and from the iterations its walk needs there, which the benchmark does not separate; it is no advantage of the EndorseRank algorithm.

C-PR walks on the union of both edge sets (10,562,390 distinct pairs, 4,111,490 nodes), but its solver keeps one matrix per layer and multiplies both in every iteration, up to about 12.2 million stored entries, 1.45 times AWP's edges. It converges in fewer iterations (82 against 94) and takes 11.56 s, 1.36 times as long as AWP. S-PR runs on the transfer layer and takes 10.88 s. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector: the 151 iterations are 56 for that solve plus 95 for the seeded transfer walk. Every algorithm that walks the transfer layer, alone or with allowance edges, pays for its size.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.2 reports the runtimes of EndorseRank and AWP at each stage of Section 4.1.1, their ratio, the edge ratio and the nodes and edges of both solver graphs.

The stages of Section 3.12.1 change the edge ratio as well as the size: the transfer graph has 2.85 times as many edges as the allowance graph at $n = 1{,}000$, 2.95 times at $n = 2{,}000$, 2.73 at $n = 5{,}000$ and 2.21 on the full cohort. If the time of a solve followed $|E|$ alone, the time ratio would follow these edge ratios; it is 2.63, 7.49, 3.96 and 3.21. At no stage does it stay close to the edge ratio. It is 0.92 times the edge ratio at $n = 1{,}000$, where AWP needs 96 iterations and EndorseRank 43, 2.54 times at $n = 2{,}000$, where AWP needs 94 and EndorseRank 30, and 1.45 times at $n = 5{,}000$ and on the full cohort, where EndorseRank needs 57 and AWP 95 and 94; EndorseRank's solves take 0.32 to 2.66 s. The iterations each walk needs on each subgraph, and

Table 4.2: Runtime of EndorseRank (ER) and AWP against the size of the evaluation set: deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`) up to the full cohort of $n = 6{,}483$ contracts. Each stage keeps the extracted edges incident to its sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per stage. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The full-cohort row is the measurement of Table 4.1, reused, not timed again.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\lvert E\rvert$	ER $\lvert V\rvert$	ER $\lvert E\rvert$	AWP $\lvert V\rvert$	AWP $\lvert E\rvert$
1,000	0.498	1.307	2.6×	2.8×	451,248	485,077	953,795	1,382,177
2,000	0.318	2.383	7.5×	2.9×	760,410	897,812	1,459,514	2,647,854
5,000	1.672	6.620	4.0×	2.7×	1,834,957	2,577,926	3,183,096	7,035,600
6,483[a]	2.657	8.530	3.2×	2.2×	2,513,982	3,808,949	3,601,727	8,426,124

not the edges alone, set the ratio. The stages are subsamples of one cohort graph, so they show how runtime grows with the edges up to the full graph and not beyond it. Figure 4.1 plots both runtimes against the stage size, with the edge count of each graph beside its point.

4.3 Same-window alignment (Claim C4)

Claim C4 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap confidence interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. Apart from the secondary part of rule A, fixed in the analysis plan (Section 3.7) and applied here to the revised scores as a re-analysis, none of the comparisons below was fixed in advance (Chapter 3).

The scores divide by the edges they walk on. The two that walk on transfers alone, AWP and C-PR ($\lambda = 0$), reach 0.736 ($[0.728, 0.744]$) and 0.700 ($[0.691, 0.709]$) on the transfer family and 0.226 ($[0.210, 0.245]$) and 0.194 ($[0.178, 0.213]$) on the allowance family; S-PR, whose restarts come from the allowance walk, reaches 0.671 ($[0.661, 0.681]$) and 0.289 ($[0.272, 0.308]$). The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.560 ($[0.548, 0.571]$) and 0.561

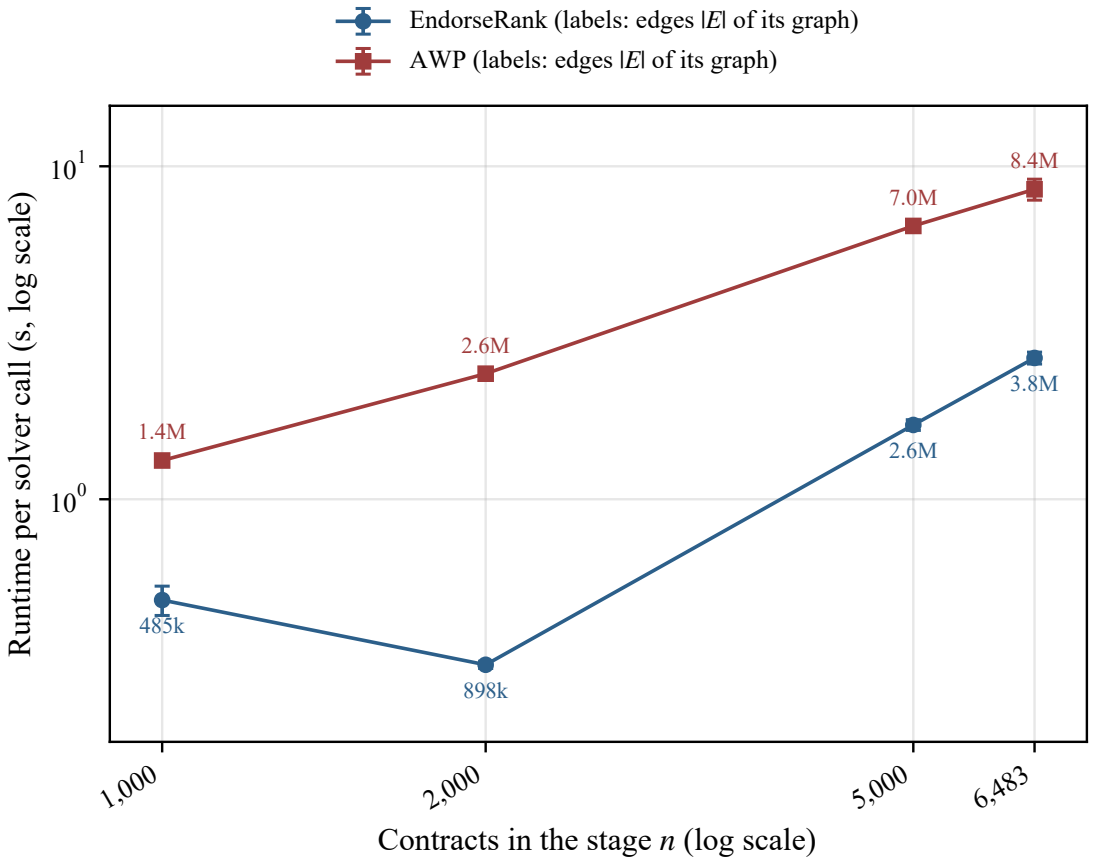


Figure 4.1: Wall-clock time of one solver call of EndorseRank and AWP against the number of contracts in each stage of Table 4.2, both axes on a log scale; the label beside each point gives the edge count $|E|$ of that graph, and error bars are the standard deviations over five timed runs.

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 6{,}483$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances in USD, each capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.8. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.196 [0.180, 0.216]	0.560 [0.548, 0.571]	0.026 [0.012, 0.041]
AWP	0.736 [0.728, 0.744]	0.226 [0.210, 0.245]	0.419 [0.408, 0.434]
C-PR ($\lambda = 1$)	0.196 [0.180, 0.216]	0.561 [0.550, 0.572]	0.028 [0.014, 0.042]
C-PR ($\lambda = 0$)	0.700 [0.691, 0.709]	0.194 [0.178, 0.213]	0.402 [0.389, 0.417]
C-PR ($\lambda = 0.25$)	0.422 [0.407, 0.438]	0.430 [0.418, 0.444]	0.157 [0.143, 0.170]
C-PR ($\lambda = 0.5$)	0.400 [0.385, 0.416]	0.449 [0.437, 0.463]	0.142 [0.128, 0.156]
C-PR ($\lambda = 0.75$)	0.377 [0.361, 0.393]	0.464 [0.452, 0.478]	0.129 [0.115, 0.143]
S-PR	0.671 [0.661, 0.681]	0.289 [0.272, 0.308]	0.398 [0.386, 0.414]

($[0.550, 0.572]$) on the allowance family and both 0.196 ($[0.180, 0.216]$) on the transfer family, and the interior mixtures of C-PR lie between the two layers. On the Sybil-stability family the scores that walk on transfers lead, AWP with 0.419 ($[0.408, 0.434]$), while EndorseRank stays at 0.026 ($[0.012, 0.041]$). Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and in-value from the transfers and in-approve degree and value from the latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's own edge type is therefore an intended-construct check and cannot count as an external test. The holdout of Section 4.6 asks the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP's largest coefficient is with transfer in-degree ($\tau = 0.742$, $[0.732, 0.752]$); with in-value it is almost as high (0.730, $[0.720, 0.740]$). That is what its weights lead one to expect: with the slope set at the median transfer (Section 3.6.1), an edge weight grows with the amounts transferred as well as with their number. In the registered analysis, with $b = 1$ on raw base units, V counted nearly

every transfer about once, and AWP followed the in-degree much more closely than the in-value (Appendix 8.6). It does not collapse to the in-degree either: propagation, time weights and restarts at active senders shift a measurable share of pairwise orderings, although out of window these shifts rank new senders below the in-degree (Section 4.6). On the Sybil-stability family AWP follows tenure (0.588, $[0.575, 0.600]$) and active months (0.616, $[0.603, 0.629]$), and it is positive on the inbound-counterparty ratio (0.055, $[0.033, 0.079]$).

EndorseRank follows the in-approve degree (0.650, $[0.639, 0.661]$) more closely than the in-approve value (0.469, $[0.454, 0.484]$). Its value weight is 0.8 at twice the median allowance of 100 dollars, so an allowance of a few hundred dollars and an unlimited one weigh nearly alike, whereas the in-approve value counts each allowance up to the cap of 19,000 dollars. Because every cohort contract received approvals from at least three owners, EndorseRank leaves few ties, so its coefficients rest on an ordering of nearly every contract (Section 4.4). C-PR ($\lambda = 1$) walks the same allowance graph with capped amounts, also restarts uniformly, and stays within 0.002 of EndorseRank on every family.

The first half of Claim C4 is supported in a narrow sense. Every score that walks on one edge type agrees most with the proxies of that edge type, with an interval well clear of zero, and the mixtures of C-PR lie between their two layers on every family. The agreement is partly built in. Section 4.7.1 locates C-PR within the window.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.199, with interval $[0.181, 0.221]$ (last row of Table 4.4), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Of the $21{,}011{,}403$ contract pairs, 96.4% are ordered by both scores, and 39.8% of those pairs are discordant.

The marginal score distributions are in Figure 4.2, and ties play almost no part in them: EndorseR-

ank, with $5{,}641$ distinct scores, ties 0.20% of the contract pairs, and AWP, with $5{,}292$, 3.38%. Under each score the largest tie block is formed by the contracts that score zero: 197 under EndorseRank and $1{,}192$ under AWP, that is, the 191 and $1{,}192$ contracts outside each graph and, under EndorseRank, 6 whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 63 further contracts share the score $c = 1.93 \times 10^{-7}$ of a node without in-edges; 62 of them are in the allowance graph only as owners, and one has received only allowances worth less than a cent, which leave its score at c to nine significant digits; every other cohort contract has received a larger allowance and scores above c. A marginal distribution cannot show how two scores order the same contracts, so Figure 4.3 plots the two rankings against each other for a random sample of $3{,}000$ contracts. Contracts that both scores rank near the top, or both near the bottom, lie close to the identity line; in between, the points spread over most of the square.

The zero blocks come from one rule of this pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping the missing contracts as isolated nodes, as a sensitivity analysis. Under uniform teleportation an isolated node scores like any node without in-edges, so the 191 contracts outside the allowance graph receive c, the lowest score that any node of the graph can have, and join the 63 contracts at c near the bottom of the order. The 6 contracts whose only pairs carry zero weight are nodes of the graph that the solver leaves out, so they keep the score zero. No pair changes direction: the pairs between the 191 and the 63 become ties, and those between the 191 and the 6 become ordered. AWP's $1{,}192$ missing contracts were handled the same way; under AWP's restarts an isolated contract that has sent nothing receives no restart mass, so it scores zero either way. Table 4.4 gives the result. No coefficient changes at the third decimal, and the agreement between the two scores stays at 0.199 ($[0.181, 0.221]$). Because the cohort was defined by approvals received, nearly every cohort contract is a spender in the allowance graph, and the rule for missing contracts decides almost nothing.

H1 is supported. The two scores order the cohort differently, and the difference does not come from ties or from the rule for missing contracts. A protocol that relies on transfer-based scores cannot assume that its ordering of contracts survives a switch to allowance-based scores or a blend with

Table 4.4: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 6{,}483$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 191 contracts for EndorseRank and 1,192 for AWP (30 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.196 [0.180, 0.216]	0.736 [0.728, 0.744]	0.196 [0.180, 0.216]	0.736 [0.728, 0.744]
Allowance	0.560 [0.548, 0.571]	0.226 [0.210, 0.245]	0.560 [0.548, 0.571]	0.226 [0.210, 0.245]
Sybil stability	0.026 [0.012, 0.041]	0.419 [0.408, 0.434]	0.026 [0.012, 0.041]	0.419 [0.408, 0.434]
EndorseRank vs. AWP	0.199 [0.181, 0.221]		0.199 [0.181, 0.221]	

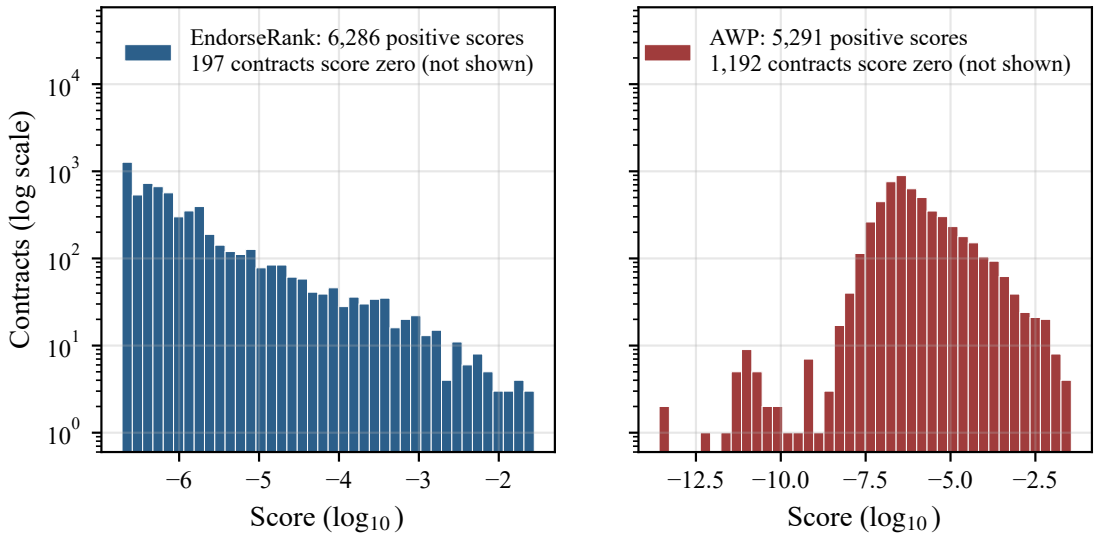


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 6{,}483$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are left out: 197 for EndorseRank and 1,192 for AWP.

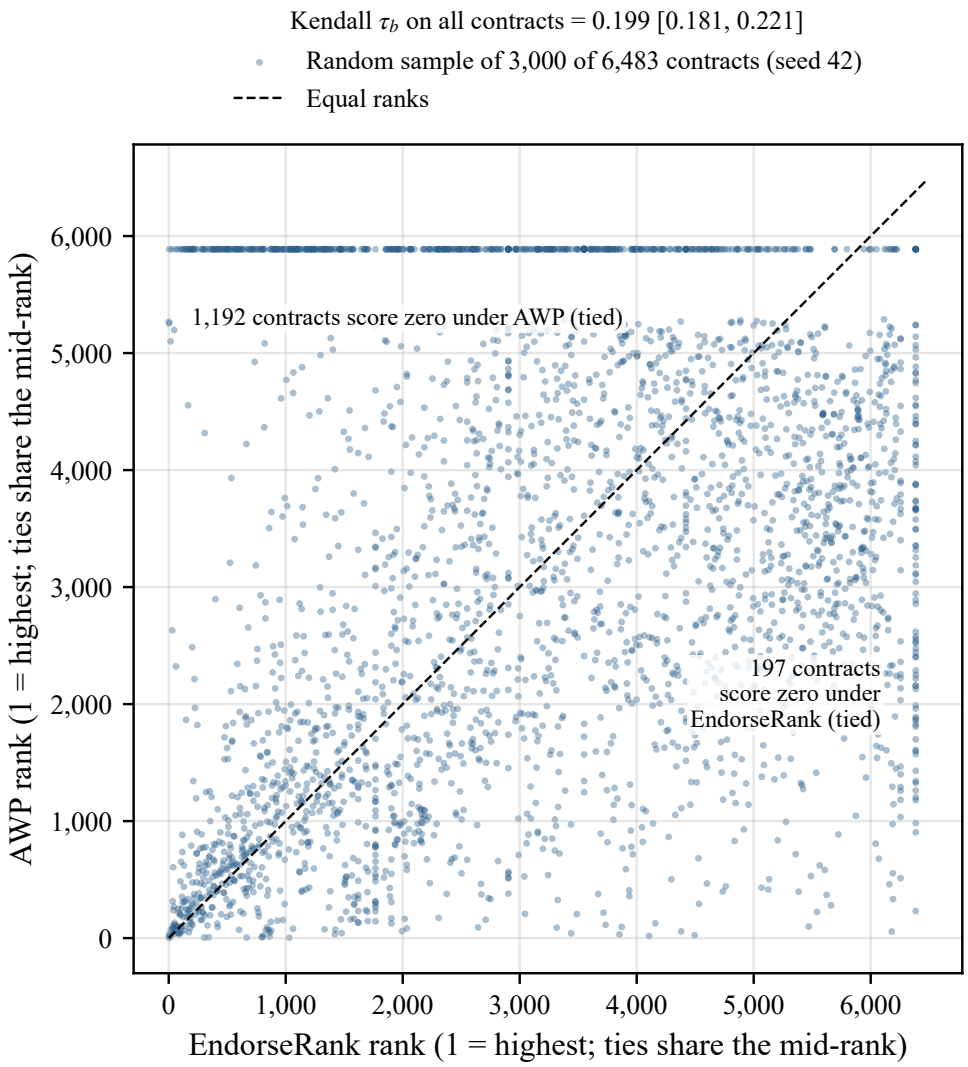


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 6,483 contracts, seed 42). Tied scores share their average rank, as in Kendall's τ_b. The column at the right edge holds the sampled contracts among the 197 that score zero under EndorseRank and the row at the top those among the 1,192 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

them. This divergence, the second half of Claim C4, is also the premise of the coupled walk: each single-layer walk orders the contracts by one kind of act, and Section 4.7 asks whether one walk over both edge types does better than either.

4.5 Robustness and sensitivity checks

Three checks, run for EndorseRank and AWP, ask whether the results hinge on one damping value, one slope of the value weight or one sample definition. The value-slope check takes the place of the plan's token check, which lost its point once every graph keeps five tokens (Section 3.12.3). The first two hold the sample fixed; the third varies the sample definition itself, so its coefficients are expected to move, and it is not evidence of robustness. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.5 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.5: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 6{,}483$ contracts); all. = allowance, tr. = transfer, Syb. = Sybil stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.1, reused, not timed again (Table 8.2 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)
0.75	0.561 [0.549, 0.573]	0.229 [0.212, 0.248]	0.196 [0.180, 0.215]	0.740 [0.732, 0.748]	0.026 [0.012, 0.041]	0.427 [0.416, 0.441]	1.986	5.956
0.85	0.560 [0.548, 0.571]	0.226 [0.210, 0.245]	0.196 [0.180, 0.216]	0.736 [0.728, 0.744]	0.026 [0.012, 0.041]	0.419 [0.408, 0.434]	2.657	8.530
0.95	0.558 [0.547, 0.570]	0.217 [0.200, 0.235]	0.197 [0.181, 0.216]	0.723 [0.715, 0.731]	0.027 [0.013, 0.041]	0.406 [0.394, 0.420]	8.113	20.805

For EndorseRank no family mean moves by more than 0.0031 across the three values. For AWP the transfer coefficient falls from 0.740 ($[0.732, 0.748]$) to 0.723 ($[0.715, 0.731]$) as d rises, and the Sybil-stability coefficient from 0.427 ($[0.416, 0.441]$) to 0.406 ($[0.394, 0.420]$). Neither score's

pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. For a spender whose owners receive no approvals, EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the owner-normalized in-strength (Section 5.3.2), and the order that this formula gives among such spenders does not depend on d. At T_{obs} the formula holds within 1% for all 5,232 such spenders among the 6,224 cohort contracts with in-edges, so the rank correlations can hardly move. Runtime grows with the number of iterations, which rises with d: EndorseRank takes 1.99, 2.66 and 8.11 s (38, 57 and 174 iterations) and AWP 5.96, 8.53 and 20.80 s (56, 94 and 269), time ratios of 3.00, 3.21 and 2.56 at an edge ratio of 2.21. Figure 4.4 plots the family means.

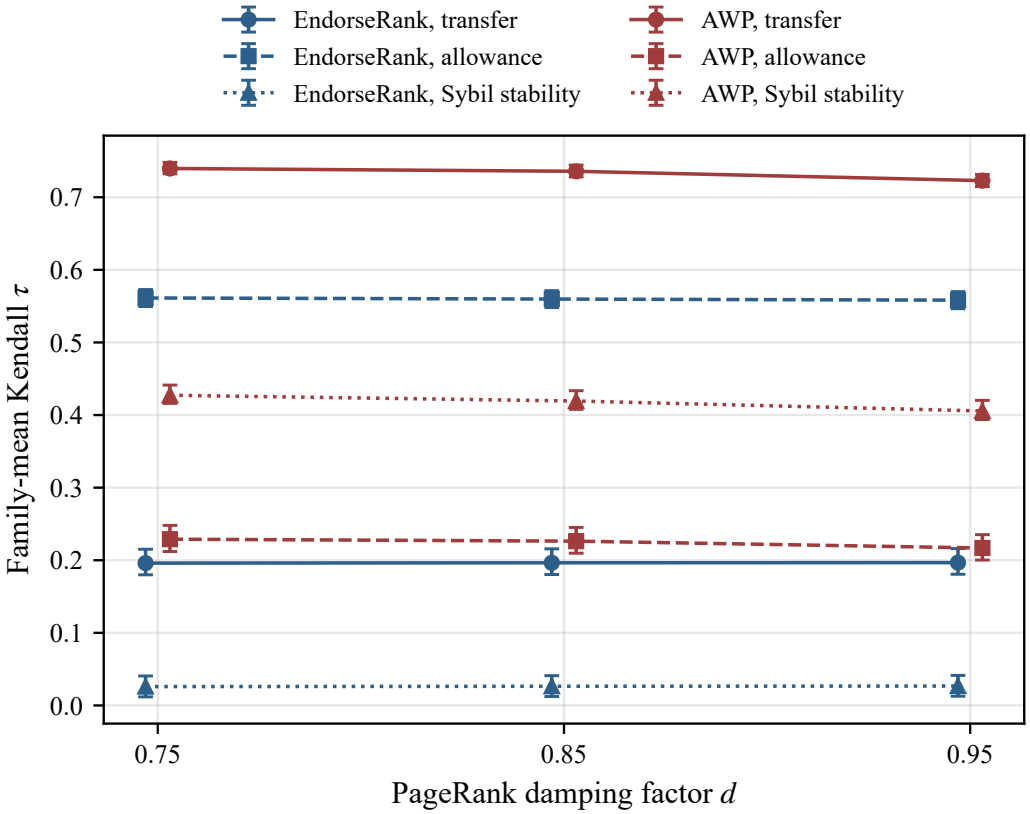


Figure 4.4: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and Sybil-stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.5; matched cohort, $n = 6{,}483$ contracts).

4.5.2 Value-slope sensitivity

Table 4.6 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.12.3), on the same graphs and proxies.

Table 4.6: Robustness: family-mean Kendall τ on the matched cohort ($n = 6{,}483$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$170 for transfers and \$100 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.979 and 0.969 at $b/10$, and 0.971 and 0.964 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Sybil		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.196	0.196	0.196	0.561	0.560	0.555	0.026	0.026	0.027
AWP	0.736	0.736	0.734	0.224	0.226	0.226	0.421	0.419	0.419

Under either slope no family mean moves by more than 0.0046, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and Sybil-stability families. The steeper slope $10b$, which puts the half weight at a tenth of the median amount and brings an edge weight closer to a count of events, lowers EndorseRank's allowance coefficient from 0.560 to 0.555 and AWP's transfer coefficient from 0.736 to 0.734; the flatter slope $b/10$ raises EndorseRank's to 0.561 and leaves AWP's at 0.736. Each variant orders the contracts almost as the main score of its method does (τ from 0.964 to 0.979). No intervals were computed for the variants.

4.5.3 Sample-definition sensitivity

Table 4.7 recomputes the family means on the subsamples of the runtime scaling.

EndorseRank moves most: its allowance coefficient is 0.665 at $n = 1{,}000$ and falls with every stage to 0.560 on the full cohort, and its transfer coefficient falls from 0.225 to 0.196. AWP stays within 0.022 of its full-cohort values on every family. No intervals were computed for the stages. None of these shifts is evidence about the scores. A stage keeps only the edges incident to its sampled contracts, so the paths that pass through unsampled cohort contracts are cut, and each

Table 4.7: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 6{,}483$ contracts), each stage scored on the edges incident to its contracts; all. = allowance, tr. = transfer, Syb. = Sybil stability. Rows are evaluated on different contract sets, so their τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. No intervals were computed for the stages. Runtimes are the measurements of Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)	AWP/ER time
1,000	0.665	0.228	0.225	0.752	0.049	0.441	0.498	1.307	2.6×
2,000	0.611	0.223	0.200	0.731	0.036	0.421	0.318	2.383	7.5×
5,000	0.577	0.229	0.201	0.735	0.033	0.420	1.672	6.620	4.0×
6,483	0.560	0.226	0.196	0.736	0.026	0.419	2.657	8.530	3.2×

score comes closer to a count of the sampled contract's own edges, which is what the proxies count. We therefore report the table as a sensitivity analysis of the sample definition (Section 3.12.4), and its values must not be compared with those of Table 4.3.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). The analysis plan fixed uniform restarts for EndorseRank and named the variant as a score to report beside it, without fixing any comparison between the two. Table 4.8 scores both versions of EndorseRank, with AWP beside them.

Table 4.8: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 6{,}483$ contracts) and τ with the two spender labels in W0 ($n = 7{,}914$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 7{,}960$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Sybil	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.196 [0.180, 0.216]	0.560 [0.548, 0.571]	0.026 [0.012, 0.041]	0.303 [0.287, 0.319]	0.272 [0.257, 0.286]	0.299 [0.284, 0.314]	0.259 [0.243, 0.275]
EndorseRank	by approving activity	0.251 [0.233, 0.271]	0.601 [0.590, 0.613]	0.056 [0.042, 0.072]	0.332 [0.315, 0.347]	0.270 [0.256, 0.285]	0.335 [0.319, 0.349]	0.236 [0.219, 0.251]
AWP	by sending activity	0.736 [0.728, 0.744]	0.226 [0.210, 0.245]	0.419 [0.408, 0.434]	0.196 [0.178, 0.213]	0.409 [0.396, 0.421]	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]

With AWP's restarts, EndorseRank's same-window allowance coefficient rises from 0.560 ($[0.548, 0.571]$) to 0.601 ($[0.590, 0.613]$), and its transfer coefficient from 0.196 ($[0.180, 0.216]$) to 0.251 ($[0.233, 0.271]$). The restart mass then goes to the owners that approve most, so the score weights a spender by how active its owners are as well as by how many there are. Out of window the variant ranks new approval pairs higher in both windows, and new senders no better in W0 and lower in W1. In W0 it reaches 0.332 ($[0.315, 0.347]$) on new approval pairs and 0.270 ($[0.256, 0.285]$) on new senders, $+0.029$ ($[0.023, 0.037]$) and -0.002 ($[-0.009, 0.005]$) against the uniform version, and in W1 0.335 ($[0.319, 0.349]$) and 0.236 ($[0.219, 0.251]$), $+0.035$ ($[0.029, 0.042]$) and -0.023 ($[-0.032, -0.013]$) against it. Table 4.8 gives the coefficients and Table 4.13 the W1 contrasts. These contrasts were computed after the labels were known. Neither version ranks new approval pairs above the in-approve degree: in W0 the variant falls below it (-0.060, $[-0.070, -0.052]$), as does the uniform version (-0.090, $[-0.100, -0.079]$; Section 4.6). Section 4.7.4 sets the variant beside C-PR.

4.6 Temporal holdout of the single-layer walks (RQ3)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the W0 spender cohort are frozen at t_1 (31 March 2026), and two labels are then counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.9 sets every score computed at t_1, the two raw-degree baselines included, against both labels. Table 4.10 gives the paired contrasts, first the two of rule A, fixed in the analysis plan before any log of the study was extracted, and then those computed after the labels were known. This section reads the single-layer walks and the raw degrees; rule A and the rows of C-PR and S-PR are taken up in Section 4.7.

The first finding answers the second part of RQ3 for the single-layer walks: neither walk improves on its raw degree, whatever its restart rule. AWP, which restarts at active senders, ranks new trans-

Table 4.9: Temporal holdout W0 on the spender cohort ($n = 7{,}914$ contracts): Kendall τ between scores frozen at 31 March 2026 and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the contract for the first time. The cohort keeps the 7,914 contracts among the 7,962 spenders with a positive latest allowance at the freeze; 39 EOAs without code and 9 EOAs with an EIP-7702 delegation are left out. 1,732 contracts received at least one new pair and 1,422 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.392 [0.377, 0.406]	0.284 [0.267, 0.299]
Transfer in-degree at t_1 (baseline)	0.203 [0.183, 0.224]	0.447 [0.435, 0.459]
EndorseRank	0.303 [0.287, 0.319]	0.272 [0.257, 0.286]
AWP	0.196 [0.178, 0.213]	0.409 [0.396, 0.421]
C-PR ($\lambda = 1$)	0.298 [0.282, 0.313]	0.267 [0.250, 0.282]
C-PR ($\lambda = 0$)	0.178 [0.160, 0.197]	0.366 [0.351, 0.378]
C-PR ($\lambda = 0.25$)	0.329 [0.312, 0.343]	0.345 [0.330, 0.357]
C-PR ($\lambda = 0.5$)	0.329 [0.312, 0.343]	0.336 [0.322, 0.348]
C-PR ($\lambda = 0.75$)	0.327 [0.311, 0.342]	0.327 [0.313, 0.339]
S-PR	0.190 [0.171, 0.209]	0.377 [0.363, 0.389]

Table 4.10: Paired bootstrap differences on the spender holdout W0 ($n = 7{,}914$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan committed on 9 October 2026 before the study extracted any data and applied here to the revised scores (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, and the increments of EndorseRank and AWP over their raw degrees; C-PR's other contrasts are in Table 4.18. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse holds, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.10.

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.329	0.298	+0.031	[0.021, 0.042]	1.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.336	0.366	-0.030	[-0.040, -0.020]	0.000
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.298	0.392	-0.094	[-0.104, -0.083]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.190	0.298	-0.108	[-0.128, -0.088]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.377	0.366	+0.011	[0.004, 0.018]	1.000
EndorseRank minus t_1 in-approve degree	new approval pairs	0.303	0.392	-0.090	[-0.100, -0.079]	0.000
AWP minus t_1 in-degree	new transfer senders	0.409	0.447	-0.038	[-0.043, -0.032]	0.000

fer senders at $\tau = 0.409$ ($[0.396, 0.421]$), below the transfer in-degree at t_1 (0.447, $[0.435, 0.459]$), a paired difference of -0.038 (interval $[-0.043, -0.032]$). The transfer layer walked with uniform restarts falls further below: C-PR ($\lambda = 0$) reaches 0.366 ($[0.351, 0.378]$) (no paired contrast was computed). On the allowance side the count wins as well. In-approve degree at t_1 predicts new approval pairs with $\tau = 0.392$ ($[0.377, 0.406]$). EndorseRank reaches 0.303 ($[0.287, 0.319]$), a paired difference of -0.090 ($[-0.100, -0.079]$), and C-PR ($\lambda = 1$) 0.298 ($[0.282, 0.313]$), a difference of -0.094 ($[-0.104, -0.083]$); with activity restarts EndorseRank also stays below the count (Section 4.5.4). Within a three-month window, a spender gains more new owners and senders the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and the local count captures this directly. In EndorseRank an owner's endorsement is shared among all the spenders it approves (Section 5.3.2), so a spender whose owners approve many other spenders ranks lower than its count alone would place it; on this label that sharing lowers the coefficient. These degree contrasts were computed after the labels were known.

Read across the whole spender cohort, each label is ranked best by scores that use its own edge type. On new approval pairs the in-approve degree (0.392, $[0.377, 0.406]$) leads, followed by the C-PR mixtures (0.327, $[0.311, 0.342]$, to 0.329, $[0.312, 0.343]$), EndorseRank and C-PR ($\lambda = 1$), while the transfer in-degree (0.203, $[0.183, 0.224]$), AWP (0.196, $[0.178, 0.213]$), S-PR and C-PR ($\lambda = 0$) stay lower. On new transfer senders the transfer in-degree and AWP lead, and the allowance-side scores stay lower and close together: EndorseRank reaches 0.272 ($[0.257, 0.286]$), C-PR ($\lambda = 1$) 0.267 ($[0.250, 0.282]$) and the in-approve degree 0.284 ($[0.267, 0.299]$), with overlapping intervals. EndorseRank gives $7{,}368$ distinct scores to the $7{,}914$ spenders, and its largest tie block holds 116. Coverage does not explain the degree comparisons, but it explains part of the comparison across edge types on new approval pairs. Of the 7,914 spenders, 1,342, about one in six, have no transfer edge up to the freeze; AWP, C-PR ($\lambda = 0$) and S-PR score them zero, and only allowance edges order them. They gained new approval pairs more often than the rest (26.7% against 20.9%) and new senders almost never (1.4% against 21.3%), so AWP's zeros lower its coefficient on the first label and raise it on the second. Table 4.11 restricts the holdout to the 6,489 spenders that had received

a transfer from another address before the freeze. There AWP still ranks new senders below the transfer in-degree (0.408, $[0.393, 0.422]$, against 0.459, $[0.445, 0.474]$; -0.051, $[-0.059, -0.043]$), and EndorseRank stays below the in-approve degree on new approval pairs (0.299, $[0.283, 0.315]$, against 0.387, $[0.370, 0.402]$; -0.087, $[-0.096, -0.078]$). On new approval pairs AWP now reaches 0.328 ($[0.312, 0.343]$), above EndorseRank and below the in-approve degree (no paired contrast was computed), so EndorseRank's lead over AWP on this label rests on the 1,425 spenders left out, which include the 1,342 that AWP scores zero. The restriction was chosen after the results were known. The row of C-PR in that table is read in Section 4.7.4.

Table 4.11: Spender holdout W0 restricted to the 6,489 of the 7,914 spender contracts that had received a transfer from another address before the freeze ($n = 6{,}489$). The other 1,425 have no inbound transfer edge from another address, and 1,342 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples; its contrasts with AWP and the in-approve degree on this subset are in Table 4.18.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.299 [0.283, 0.315]	0.315 [0.300, 0.331]
AWP	0.328 [0.312, 0.343]	0.408 [0.393, 0.422]
C-PR ($\lambda = 0.5$)[a]	0.336 [0.321, 0.351]	0.372 [0.359, 0.386]
In-approve degree at t_1	0.387 [0.370, 0.402]	0.335 [0.318, 0.350]
Transfer in-degree at t_1	0.341 [0.325, 0.357]	0.459 [0.445, 0.474]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.087$ [-0.096, -0.078]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = -0.051$ [-0.059, -0.043]	

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these spender results bears on credit or loss; the trader labels below are built from neither edge type but are not credit labels either. Both labels are also sparse. Of the 7,914 spenders, 1,732 gained a new approval pair and 1,422 a new sender; every coefficient rests largely on how these subsets are ordered.

Other candidate labels were computed on the holdout, and they are reported here without being offered as support. Revocation counts, which for some tokens include allowances spent in full (Section 3.7), and the drain heuristic both rise with EndorseRank ($\tau = 0.343$, interval $[0.330, 0.359]$, and 0.295, interval $[0.279, 0.312]$) and with every other score, and revocations rise most steeply with

the raw in-approve degree (0.384, $[0.366, 0.399]$). For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 applies the W0 freeze to the GMX V2 contract traders and is reported as exploratory; its liquidation labels are left for the comparison of Section 4.8. Table 4.12 gives every score on the four profit labels of the 87 contract traders with at least three closes from April to June.

The walks that use transfer edges, the mixtures included, rank the number of profitable closes and the realized gain with intervals above zero: AWP reaches 0.166 ($[0.015, 0.308]$) and 0.197 ($[0.044, 0.332]$), and C-PR ($\lambda = 0$), the three mixtures and S-PR 0.163 ($[0.012, 0.289]$) to 0.172 ($[0.024, 0.300]$) and 0.273 ($[0.121, 0.397]$) to 0.297 ($[0.133, 0.427]$). The intervals of EndorseRank, its variant with activity restarts, C-PR ($\lambda = 1$) and both raw degrees contain zero, and the transfer in-degree stays low on both labels (0.069, $[-0.076, 0.210]$, and 0.074, $[-0.090, 0.229]$). The paired contrasts separate the two sides only on realized gain, for the C-PR layers (-0.257, $[-0.447, -0.071]$), and not on profitable closes (EndorseRank minus AWP -0.046, $[-0.205, 0.113]$; Table 8.11). Both labels grow with how much a contract trades. On the two shares, which do not, every coefficient is negative. On the non-loss share every interval lies below zero, from -0.167 ($[-0.297, -0.025]$) for EndorseRank to -0.288 ($[-0.450, -0.126]$) for the in-approve degree; on the profitable share none does. Part of this is built into the non-loss share (Section 4.8). None of the scores ranks how well a contract trades. The registered window gives C-PR a trader label of its own.

Table 4.12: W0 trader run: Kendall τ between scores frozen at 31 March 2026 and GMX V2 labels counted from 1 April 2026 to 30 June 2026, on the $n = 87$ GMX V2 contract accounts in the freeze-date graph with at least three closes in that window, 39 of which had received an approval by the freeze. The run is exploratory; the liquidation labels of W0 were left for the comparison (Table 4.19). Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	0.067 [-0.117, 0.225]	0.048 [-0.131, 0.222]	-0.160 [-0.338, 0.020]	-0.288 [-0.450, -0.126]
Transfer in-degree at t_1 (baseline)	0.069 [-0.076, 0.210]	0.074 [-0.090, 0.229]	-0.117 [-0.275, 0.036]	-0.214 [-0.367, -0.062]
EndorseRank	0.120 [-0.032, 0.260]	0.058 [-0.092, 0.196]	-0.047 [-0.193, 0.090]	-0.167 [-0.297, -0.025]
AWP	0.166 [0.015, 0.308]	0.197 [0.044, 0.332]	-0.121 [-0.264, 0.032]	-0.275 [-0.413, -0.145]
C-PR ($\lambda = 1$)	0.077 [-0.075, 0.233]	0.039 [-0.128, 0.197]	-0.076 [-0.240, 0.072]	-0.186 [-0.319, -0.045]
C-PR ($\lambda = 0$)	0.166 [0.020, 0.300]	0.297 [0.133, 0.427]	-0.088 [-0.247, 0.062]	-0.233 [-0.375, -0.102]
C-PR ($\lambda = 0.25$)	0.163 [0.012, 0.289]	0.289 [0.134, 0.414]	-0.082 [-0.234, 0.064]	-0.225 [-0.373, -0.089]
C-PR ($\lambda = 0.5$)	0.172 [0.024, 0.300]	0.287 [0.131, 0.413]	-0.080 [-0.227, 0.062]	-0.222 [-0.365, -0.084]
C-PR ($\lambda = 0.75$)	0.166 [0.021, 0.290]	0.273 [0.121, 0.397]	-0.076 [-0.228, 0.066]	-0.218 [-0.359, -0.081]
S-PR	0.168 [0.018, 0.301]	0.282 [0.118, 0.407]	-0.060 [-0.208, 0.092]	-0.203 [-0.354, -0.057]

4.6.2 The single-layer walks in the registered window W1

The registered window W1 (Section 3.7.2) freezes the scores at T_{obs}, 30 June 2026, and counts the same two spender labels from July to September 2026. Its registration and rule B concern C-PR and are reported in Section 4.7.3. At the freeze, 7,960 contract spenders held a positive latest allowance; 1,495 of them gained a new approval pair in the label window and 1,371 a new sender. Table 4.13 adds EndorseRank and its variant with activity restarts, which the registration does not name. None of their rows enters a rule, and the table's contrasts were computed after the labels were known, those of the variant from the saved W1 scores.

The W0 pattern holds for both single-layer walks in this window too. The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.381, $[0.365, 0.396]$, against 0.299, $[0.284, 0.314]$; -0.082, $[-0.090, -0.073]$), and the transfer in-degree again ranks new senders above AWP (0.439, $[0.426, 0.454]$, against 0.398, $[0.386, 0.412]$; -0.041, $[-0.047, -0.034]$). With activity restarts EndorseRank still ranks new approval pairs below the degree (-0.046, $[-0.055, -0.038]$), as it did in W0. On the liquidation-free close rate of the W1 traders EndorseRank reaches 0.297 ($[0.174, 0.418]$).

4.7 The coupled walk C-PR (RQ4, RQ5)

C-PR is the method of this thesis: one random walk over the US-dollar allowance layer and the US-dollar transfer layer, which at each node mixes the two transition rows with weight λ (Section 3.6.3). The analysis plan fixed one decision rule, and its object is C-PR: rule A asks the coupled walk to beat each of its layers walked alone (RQ4). The registration of W1 added rule B, which asks it to improve on its transfer layer on new data (RQ5). Both rules compare C-PR with its own layers, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), because only that contrast changes the coupling and nothing else; C-PR also differs from EndorseRank and AWP in its amount weights and its uniform restarts. Both rules are applied to the revised scores as a re-analysis. This section places C-PR within the window, then reports rule A in W0 and rule B in W1, and ends with its contrasts with every other

Table 4.13: EndorseRank and its variant with activity restarts on the cohorts and labels of the window W1 (spenders $n = 7{,}960$ contracts, traders $n = 70$). Neither these rows nor the contrasts below them are part of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.15, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.10, and those of C-PR in Table 4.18. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 7{,}960$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.381 [0.365, 0.396]	0.269 [0.250, 0.285]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.165 [0.143, 0.186]	0.439 [0.426, 0.454]	0.253 [0.116, 0.363]
EndorseRank	0.299 [0.284, 0.314]	0.259 [0.243, 0.275]	0.297 [0.174, 0.418]
EndorseRank, restarts by approving activity	0.335 [0.319, 0.349]	0.236 [0.219, 0.251]	0.286 [0.160, 0.404]
AWP	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]	0.263 [0.140, 0.379]
C-PR ($\lambda = 0.5$)	0.316 [0.300, 0.330]	0.324 [0.308, 0.337]	0.282 [0.162, 0.397]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.082$ [-0.090, -0.073]		
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = -0.041$ [-0.047, -0.034]		
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = -0.046$ [-0.055, -0.038]		
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.035$ [0.029, 0.042]		
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = -0.023$ [-0.032, -0.013]		

score of the study, which decide nothing.

4.7.1 The coupled walk within the window

Two hybrids were defined in Section 3.6.3. C-PR mixes the US-dollar allowance and transfer layers in each node's transition row with weight λ, and its ablation S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.14 gives their paired contrasts in this window: first the eight of Section 3.10 against the two layers walked alone, then six against EndorseRank and AWP, two of them computed afterwards from the saved scores. RQ4 and RQ5 are decided out of window. Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

C-PR lies between its two layers on every family and stays nearer to its allowance layer. At the primary setting it reaches 0.449 ($[0.437, 0.463]$) on the allowance family, 0.112 below C-PR ($\lambda = 1$) ($[-0.124, -0.100]$), and 0.400 ($[0.385, 0.416]$) on the transfer family, 0.300 below C-PR ($\lambda = 0$) ($[-0.318, -0.281]$). As λ rises from 0.25 to 0.75 it moves toward the allowance layer on all three

Table 4.14: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 6{,}483$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.3. The secondary part of rule A, fixed in the analysis plan and applied here to the revised scores, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the Sybil-stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples: C-PR against EndorseRank on the transfer family and against AWP on the allowance family, the family that each single-layer walk follows least.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.400	0.700	-0.300	[-0.318, -0.281]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.449	0.561	-0.112	[-0.124, -0.100]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.142	0.402	-0.260	[-0.278, -0.241]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.142	0.028	+0.115	[0.104, 0.123]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.671	0.700	-0.029	[-0.036, -0.023]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.289	0.561	-0.272	[-0.290, -0.253]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.398	0.402	-0.004	[-0.008, 0.000]	0.037
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.398	0.028	+0.371	[0.351, 0.390]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.400	0.736	-0.336	[-0.354, -0.316]	0.000
C-PR allowance minus EndorseRank allowance	0.449	0.560	-0.111	[-0.122, -0.099]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.142	0.419	-0.277	[-0.294, -0.260]	0.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.142	0.026	+0.116	[0.105, 0.125]	1.000
C-PR transfer minus EndorseRank transfer[a]	0.400	0.196	+0.204	[0.191, 0.214]	1.000
C-PR allowance minus AWP allowance[a]	0.449	0.226	+0.223	[0.206, 0.240]	1.000

families. AWP ranks the transfer family better than C-PR (-0.336, $[-0.354, -0.316]$), and EndorseRank the allowance family (-0.111, $[-0.122, -0.099]$). On the family that each single-layer walk follows least, C-PR leads that walk: it ranks the transfer family 0.204 above EndorseRank ($[0.191, 0.214]$) and the allowance family 0.223 above AWP ($[0.206, 0.240]$). A node with out-edges in only one layer follows that layer whatever λ is (Equation (3.9)); λ acts only at the nodes with out-edges in both layers. Almost every node of the allowance layer is an owner with out-edges, so the allowance rows enter the walk wherever an owner also sent a transfer that the extraction kept. On the Sybil-stability family C-PR exceeds its allowance layer ($+0.115$, $[0.104, 0.123]$) and lies below its transfer layer (-0.260, $[-0.278, -0.241]$).

On every family, S-PR falls below the layer that leads that family or is level with it. Restarting the transfer walk from the allowance walk drops the transfer coefficient to 0.671 ($[0.661, 0.681]$; $\Delta\tau = -0.029$, $[-0.036, -0.023]$, against C-PR ($\lambda = 0$)), raises the allowance coefficient only as far as 0.289 ($[0.272, 0.308]$; $\Delta\tau = -0.272$, $[-0.290, -0.253]$, against C-PR ($\lambda = 1$)), and leaves Sybil-adjusted stability level with the transfer layer (-0.004, $[-0.008, 0.000]$). A restart vector changes where the walk lands after each teleport. It carries part of the allowance signal into the transfer walk, but less than walking the allowance edges does. This is the same-window counterpart of the finding that Section 4.7.2 reaches out of window.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on Sybil-adjusted stability. At $\lambda = 0.5$, C-PR fails all three parts, as in the registered analysis (Appendix 8.6). Its transfer coefficient (0.400) lies below the interval of C-PR ($\lambda = 0$) ($[0.691, 0.709]$), its allowance coefficient (0.449) lies below that of C-PR ($\lambda = 1$) ($[0.550, 0.572]$), and on Sybil-adjusted stability it exceeds only the allowance layer. The paired contrasts of Table 4.14, reported alongside the rule as a stricter check, point the same way. In the same-window tables the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family. The holdout takes up the remaining question of whether the mixture has any predictive content that the single layers lack.

4.7.2 Rule A in the window W0

The coupled walk fails rule A out of window, although in an exploratory comparison it improves on its transfer layer walked alone. That rule, applied here to the revised scores as a re-analysis, asked C-PR to do no worse than C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders in W0 (first block of Table 4.10). At $\lambda = 0.5$, C-PR lies above the allowance walk on new approval pairs ($\Delta\tau = +0.031$, interval $[0.021, 0.042]$), so the first condition holds, and below the transfer walk on new transfer senders ($\Delta\tau = -0.030$, interval $[-0.040, -0.020]$), so the second fails, and with it the primary part, as in the registered analysis (Table 8.12). Against the transfer walk alone, a comparison the rule did not include, C-PR does better. On new approval pairs C-PR reaches 0.329 ($[0.312, 0.343]$) against 0.178 ($[0.160, 0.197]$), a paired difference of $+0.151$ (interval $[0.130, 0.170]$, on the same resamples). The transfer walk, like AWP, scores zero the spenders without a transfer edge, and C-PR was not contrasted with it on the other spenders, so part of this gain may come from ordering those spenders. This contrast was computed after the labels were known; in the registered analysis the same contrast became the hypothesis F1 of the registration of W1 (Section 4.7.3). Across the three values of λ, C-PR is flat on new approval pairs (0.329, $[0.312, 0.343]$, at $\lambda = 0.25$; 0.327, $[0.311, 0.342]$, at $\lambda = 0.75$), and its point estimate on new transfer senders falls as λ rises, from 0.345 ($[0.330, 0.357]$) to 0.327 ($[0.313, 0.339]$) (no paired contrast was computed, and λ was not tuned).

The seeded ablation shows where the allowance signal has to sit. S-PR ranks the flow label slightly above the transfer walk ($\Delta\tau = +0.011$, interval $[0.004, 0.018]$) and falls below the allowance walk on the endorsement label ($\Delta\tau = -0.108$, $[-0.128, -0.088]$), ranking new approval pairs only a little above the transfer walk (0.190, $[0.171, 0.209]$, against 0.178). When transfer edges decide where the walk goes, a restart distribution derived from endorsements changes little, although, like the transfer walk, S-PR scores zero the spenders without a transfer edge, so part of this gap may come from coverage. On the spender holdout, the allowance signal survives when allowance edges carry the walk, fully in the allowance walk and mixed in C-PR, and most of it is lost when they

only decide where the walk restarts.

4.7.3 Rule B in the registered window W1

The registration of Section 3.7.2 was committed to the public repository as commit `0d3a057` at 16:49:32 UTC on 9 October 2026, after the W0 results of the registered analysis were known and before the study extracted the July–September logs. It fixed rule B with the margin $\delta = 0.01$, the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040), rounded up. The extraction manifest records no use of the testing override (Appendix 8.1). The July–September logs held 6,805 GMX V2 closes by contract accounts, all of which decoded. The revised scores were evaluated afterwards, with 400 paired resamples and seed 42, and rule B and R1–R3 are applied to them as a re-analysis; the registered analysis reached the same verdicts except on R1 (Appendix 8.6).

The spender cohort is that of Section 4.6.2. Seventy GMX V2 contract traders in the freeze-date graph closed at least three positions in the label window; 9 of them had at least one liquidation, and 1 had every close liquidated. Table 4.15 gives every score of rule B on its three labels, and Table 4.16 the contrasts with their verdicts.

The rule is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.316$ ($[0.300, 0.330]$) against 0.146 ($[0.126, 0.168]$) for its transfer layer walked alone, a difference of $+0.170$ (interval $[0.149, 0.189]$), so the W0 gain on authorization recurs. The in-approve degree at the freeze ranks this label at 0.381 ($[0.365, 0.396]$), above C-PR's point estimate, as it did in W0 (0.392 against 0.329), and C-PR lies below it in both windows (Section 4.7.4); F1 shows a gain over the transfer walk, part of which may come from the spenders that walk scores zero, not a gain over the approval count. F2 fails. On new transfer senders C-PR reaches 0.324 ($[0.308, 0.337]$) against 0.348 ($[0.335, 0.363]$); the difference is -0.025 ($[-0.038, -0.015]$), and the whole interval lies below the margin of -0.01. F3 fails narrowly. On the liquidation-free close rate of the traders C-PR reaches 0.282 ($[0.162, 0.397]$) and the transfer walk 0.282 ($[0.162, 0.394]$), and the difference is $+0.000$ ($[-0.010, 0.010]$), whose lower end lies just below -0.01. The rule needed all three

Table 4.15: Window W1: Kendall τ between scores frozen at 30 June 2026 and the three labels of rule B, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 7{,}960$ contract spenders holding a positive latest allowance at the freeze; 1,495 of them gained a new approval pair and 1,371 a new sender. The traders are the $n = 70$ GMX V2 contract accounts in the freeze-date graph with at least three closes in the label window; 9 of them had at least one liquidation and 1 had only liquidations. C-PR ($\lambda = 0$) is the comparator of rule B and AWP that of its first sensitivity analysis (Table 4.16); EndorseRank is in Table 4.13. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 7{,}960$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.381 [0.365, 0.396]	0.269 [0.250, 0.285]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.165 [0.143, 0.186]	0.439 [0.426, 0.454]	0.253 [0.116, 0.363]
AWP	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]	0.263 [0.140, 0.379]
C-PR ($\lambda = 1$)	0.295 [0.280, 0.310]	0.257 [0.241, 0.274]	0.302 [0.177, 0.425]
C-PR ($\lambda = 0$)	0.146 [0.126, 0.168]	0.348 [0.335, 0.363]	0.282 [0.162, 0.394]
C-PR ($\lambda = 0.25$)	0.315 [0.299, 0.328]	0.332 [0.317, 0.346]	0.282 [0.162, 0.397]
C-PR ($\lambda = 0.5$)	0.316 [0.300, 0.330]	0.324 [0.308, 0.337]	0.282 [0.162, 0.397]
C-PR ($\lambda = 0.75$)	0.316 [0.298, 0.330]	0.315 [0.299, 0.329]	0.284 [0.160, 0.398]
S-PR	0.162 [0.140, 0.184]	0.367 [0.354, 0.381]	0.300 [0.175, 0.411]

hypotheses, so, as the registration said for this case, the thesis makes no claim that C-PR improves on a walk over the transfers alone.

The registration warned that F3 could pass with both coefficients near zero. Neither is near zero here: every score of Table 4.15 ranks this label with an interval above zero, from 0.253 ($[0.116, 0.363]$) for the transfer in-degree at the freeze to 0.394 ($[0.261, 0.524]$) for the in-approve degree, the highest point estimate; of the walks only EndorseRank and C-PR were contrasted with it (Sections 4.7.4 and 4.8).

Of the results reported beside the decision, F4 fails: on the liquidation-free close rate C-PR is not better than the transfer walk. F5 fails, since C-PR and the allowance walk cannot be told apart on that label (-0.019, $[-0.123, 0.089]$). F6, the primary part of rule A, fails again: C-PR lies above the allowance walk on new approval pairs ($+0.021$, $[0.009, 0.032]$), which meets the no-worse condition, but below the transfer walk on new senders. As in W0, C-PR's coefficients are flat across λ on new approval pairs and fall on new senders as λ rises. S-PR ranks new senders

Table 4.16: Window W1: the paired contrasts F1–F6 of rule B with their verdicts, and the two sensitivity analyses (rule B and its sensitivity analyses were registered for the raw-unit analysis; here they are applied to the revised scores). $\Delta\tau = \tau(a) - \tau(b)$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. F1, F2, F6a and F6b are on the spenders ($n = 7{,}960$ contracts), F3–F5 on the traders ($n = 70$), as in Table 4.15. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	$\tau(a)$	$\tau(b)$	$\Delta\tau$ [95% CI]	Verdict
Rule B: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.316	0.146	+0.170 [0.149, 0.189]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.324	0.348	-0.025 [-0.038, -0.015]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.282	0.282	+0.000 [-0.010, 0.010]	fails
Decision: F1, F2 and F3 must all hold							not met
Reported, not part of the decision							
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.282	0.282	+0.000 [-0.010, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.282	0.302	-0.019 [-0.123, 0.089]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.316	0.295	+0.021 [0.009, 0.032]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.324	0.348	-0.025 [-0.038, -0.015]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.316	0.157	+0.159 [0.139, 0.178]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.324	0.398	-0.075 [-0.085, -0.065]	fails
F3	C-PR vs. AWP	liquidation-free close rate	non-inferiority, -0.01	0.282	0.263	+0.019 [-0.026, 0.064]	fails
F4	C-PR vs. AWP	liquidation-free close rate	superiority	0.282	0.263	+0.019 [-0.026, 0.064]	fails
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.316	0.148	+0.168 [0.147, 0.187]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.324	0.349	-0.025 [-0.039, -0.016]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.282	0.282	+0.000 [-0.010, 0.010]	fails
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.282	0.282	+0.000 [-0.010, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.282	0.322	-0.040 [-0.116, 0.035]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.316	0.295	+0.021 [0.009, 0.032]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.324	0.349	-0.025 [-0.039, -0.016]	fails

at 0.367 ($[0.354, 0.381]$), above the transfer walk's point estimate, and new approval pairs at 0.162 ($[0.140, 0.184]$), below the allowance walk (0.295, $[0.280, 0.310]$); no paired contrast of S-PR was registered for W1.

Neither registered sensitivity analysis changes the verdict (Table 4.16). With AWP as the comparator, F1 still holds ($+0.159$, $[0.139, 0.178]$), a lead that comes largely, but not only, from the spenders AWP scores zero (Section 4.7.4), and F2 fails clearly: AWP ranks new senders at 0.398 ($[0.386, 0.412]$), and C-PR's difference to it is -0.075 ($[-0.085, -0.065]$), an interval that lies entirely below the margin. F3 and F4 fail ($+0.019$, $[-0.026, 0.064]$). On 70 traders this interval is 0.090 wide, so F3 could have held only with a lead of about 0.035; its failure leaves a loss of 0.026 possible and does not show that C-PR ranks the label worse. With the contracts outside a graph kept as isolated nodes, every verdict stays as it was. The point estimates change by at most 0.002, except on F5: the rule lifts the traders outside the allowance graph from zero to the score of a node without in-edges, the allowance walk's coefficient on the liquidation-free close rate rises from 0.302 to 0.322, and C-PR's difference to it widens to -0.040 ($[-0.116, 0.035]$).

Table 4.17 gives the W1 trader labels that do not enter the decision, with the 2,000 resamples of Section 4.8. On the two count labels the walks that use transfer edges again lead, while the transfer in-degree ranks both with intervals that just contain zero (0.156, $[-0.012, 0.306]$; 0.142, $[-0.018, 0.278]$), and the intervals of C-PR at $\lambda = 0.5$ and 0.75 on profitable closes, and those of the three mixtures and S-PR on realized gain, contain zero. AWP has the highest point estimates on profitable closes (0.264, $[0.103, 0.408]$) and realized gain (0.280, $[0.111, 0.435]$); its lead over C-PR ($\lambda = 0$) is resolved on neither (C-PR ($\lambda = 0$) minus AWP -0.042, $[-0.087, 0.005]$, and -0.052, $[-0.108, 0.002]$). The allowance-side scores stay near zero on both. On the no-liquidation flag every score is positive, the in-approve degree highest (0.407, $[0.268, 0.541]$). On both shares the intervals of the in-approve degree, C-PR ($\lambda = 1$), EndorseRank and its variant with activity restarts lie below zero, for the in-approve degree at -0.294 ($[-0.468, -0.106]$) on the non-loss share, while AWP and C-PR ($\lambda = 0$) are positive with intervals that contain zero.

Table 4.17: Window W1, traders: Kendall τ between scores frozen at 30 June 2026 and the other GMX V2 labels counted from 1 July 2026 to 30 September 2026, on the $n = 70$ contract accounts with at least three closes in that window. No liquidation is 1 when none of the account's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter rule B. The intervals use the resamples of Table 4.19. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Score at 30 June 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.407 [0.268, 0.541]	0.042 [-0.158, 0.243]	0.033 [-0.164, 0.230]	-0.221 [-0.406, -0.034]	-0.294 [-0.468, -0.106]
Transfer in-degree at the freeze (baseline)	0.266 [0.125, 0.395]	0.156 [-0.012, 0.306]	0.142 [-0.018, 0.278]	-0.059 [-0.225, 0.099]	-0.071 [-0.225, 0.082]
EndorseRank	0.303 [0.172, 0.422]	0.063 [-0.124, 0.253]	-0.011 [-0.211, 0.187]	-0.184 [-0.341, -0.014]	-0.265 [-0.411, -0.107]
EndorseRank, restarts by approving activity	0.292 [0.155, 0.415]	0.049 [-0.129, 0.237]	-0.019 [-0.195, 0.166]	-0.200 [-0.345, -0.037]	-0.275 [-0.418, -0.123]
AWP	0.265 [0.139, 0.383]	0.264 [0.103, 0.408]	0.280 [0.111, 0.435]	0.062 [-0.101, 0.213]	0.049 [-0.090, 0.185]
C-PR ($\lambda = 1$)	0.308 [0.175, 0.428]	0.009 [-0.167, 0.192]	-0.066 [-0.249, 0.120]	-0.208 [-0.370, -0.030]	-0.287 [-0.443, -0.121]
C-PR ($\lambda = 0$)	0.286 [0.166, 0.400]	0.222 [0.072, 0.366]	0.228 [0.063, 0.388]	0.058 [-0.108, 0.210]	0.029 [-0.117, 0.170]
C-PR ($\lambda = 0.25$)	0.284 [0.163, 0.399]	0.166 [0.011, 0.309]	0.168 [-0.002, 0.331]	0.008 [-0.157, 0.162]	-0.023 [-0.171, 0.117]
C-PR ($\lambda = 0.5$)	0.283 [0.162, 0.398]	0.129 [-0.018, 0.271]	0.131 [-0.041, 0.297]	-0.025 [-0.185, 0.129]	-0.062 [-0.210, 0.080]
C-PR ($\lambda = 0.75$)	0.284 [0.163, 0.398]	0.103 [-0.049, 0.254]	0.099 [-0.076, 0.267]	-0.051 [-0.212, 0.107]	-0.102 [-0.253, 0.047]
S-PR	0.304 [0.177, 0.418]	0.177 [0.033, 0.310]	0.146 [-0.031, 0.321]	0.009 [-0.143, 0.153]	-0.027 [-0.169, 0.111]

4.7.4 C-PR beside every other score

Rules A and B leave open how C-PR stands against the other scores of the study. Table 4.18 sets it against each of them and against the two raw degrees, on the spender labels of both windows and on the liquidation-free close rate of the W1 traders, with the resamples of the rules. Only the cells marked in the table belong to a rule; every other contrast was computed after the labels were known and decides nothing.

Table 4.18: Paired contrasts of C-PR ($\lambda = 0.5$) with every other score of this study and with the two raw degrees, on the spender labels of W0 ($n = 7{,}914$ contracts) and W1 ($n = 7{,}960$) and on the liquidation-free close rate of the W1 traders ($n = 70$). Each cell is $\Delta\tau = \tau_{\text{C-PR}} - \tau_b$ over its 95% interval; within a cohort every cell uses the same resamples. Cells marked [A] are the contrasts of rule A, fixed in the analysis plan; [B] marks those named in the registration of W1 (rule B with F5 and F6a) and [S] those of its first sensitivity analysis, with AWP as the comparator. These rules were fixed for the raw-unit analysis and are applied here to the revised scores. Every other cell was computed after the labels were known and belongs to no rule. The lower block repeats the contrasts with AWP and the in-approve degree on the spenders that had received a transfer from another address before the freeze, the spenders AWP can order; that restriction was also chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	W0 spenders		W1 spenders		W1 traders
C-PR minus	New approval pairs	New transfer senders	New approval pairs	New transfer senders	Liquidation-free close rate
C-PR ($\lambda = 1$)	+0.031[A] [0.021, 0.042]	+0.069 [0.057, 0.080]	+0.021[B] [0.009, 0.032]	+0.066 [0.054, 0.078]	-0.019[B] [-0.123, 0.089]
C-PR ($\lambda = 0$)	+0.151 [0.130, 0.170]	-0.030[A] [-0.040, -0.020]	+0.170[B] [0.149, 0.189]	-0.025[B] [-0.038, -0.015]	+0.000[B] [-0.010, 0.010]
EndorseRank	+0.026 [0.016, 0.037]	+0.064 [0.052, 0.075]	+0.017 [0.006, 0.029]	+0.064 [0.051, 0.076]	-0.015 [-0.116, 0.094]
EndorseRank (activity restarts)	-0.003 [-0.014, 0.008]	+0.066 [0.055, 0.078]	-0.019 [-0.029, -0.008]	+0.087 [0.075, 0.102]	-0.004 [-0.102, 0.105]
AWP	+0.133 [0.114, 0.153]	-0.073 [-0.083, -0.063]	+0.159[S] [0.139, 0.178]	-0.075[S] [-0.085, -0.065]	+0.019[S] [-0.026, 0.064]
S-PR	+0.139 [0.121, 0.156]	-0.041 [-0.049, -0.032]	+0.154 [0.134, 0.174]	-0.044 [-0.052, -0.037]	-0.017 [-0.094, 0.048]
In-approve degree at the freeze	-0.063 [-0.076, -0.052]	+0.052 [0.038, 0.065]	-0.065 [-0.077, -0.054]	+0.055 [0.040, 0.071]	-0.112 [-0.194, -0.026]
Transfer in-degree at the freeze	+0.126 [0.104, 0.145]	-0.111 [-0.121, -0.100]	+0.151 [0.130, 0.173]	-0.115 [-0.128, -0.105]	+0.030 [-0.021, 0.099]
Spenders that had received a transfer before the freeze (W0 $n = 6{,}489$, W1 $n = 6{,}567$)					
AWP	+0.008 [-0.002, 0.020]	-0.036 [-0.044, -0.026]	+0.022 [0.011, 0.033]	-0.037 [-0.046, -0.028]	—
In-approve degree at the freeze	-0.051 [-0.064, -0.037]	—	-0.053 [-0.066, -0.041]	—	—

Against EndorseRank, the walk over allowances alone, C-PR ranks both spender labels higher in

both windows: new approval pairs by 0.026 ($[0.016, 0.037]$) in W0 and 0.017 ($[0.006, 0.029]$) in W1, and new senders by 0.064 ($[0.052, 0.075]$) and 0.064 ($[0.051, 0.076]$). On the W1 traders it cannot be told apart from EndorseRank (-0.015, $[-0.116, 0.094]$). The variant of EndorseRank with activity restarts is level with C-PR on new approval pairs in W0 (-0.003, $[-0.014, 0.008]$) and ranks them higher in W1 (-0.019, $[-0.029, -0.008]$); on new senders C-PR leads it in both windows ($+0.066$, $[0.055, 0.078]$, and $+0.087$, $[0.075, 0.102]$). Within the window C-PR leads the variant on the transfer and Sybil-stability families (0.400 and 0.142 against 0.251 and 0.056) and trails it on the allowance family (0.449 against 0.601; Tables 4.3 and 4.8).

Against AWP, the walk over transfers alone, C-PR ranks new approval pairs higher ($+0.133$, $[0.114, 0.153]$, in W0 and $+0.159$, $[0.139, 0.178]$, in W1) and new senders lower (-0.073, $[-0.083, -0.063]$, and -0.075, $[-0.085, -0.065]$). Its lead on approvals comes largely from the spenders that AWP cannot order. AWP scores zero every spender without a transfer edge up to the freeze, 1,342 of them in W0, and those spenders gain new approval pairs more often than the rest (Section 4.6). Among the spenders that had received a transfer from another address before the freeze, C-PR and AWP are level on new approval pairs in W0 ($+0.008$, $[-0.002, 0.020]$) and C-PR leads in W1 ($+0.022$, $[0.011, 0.033]$), while AWP ranks new senders higher in both windows (-0.036, $[-0.044, -0.026]$, and -0.037, $[-0.046, -0.028]$) (lower block of Table 4.18; this restriction, too, was chosen after the results were known). Out of window, what C-PR adds to a walk over transfers is mainly an order among the contracts that this walk leaves tied, at the price of a worse order of later inflow among those it can rank.

Against the raw degrees, C-PR does not rank either label above the count of the same edge type. On new approval pairs it lies below the in-approve degree in both windows (-0.063, $[-0.076, -0.052]$, and -0.065, $[-0.077, -0.054]$); on new senders it is below the transfer in-degree in both windows (-0.111, $[-0.121, -0.100]$, and -0.115, $[-0.128, -0.105]$). Against its ablation S-PR it ranks new approval pairs higher ($+0.139$, $[0.121, 0.156]$, and $+0.154$, $[0.134, 0.174]$) and new senders lower (-0.041, $[-0.049, -0.032]$, and -0.044, $[-0.052, -0.037]$). On the liquidation-free close rate of the 70 traders C-PR ranks below the in-approve degree (-0.112, $[-0.194, -0.026]$), and no other

contrast of C-PR excludes zero.

Taken together, C-PR ranks every spender label above EndorseRank in both windows, and it ranks later approvals above AWP, largely through the contracts that AWP leaves tied. It stays below the raw count of each edge type on that edge type's label in both windows, below AWP on later senders, below EndorseRank with activity restarts on later approvals in W1, and below the in-approve degree on the traders' liquidation-free close rate. None of these contrasts was fixed in advance.

4.8 Outcomes built from neither edge type

The GMX trader labels are the only outcomes in these data built from neither edge type. The comparison of Section 3.7.3 was registered with rule B, before the study extracted the July–September logs and before it scored the liquidation labels of W0. Here its contrasts, rules and wording are applied to the revised scores as a re-analysis, with W1 as the primary window and W0 as the secondary one. Intervals come from 2,000 paired resamples and are 95% unless marked 97.5%, the Bonferroni interval of the decision contrasts P and Q. Table 4.19 gives the contrasts on the liquidation-free close rate, Table 4.20 the traders that receive approvals, and Table 8.11 (Appendix 8.5.4) all six trader labels.

R1 holds. In W1 the in-approve degree at the freeze reaches $\tau = 0.394$ ($[0.265, 0.518]$) on the liquidation-free close rate and the transfer in-degree 0.253 ($[0.117, 0.379]$), so P, their difference, is $+0.141$ (97.5% interval $[0.026, 0.256]$), an interval above zero. Stratified by the number of closes, P is $+0.145$ ($[0.040, 0.328]$). In W0, P is $+0.095$ (97.5% interval $[-0.041, 0.263]$), positive but not resolved; R1 asked for both W1 intervals above zero, so, in the registered wording, there is a count-level lead of allowance over transfer on later liquidation avoidance. R2 does not hold. Q, EndorseRank with AWP's activity restarts minus AWP (the like-for-like PageRank pair), is $+0.023$ in W1 and -0.012 in W0, and both 97.5% intervals include zero ($[-0.071, 0.127]$ and $[-0.177, 0.168]$), so the PageRank versions are not resolved. Neither are C-PR's layers ($+0.019$, $[-0.093, 0.128]$, in W1), EndorseRank against AWP ($+0.034$, $[-0.067, 0.134]$) or C-PR ($\lambda = 0$)

Table 4.19: Allowance-side against transfer-side scores on the liquidation-free close rate of the GMX V2 contract traders with at least three closes in the label window, a label built from neither edge type ($n = 70$ traders in W1, $n = 87$ in W0). The pairs, the decision contrasts P and Q with their Bonferroni 97.5% intervals and the stratified rows were registered before the study extracted the July–September logs, for the raw-unit analysis, and are applied here to the revised scores; W1 is the primary window and W0 the secondary one. Paired contract bootstrap with 2,000 resamples (seed 42). The stratified rows combine Kendall's τ_b within strata of the number of label-window closes (3–4, 5–9, 10–24, 25 or more), weighted by the number of contract pairs. Registered rules applied to the revised scores: R1 (count-level lead) met; R2 (PageRank-level lead) not met; R3 (specificity) applies.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI	97.5% CI
Window W1, July–September 2026 ($n = 70$ traders)					
In-approve degree minus transfer in-degree	0.394	0.253	0.141	[0.037, 0.240]	[0.026, 0.256]
EndorseRank (activity restarts) minus AWP	0.286	0.263	0.023	[-0.062, 0.113]	[-0.071, 0.127]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.302	0.282	0.019	[-0.093, 0.128]	–
EndorseRank minus AWP	0.297	0.263	0.034	[-0.067, 0.134]	–
C-PR ($\lambda = 0$) minus AWP	0.282	0.263	0.019	[-0.031, 0.066]	–
EndorseRank minus in-approve degree	0.297	0.394	-0.097	[-0.174, -0.009]	–
In-approve degree minus transfer in-degree, stratified by closes	0.381	0.236	0.145	[0.040, 0.328]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.273	0.238	0.034	[-0.096, 0.171]	–
Window W0, April–June 2026 ($n = 87$ traders)					
In-approve degree minus transfer in-degree	0.345	0.249	0.095	[-0.027, 0.234]	[-0.041, 0.263]
EndorseRank (activity restarts) minus AWP	0.242	0.254	-0.012	[-0.153, 0.149]	[-0.177, 0.168]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.237	0.294	-0.057	[-0.209, 0.092]	–
EndorseRank minus AWP	0.236	0.254	-0.019	[-0.173, 0.144]	–
C-PR ($\lambda = 0$) minus AWP	0.294	0.254	0.040	[-0.023, 0.098]	–
EndorseRank minus in-approve degree	0.236	0.345	-0.109	[-0.196, -0.018]	–
In-approve degree minus transfer in-degree, stratified by closes	0.342	0.253	0.089	[-0.025, 0.252]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.229	0.265	-0.036	[-0.186, 0.145]	–

against AWP ($+0.019$, $[-0.031, 0.066]$), in either window. In a contrast the registration did not name, EndorseRank falls below the approval count, by 0.097 ($[-0.174, -0.009]$) in W1 and 0.109 ($[-0.196, -0.018]$) in W0.

The other labels do not enter the decision (Table 8.11). On the no-liquidation flag, P is $+0.141$ ($[0.044, 0.232]$) in W1 and $+0.109$ ($[-0.022, 0.257]$) in W0. On profitable closes and realized gain, which grow with activity, the PageRank pairs favor the transfer side in W1, with intervals below zero for Q (-0.215, $[-0.399, -0.042]$, and -0.299, $[-0.457, -0.146]$) and for EndorseRank against AWP (-0.200, $[-0.390, -0.023]$, and -0.292, $[-0.460, -0.134]$); P points the same way without being resolved. R3 applies. In W1 the in-approve degree ranks below the transfer in-degree on the profitable share (-0.162, $[-0.295, -0.025]$) and the non-loss share (-0.222, $[-0.328, -0.106]$), so, in the registered wording, the signal concerns forced-liquidation avoidance, not trading success. Part of the non-loss gap is built in: the pipeline counts a liquidated close as a close that was not a loss, so a trader with liquidations gains non-loss share, and R3 therefore rests mainly on the profitable share, which counts a liquidated close as not profitable.

The count-level comparison rests on few traders. In W1 the highest in-approve degree among the traders is 1, so P in effect compares the 37 recipients with the 33 other traders; in W0 the maximum is 3, and 39 of the 87 traders are recipients. Every recipient avoided liquidation in both windows (37 of 37 and 39 of 39), against 72.7% of the other traders in W1 and 75.0% in W0. Recipients also closed more positions (median 24 against 11 in W1, 30 against 10 in W0) and had lower profitable and non-loss shares (mean non-loss share 0.433 against 0.644 in W1), the pattern behind R3. Without the 10 recipients of highest in-approve degree, P is $+0.155$ ($[0.032, 0.276]$) in W1 and $+0.088$ ($[-0.048, 0.240]$) in W0.

4.9 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. C1–C5 are the primary claims; C2 and C3 rest on contrasts com-

Table 4.20: GMX V2 contract traders with a positive in-approve degree at the freeze (recipients) against all other labelled traders, in both windows ($n = 70$ traders in W1, $n = 87$ in W0). The comparison of recipients with the other traders and the influence check were registered with the contrasts of Table 4.19 and are applied here to the revised scores. The highest in-approve degree among the traders is 1 in W1 and 3 in W0, so the ten recipients left out by the influence check are taken from tied values in node order. The influence check is a paired contract bootstrap on the remaining traders (2,000 resamples, seed 42).

	Recipients	Other traders
Window W1, July–September 2026 ($n = 70$ traders)		
Traders	37	33
Share with no liquidation	1.000	0.727
Mean liquidation-free close rate	1.000	0.878
Mean profitable share	0.325	0.479
Mean non-loss share	0.433	0.644
Median closes in the label window	24.0	11.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.155$ [0.032, 0.276]		
Window W0, April–June 2026 ($n = 87$ traders)		
Traders	39	48
Share with no liquidation	1.000	0.750
Mean liquidation-free close rate	1.000	0.919
Mean profitable share	0.359	0.474
Mean non-loss share	0.401	0.607
Median closes in the label window	30.0	10.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.088$ [-0.048, 0.240]		

puted after the labels were known, and C6 on the comparison on outcomes built from neither edge type. The verdicts of rules A and B in C1 and of R1–R3 in C6 apply the registered rules to the revised scores, as a re-analysis; the registered analysis reached the same verdicts except on R1 (Appendix 8.6).

C1. Under its rules, C-PR, a fixed-λ mixture of two dollar-weighted layers with uniform restarts, does not beat its two layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs ($+0.031$, $[0.021, 0.042]$) but worse than the transfer walk on new senders (-0.030, $[-0.040, -0.020]$), and its same-window part fails as well (Tables 4.10 and 4.14). Rule B is not met (Table 4.16): the W0 gain over the transfer walk on new approval pairs, found after the labels were known, recurs in W1 (F1, $+0.170$, $[0.149, 0.189]$), but C-PR falls below the transfer walk on new senders by more than the margin of 0.01 (F2, -0.025, $[-0.038, -0.015]$), and F3 fails narrowly ($+0.000$, $[-0.010, 0.010]$). Against AWP, in the first sensitivity analysis, F2 fails clearly (-0.075, $[-0.085, -0.065]$). S-PR ranks new senders slightly above the transfer walk and new approval pairs well below the allowance walk, so allowance edges help mainly when they carry the walk; Section 4.7.1 reports where the hybrids stand within the window.

C2. Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows (new approval pairs $+0.026$, $[0.016, 0.037]$, and $+0.017$, $[0.006, 0.029]$; new senders $+0.064$, $[0.052, 0.075]$, and $+0.064$, $[0.051, 0.076]$) and leads it within the window on the transfer and Sybil-stability families. It ranks new approval pairs above AWP ($+0.133$, $[0.114, 0.153]$; $+0.159$, $[0.139, 0.178]$), largely through the spenders that AWP scores zero: among the spenders that had received a transfer, the two are level in W0 and C-PR leads by 0.022 ($[0.011, 0.033]$) in W1. C-PR ranks new senders below AWP and below the transfer in-degree in both windows, new approval pairs below the in-approve degree in both windows (-0.063, $[-0.076, -0.052]$; -0.065, $[-0.077, -0.054]$) and below EndorseRank with activity restarts in W1 (-0.019, $[-0.029, -0.008]$), and the liquidation-free close rate of the W1

traders below the in-approve degree (-0.112, $[-0.194, -0.026]$) (Table 4.18). None of these contrasts was fixed in advance.

C3. Out of window neither single-layer walk beats the raw degree it smooths. AWP ranks later new transfer senders below the transfer in-degree in both windows (W0 -0.038, $[-0.043, -0.032]$; W1 -0.041, $[-0.047, -0.034]$), and EndorseRank ranks later new approval pairs below the in-approve degree in both (W0 -0.090, $[-0.100, -0.079]$; W1 -0.082, $[-0.090, -0.073]$) (Tables 4.10 and 4.13); in W0 neither comparison is a matter of coverage (Table 4.11), and no such restriction of these two contrasts was computed for W1. With activity restarts EndorseRank also stays below its count (Section 4.5.4), so on these labels propagation adds nothing to the count of the same edge type, whatever the restart rule. On the profit labels of the contract traders of W0 the paired contrasts separate the two sides only on realized gain, where the transfer layer walked alone ranks above the allowance layer (C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$) -0.257, $[-0.447, -0.071]$), not on profitable closes (EndorseRank minus AWP -0.046, $[-0.205, 0.113]$), and no score ranks the share of closes that were profitable (Tables 4.12 and 8.11). These comparisons were computed after the labels were known.

C4. In the same window each single-layer walk agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.560, $[0.548, 0.571]$) and AWP with the transfer family (0.736, $[0.728, 0.744]$), and the mixtures of C-PR lie between their two layers (Table 4.3). These tests measure internal consistency, and part of the agreement is built in; external validity is outside their scope. The two walks order the cohort differently: their agreement, $\tau_b = 0.199$ ($[0.181, 0.221]$), has an interval that excludes zero and one; 96.4% of the contract pairs are ordered by both scores, and 39.8% of those are discordant. Neither result comes from ties or from the rule for contracts outside a graph (Table 4.4, Figure 4.3).

C5. On identical contract sets, the cost of each score's solver call grows with the size of its graph, since each iteration costs time in $|E|$ and $|V|$, and with the iterations its walk needs. The allowance graph has 45.2% of the transfer graph's edges, and AWP takes 3.21 times as long

as EndorseRank at an edge ratio of 2.21 (Table 4.1). On the subsamples the time ratio follows the edge ratio at no stage, and the stages do not test edge counts beyond those of the cohort graph (Table 4.2). The low cost is no advantage of the algorithm: C-PR and S-PR walk the transfer layer and take longer than AWP, C-PR 1.36 times as long.

C6. On later liquidation avoidance of the GMX V2 contract traders, the one outcome built from neither edge type that bears on counterparty risk, the approval count at the freeze leads the transfer count in the primary window W1 (P $= +0.141$, 97.5% interval $[0.026, 0.256]$), while in W0 P is positive but not resolved ($+0.095$, 97.5% interval $[-0.041, 0.263]$). The PageRank versions are not resolved (Q $= +0.023$, 97.5% interval $[-0.071, 0.127]$). In W1 the approval count ranks traders lower than the transfer count on the shares of closes that were profitable or not losing, so, in the wording of R3, the signal concerns forced-liquidation avoidance, not trading success; in W0 neither difference is resolved (Table 8.11). The samples are small, 70 and 87 contract traders, and every interval is wide (Section 4.8).

The same-window tables and the rank divergence answer RQ1, the benchmark tables RQ2, the holdouts of the single-layer walks with Table 4.18 RQ3, and Section 4.7 RQ4 and RQ5; C6 adds no research question. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and Table 4.19, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the research questions and objectives of Chapter 1.

5 Discussion

Chapter 4 reported the measurements taken on Arbitrum One. Here each research question and each objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. Unless noted otherwise, same-window statements concern the matched cohort of contracts and holdout statements the spender cohorts.

5.1 Findings by research question

5.1.1 RQ1: The two single-layer walks

RQ1 asks whether EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, order the same contracts in different ways, and how closely each score agrees with contemporaneous allowance and transfer checks. They order the contracts differently: the interval of their agreement excludes both zero and one (Table 4.4). The divergence does not come from ties or from the rule for missing contracts: EndorseRank leaves few contract pairs tied, and keeping the contracts missing from a graph as isolated nodes leaves the agreement unchanged (Section 4.4). The divergence follows from how the two graphs are built. The endorsement graph records the latest approvals and has fewer than half of the edges of the transfer graph, which records every transfer (T. Do et al., 2023), and the two methods restart differently. PageRank places stationary mass according to the edges it is given (Langville & Meyer, 2006; Page et al., 1999), so a contract that receives many transfers from active senders but few approvals ranks high under AWP and modestly under EndorseRank, and the reverse holds for a contract that many owners approve without being a transfer hub. H1 is supported, largely by construction, since two walks over different graphs with different restarts were not expected to agree. In practice one score cannot stand in for the other: an operator who adopts EndorseRank for its speed gets a different ranking and not a faster

approximation of AWP. The divergence is also the premise of C-PR: if the two walks ordered the contracts alike, one walk over both could add little to either.

In the same window each score follows the edges it walks on: the walks over transfers agree most with the transfer family and much less with the allowance family, the walks over allowances agree most with the allowance family, and every interval on a score's own family excludes zero (Table 4.3).

Circularity bounds H2. Every proxy in a family is computed from the same events as the matching graph, so the agreement is an intended-construct check in the sense of Cronbach and Meehl (1955) and not an external test. Part of it is built in: AWP, whose edges weigh each transfer by its dollar value, follows the summed value of inbound transfers almost as closely as their number (Table 8.7). Ties do not bound H2 either: the isolated-node rule changes no coefficient (Table 4.4). In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. None of these comparisons was fixed in advance, and the paired differences between EndorseRank and AWP, which carry no claim, are in Appendix 8.5.

5.1.2 RQ2: Computational cost

RQ2 asks what each score costs to compute on the same sample, and how much walking both edge sets adds. EndorseRank is the cheapest score in time, memory and iterations, and the two hybrids, which walk the transfer layer, take longer than AWP (Table 4.1), so walking both edge sets costs more than walking either. Each iteration costs time in $|E|$ and in $|V|$ (Langville & Meyer, 2006), and AWP needs more iterations than EndorseRank; its time ratio lies well above the edge ratio at $d = 0.85$ and moves with d (Section 4.2), so the cost grows with the size of each graph and with the iterations its walk needs, which the benchmark does not separate.

The scaling tables qualify this result more than they extend it (Table 4.2). Each stage is a subsample of the cohort that keeps the edges incident to its contracts, so the stages never test a graph larger than the cohort graph (Section 4.2.1). The damping and value-slope checks leave the alignment pattern as it was (Tables 4.5 and 4.6). H3 is supported on the matched cohort, again largely by

construction, since both scores run the same solver; on the subsamples EndorseRank's lead in speed holds at every stage, but the time ratio follows the edge ratio at none, because the iterations each walk needs differ from stage to stage. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ3–RQ5 and Section 5.1.6.

5.1.3 RQ3: Temporal holdout and the raw degrees

RQ3 has two clauses. The first asks whether scores frozen at the end of March 2026 predict new owner–spender approvals and new transfer senders on the spender cohort over the following three months. Every score predicts both labels, and the edge type decides how well (Table 4.9). On new approval pairs the in-approve degree leads, followed by the interior mixtures, EndorseRank and the allowance layer, while the scores built on transfers alone stay lower, partly because they score zero the one spender in six without a transfer edge, and these gained new approval pairs more often. On the other spenders AWP ranks this label above EndorseRank and below the in-approve degree in W0, without a paired contrast, level with C-PR in W0 and below it in W1 (Tables 4.11 and 4.18; restrictions chosen after the labels were known). On new senders the transfer in-degree leads, ahead of AWP, and the scores built on allowances alone rank lowest. W1 repeats this pattern among the same scores (Tables 4.15 and 4.13). The evidence concerns spenders; Section 5.1.6 turns to the traders.

The second clause asks whether any PageRank beats the raw degree it smooths, and none does (Tables 4.10 and 4.13). AWP ranks later new senders below the transfer in-degree in both windows, with intervals below zero, and in W0 also on the spenders that had received a transfer before the freeze, so its zeros do not explain the gap (Table 4.11). EndorseRank, with the uniform restarts of the analysis plan, ranks new approval pairs below the in-approve degree in both windows, and so does its variant with AWP's activity restarts, by less (Section 4.5.4); the transfer layer of C-PR, walked with uniform restarts, stays further below the transfer in-degree, without a paired contrast (Section 4.6, Table 4.15). On these labels the restart rule changes how far a walk falls short of its count, not whether it does. Known results lead one to expect the count to be hard to beat: in grow-

ing networks new links attach preferentially to nodes that already have many (Barabási & Albert, 1999; Newman, 2001), which makes degree a standard link-prediction baseline (Liben-Nowell & Kleinberg, 2007), also on Ethereum transaction networks (D. Lin et al., 2022), and where degree correlations are weak, as on the Web, PageRank is on average proportional to in-degree (Fortunato et al., 2008). For most spenders EndorseRank is a reweighted approval count (Section 5.3.2). This does not undo the first clause, since the counts are statistics of the same edges. The coupled walk answers the second clause no better: C-PR ranks neither label above the count of the same edge type, lying below the in-approve degree on new approval pairs and below the transfer in-degree on new senders in both windows (Table 4.18). All of these contrasts were computed after the labels were known.

The labels lie outside the window but belong to the same constructs as the scores, and neither contains any element of profit or repayment. RQ3 therefore shows the temporal predictive validity of each construct, with the raw degree as the baseline that a PageRank must beat, and leaves credit risk aside; the liquidation label of W1 (Section 3.7.2), on which Section 5.1.6 compares the edge types, comes closest to it.

5.1.4 RQ4: The coupled walk against its layers

RQ4 and RQ5 concern the method of the thesis, and for C-PR, the coupling the plan fixed, both get a negative answer, each under its own rule, applied here, like R1–R3 in Section 5.1.6, to the revised scores as a re-analysis; the registered analysis reached the same verdicts on both rules (Appendix 8.6). RQ4 asks whether one walk over both layers ranks contracts better than either layer walked alone. Under rule A, fixed in the analysis plan, it does not. On W0 C-PR is no worse than the allowance layer on new approval pairs but falls behind the transfer layer on new senders (Table 4.10), and in the same window the secondary rule fails on all three families (Section 4.7.1). The rule was fixed before any coupled score existed, so λ could not have been tuned to it, although the revised C-PR's dollar weights, allowance cap and score windows were set after the registered results were known. It limits an obvious next step: the plan's convex mixture of the two edge

types at the transition level gained on new approval pairs over the allowance layer and lost on new senders against the transfer layer.

The ablation and the layer weights explain this answer. S-PR keeps the transfer walk but restarts it from the allowance walk; it ranks the flow label as well as the transfer walk or slightly better and loses most of the endorsement signal, since a teleportation vector derived from endorsements changes little when transfer edges decide where the walk goes, and since S-PR scores only the transfer-layer nodes, part of that loss is coverage. In C-PR the allowance edges carry part of the walk, so the allowance signal survives on the spender holdout: every interior mixture is above the allowance layer on new approval pairs and below the transfer layer on new senders (Tables 4.9 and 4.10). Its layers carry dollar amounts without a bound, apart from the cap on allowances, and restart uniformly, whereas AWP bounds the weight of each transfer and restarts at active senders; AWP ranks new senders better than C-PR in both windows, and walked alone C-PR's transfer layer already ranks them below AWP (Tables 4.9 and 4.15), so part of that gap comes from the weights and restarts, not from the coupling. Other couplings are not evaluated in this thesis (Section 5.9).

5.1.5 RQ5: Allowance edges added to a transfer walk

RQ5 asks whether the walk improves on a walk over the transfers alone. An exploratory comparison on W0 said that it did on new approval pairs, but it was made after the labels were known. Rule B, registered before the study extracted the July–September logs, decides the question, and its answer is no: C-PR again beat the transfer layer on new approval pairs, but non-inferiority was shown neither on new senders nor on the traders' liquidation-free close rate, and the rule needed all three conditions (Section 4.7.3). In the first sensitivity analysis, with AWP as the comparator, F1 holds and F2 and F3 fail: C-PR ranks new approval pairs higher and new senders lower, and the close rate does not separate the two (Table 4.16). The rule was registered before the study extracted its labels and fails on two of its three conditions. W1 establishes something narrower: adding allowance edges to C-PR's transfer walk ranks new approval pairs better on new data, partly through the spenders the walk scores zero, at a loss on new senders larger than the margin of the registration;

whether C-PR beats its transfer layer on the other spenders was not computed.

Beside the rules, the contrasts of Table 4.18 show where that information comes from and what it costs. C-PR ranks both spender labels above EndorseRank in both windows, as it does above its own allowance layer, so walking the transfers as well improves on a walk over allowances alone on every spender label, on later inflow most. Against AWP, the gain on later approvals comes largely from the spenders that the transfer walk scores zero because they have no transfer edge; among the spenders it can rank, C-PR and AWP are level on later approvals in W0 and C-PR leads in W1, while AWP orders later senders better in both windows. What the allowance layer adds to a transfer walk out of window is therefore mainly coverage, an order among the contracts that the transfer walk leaves tied, with a small gain on later approvals among the others in W1. EndorseRank with activity restarts, an allowance walk that restarts at approving owners, is level with C-PR on later approvals in W0 and ranks them better in W1; C-PR leads it on later senders in both windows and within the window on the transfer and Sybil-stability families (Section 4.7.4). No PageRank of the study, the coupled one included, ranks later arrivals above the raw count of the same edge type. These contrasts were computed after the labels were known and decide nothing.

5.1.6 Allowance against transfer on outcomes built from neither edge type

Unlike the spender labels of RQ3–RQ5, which favor the scores built on their own edge type, the GMX trader outcomes are built from neither edge type and were measured after the scores were frozen. The comparison of the two edge types on them (Section 3.7.3) adds no research question and is reported as Claim C6 (Section 4.8).

This comparison bears on counterparty risk, the use that motivated the thesis (Chapter 1). For a protocol that extends leverage or credit to a contract account, a counterparty that is often force-liquidated is a riskier one, and forced liquidation is the nearest outcome to default in these data. On later liquidation avoidance in W1, the primary window, the difference P between the approval count and the transfer count at the freeze has a Bonferroni 97.5% interval above zero (Table 4.19), as does the interval stratified by the number of closes, so R1 is met: in the registered wording,

there is a count-level lead of allowance over transfer on later liquidation avoidance. In W0, the secondary window, P was positive but not resolved. The transfer count also ranks this outcome (Table 4.15), and the transfer scores rank as well the outcomes that grow with activity, the number of profitable closes and the realized gain (Table 4.17): they record how much a contract trades.

Four limits bound this reading. First, R2 is not met: the like-for-like PageRank pair, EndorseRank with AWP's restarts against AWP, is not resolved in either window, and EndorseRank again falls below the approval count it smooths (Table 4.19). Second, the comparison rests on a few dozen approval-receiving traders per window, all of whom avoided liquidation, against about three in four of the others (Table 4.20), and every interval is wide. Third, every trader is a contract, so account type cannot explain the pattern, but the role of a contract may: nearly all recipients were approved by one contract (Section 3.7.3), so they form one cluster, and they close more positions, which fits a family of managed or pooled trading contracts. Fourth, R3 applies in W1: the same signal is lower on the share of closes that were profitable, so the construct it ranks is forced-liquidation avoidance, not trading success; its lower non-loss share is partly built in, because the pipeline counts a liquidated close as not a loss.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence.

Eight findings outside the objectives also shape the conclusions. Five limit the claims: the value slopes chosen in the revision were checked only in the same window (Section 5.8.1); the subsample stages never ran on a graph larger than the cohort graph (Section 5.1.2); on the trader cohorts no score ranks the share of profitable closes with an interval above zero (Section 4.6.1); EndorseRank's point estimates on both spender labels lie slightly above those of the allowance layer of C-PR, without a paired contrast, so how approvals are weighted may matter (Section 5.8.1); and C-PR's lead over AWP on new approval pairs comes largely from the spenders that AWP scores zero

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of the τ between EndorseRank and AWP contains neither 0 nor 1; a score agrees with a construct when its interval on that family excludes 0, and agreement with the construct of a score's own edges is an intended-construct check	Table 4.4, last row: the interval $[0.181, 0.221]$ lies strictly inside $(0, 1)$. Table 4.3: every walk over transfers alone on the transfer family (0.700 $[0.691, 0.709]$ to 0.736 $[0.728, 0.744]$) and both walks over allowances on the allowance family (0.560 $[0.548, 0.571]$ and 0.561 $[0.550, 0.572]$); every interval excludes 0; the proxies come from the same events as the graphs	Met (largely by construction)
O2	Runtime ratio given alongside the edge ratio, under a single fixed protocol	Table 4.1: mean, SD, memory, iterations, $\|V\|$ and $\|E\|$ on the full graphs; Table 4.2: mean time, $\|V\|$ and $\|E\|$ per stage; time ratio 3.21 beside an edge ratio of 2.21, and 2.56 to 3.21 across damping values (Table 4.5); C-PR and S-PR slower than AWP; one solver for all scores	Met (largely by construction)
O3	A score predicts a label when its interval excludes 0; a PageRank improves on a degree only when the interval of the paired difference lies above 0	Tables 4.9, 4.15 and 4.13: every score predicts both labels in W0 and W1. Tables 4.10 and 4.13: AWP minus transfer in-degree -0.038 $[-0.043, -0.032]$ (W0) and -0.041 $[-0.047, -0.034]$ (W1); EndorseRank minus in-approve degree -0.090 $[-0.100, -0.079]$ (W0) and -0.082 $[-0.090, -0.073]$ (W1); EndorseRank with activity restarts minus in-approve degree -0.060 $[-0.070, -0.052]$ (W0) and -0.046 $[-0.055, -0.038]$ (W1). Table 4.18: C-PR minus in-approve degree on new approval pairs -0.063 $[-0.076, -0.052]$ (W0) and -0.065 $[-0.077, -0.054]$ (W1), minus transfer in-degree on new senders -0.111 $[-0.121, -0.100]$ and -0.115 $[-0.128, -0.105]$; all computed after the labels were known	First clause met; second met by no PageRank
O4	Rule A of 9 October 2026: on W0, C-PR does no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, its point estimate lies within the leading layer's interval on the transfer and allowance families, and it exceeds both layers on Sybil stability	Revised scores, re-analysis. Table 4.10: no worse holds ($+0.031$ $[0.021, 0.042]$), better fails (-0.030 $[-0.040, -0.020]$). Same window: transfer (0.400) lies outside the interval of C-PR ($\lambda = 0$) and allowance (0.449) outside that of C-PR ($\lambda = 1$) (Table 4.3); Sybil stability lies between the layers (Table 4.14). All three parts fail (Section 4.7.1)	Not met, as in the registered analysis (Table 8.12)
O5	Rule B, registered on 9 October 2026: on W1, C-PR beats C-PR ($\lambda = 0$) on new approval pairs and is not worse by more than 0.01 (lower 95% bound above -0.01) on new senders and on the traders' liquidation-free close rate	Revised scores, re-analysis. Table 4.16: new approval pairs $+0.170$ $[0.149, 0.189]$ holds; new senders -0.025 $[-0.038, -0.015]$, wholly below -0.01, fails; liquidation-free close rate $+0.000$ $[-0.010, 0.010]$, lower bound below -0.01, fails. Sensitivity 1, AWP as comparator: $+0.159$ $[0.139, 0.178]$ holds, -0.075 $[-0.085, -0.065]$ and $+0.019$ $[-0.026, 0.064]$ fail; post hoc comparisons in Table 4.18	Not met, as in the registered analysis (Table 8.12)

(Section 4.7.4). Three add to the claims: C-PR ranks both spender labels above EndorseRank in both windows (Section 4.7.4); every trader contract that had received an approval, nearly all from one contract, avoided liquidation in both windows, in small samples (Section 5.1.6); and EndorseRank matches a closed form for every depth-one spender (Section 5.3.2).

5.3 Theoretical implications

The results bear on three areas of theory: propagation semantics on financial graphs, the construct validity of on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network (Page et al., 1999). H.-D. Do and Do (2023) carried this reading over to Ethereum transactions, and AWP (T. Do et al., 2023) refined it with value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner who lets a spender act for them issues a directed endorsement of spending authority (Vogelsteller & Buterin, 2015). The RQ1 divergence shows that changing what the edges mean, and where the walk restarts, changes the ranking even on the same nodes, so graph-based reputation is a family of methods, each valid relative to the relationships its edges encode. Peer-to-peer and web trust systems reached the same conclusion from the other side: in EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) the seed set and the edge definitions largely decide the outcome. EndorseRank's edge is native to a token standard, where theirs belonged to a message-passing overlay or a hyperlink graph. C-PR takes the next step and lets one walk follow both relations of the same addresses, so that a contract's score can draw on the payments it receives and the spending rights it is granted at once.

The holdouts add a qualification that the propagation metaphor hides. Changing the edge type changed what the score could anticipate: on new approval pairs the scores built on allowance edges led, partly through the spenders that walks over transfers score zero, and on new senders the

walks over transfers led, in both windows. No walk added to the count of its own edges out of window: AWP, EndorseRank with either restart rule and the layers of C-PR all ranked their labels below the raw degree (Section 5.1.3). On these data the choice of edge decides which spenders a score can order, and the count of that edge orders them best. When the two layers share one walk, the allowance signal survives where allowance edges carry the walk and is mostly lost where they only seed its restarts (Section 5.1.4), and the coupled walk ranks every spender label above the walk over allowances alone while staying below the walk over transfers on later senders (Section 5.1.5).

5.3.2 Depth-one spenders: what a walk adds to a count

The allowance graphs keep only the approvals of the five tokens that a cohort contract grants or receives (Section 3.6), so most nodes are pure owners, with out-edges only, or pure spenders, with in-edges only. A cohort contract can be both, so the graph also has paths of more than one step. For a spender whose owners receive no approvals, a walk with uniform restarts has a closed form.

Proposition. Let PageRank with transition matrix P be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c, and every node v whose in-neighbors all lack in-edges has

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u P_{uv}, \tag{5.1}$$

the sum of the transition probabilities into v. In EndorseRank $P_{uv} = w(u,v)/o(u)$, so $\delta(v)$ is the owner-normalized in-strength of v: each owner's endorsement is divided over all the spenders it approves. In C-PR P_{uv} is the coupled transition of Equation (3.9), and an in-neighbor without in-edges is one that has none in either layer.

Derivation. At the fixed point, $r_v = (1 - d + d\,r_D)/|V| + d\sum_u r_u\,P_{uv}$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1 - d + d\,r_D)/|V|$, the constant c of Section 5.6.3. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On EndorseRank's graph at T_{obs}, 5,232 of the 6,224 cohort spenders have depth one, and EndorseRank as computed in this thesis lies within 1% of Equation (5.1) for every one of them; of the deeper spenders, which have an owner that itself receives approvals, 21.9% do. Over all spenders the share is 87.5%, and over the W0 spenders at t_1 it is 70.8%. Over the cohort spenders EndorseRank's Kendall τ with δ is 0.908, a point estimate (scripts `run_same_window.py` and `run_holdout.py`). These shares were not computed for the union graph of C-PR, where an owner that also received a transfer has in-edges and its spenders are therefore not of depth one.

Two consequences follow. First, for a depth-one spender EndorseRank only reweights the approval count, counting an owner that approves many spenders for less at each; besides the time and value weights, that normalization is all that separates it from the count, which ranks new approval pairs better (Section 5.1.3). For these spenders the negative answer to the second clause of RQ3 for EndorseRank thus has an analytic basis: there was no propagated information for the walk to add. With activity restarts, dangling mass follows the restarts, so for a depth-one spender the variant is a one-step sum over its owners weighted by their restart mass: it differs from the count only in how owners are weighted, not through longer paths, and it too ranks new approval pairs below the count. Second, the graph does have deeper paths, and for most deeper spenders the score departs from the closed form, so multi-hop propagation is present in this graph, but over all spenders EndorseRank did not beat the raw count out of window, and the deeper spenders were not tested apart. The ego network cuts longer paths (Section 5.8.3), so testing propagation further would need a deeper sample, such as the two-hop neighborhood of the cohort or a chain-wide allowance graph, or restarts seeded at vetted contracts.

5.3.3 Multi-construct domain alignment

The checks were designed to separate transfer centrality from allowance delegation, and each score aligns with the construct of the edges it walks on, the mixtures falling between their layers. Classical construct-validity theory would predict this pattern for instruments that measure different constructs (Cronbach & Meehl, 1955), and the integrated view of Messick (1995) goes further, counting the

consequences of using a score as part of its validity. The same theory says what such a pattern cannot settle: when two instruments measure different constructs, the one that comes out ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. Outcomes built from neither edge type are the exception, since neither instrument was built for them, which is why the thesis compares the edge types there (Section 5.1.6); even there, which edge type leads depends on the outcome.

5.3.4 Reputation under cheap pseudonyms

Identities are cheap to create on public ledgers (Nakamoto, 2008), and under Ethereum's account model a new address needs no registration step at all (Wood, 2025). Any reputation system that rewards inbound transfers or allowances is therefore exposed to Sybil strategies, in which an attacker creates edges among addresses it controls (Douceur, 2002). Section 5.6 formalizes the cost of attacking an allowance graph, where a closed farm of addresses that approve one another needs, unlike transfer wash trading, no token balance and pays little beyond gas.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers have to choose between transfer-hub detection, authorization-aware reputation and a walk that combines the two. The measured results support choosing the score by scenario over recommending a single one.

5.4.1 The coupled walk and combined use

Whether to combine the two signals is a question about deployment and about method, and the evidence answers the two differently. Folding the two edge types into one walk did not pay against the published transfer walk on these data: C-PR costs more than AWP, more than the EndorseRank and AWP solves together (Table 4.1), trails it on new senders in both windows and, among the contracts AWP can rank, gains on new approval pairs only in W1 and only slightly (Sections 5.1.5 and 4.7.4), so an operator free to choose should keep the two scores apart, running EndorseRank

or the approval count for authorization-aware monitoring and AWP for transfer-hub audits, at the cost of both solves. For an operator tied to a single transfer walk weighted by dollar amounts, adding allowance edges bought authorization information that held on new data, chiefly by ranking contracts that the transfer walk leaves at zero, at a loss on inflow larger than 0.01, so rule B is not met.

5.4.2 Transfer-hub and flow monitoring

A protocol that describes current flow (transfer-hub search by market makers and liquidity providers, routing analysis, or transfer-based compliance heuristics) can read it from in-degree and in-value directly. AWP (T. Do et al., 2023) agrees with present inflow (Table 4.3) and out of window ranked later new senders above every other PageRank but below the raw transfer in-degree in both windows (Tables 4.9 and 4.13), so on these data a protocol that wants to anticipate flow has no reason to use it over the count.

5.4.3 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most (token vaults, allowance-gated integrations, or risk engines that treat approvals as acts of trust), protocols have reason to use allowance edges. The cheapest form is the spender's in-approve degree, which needs no solver and predicted new approvals better than EndorseRank, and than every other PageRank of the study, in both windows (Tables 4.10 and 4.13). EndorseRank, its propagated version, agrees most with the allowance checks (Table 4.3) and solves faster than AWP (Table 4.1), but for most spenders it reflects the endorsing owners only through how many spenders each approves (Section 5.3.2), a property not shown to improve any label, and under uniform teleportation it is open to cheap inflation by closed farms (Section 5.6). The variant with activity restarts came closer to the count after the labels were known but stayed below it in both windows, so on these data no propagated allowance score improves on the count.

5.4.4 Industry adoption scenarios

The scenarios below say for whom each score is useful and for whom it is not.

- DEX operators. A protocol that sets fee tiers or monitoring priority by user needs a score for its users, and its users are mostly externally owned accounts, which this thesis does not score (Section 3.2). The scores can instead rank the routers, vaults and pools that the DEX integrates with.
- Lending and margin protocols. Lending markets want to put more of their capital to use without cutting collateral (Bartoletti et al., 2021), and the stress of 2020 showed the cost of updating risk parameters late (Gudgeon, Perez, et al., 2020). When the counterparty is a contract, their screening question is whether it will be force-liquidated. On that outcome a lead of the approval count over the transfer count was resolved in the primary window but not in W0, the PageRank pair is unresolved, and the signal, lower on trading success, rests on a few dozen contract traders per window (Section 5.1.6). A screen built on it would need a registered test on a larger sample. The scores are not tools for predicting default.
- Wallet and security tooling. Users and integrators want to know whether a spender contract is widely authorized. A badge that shows how many owners have approved a spender, or its EndorseRank, makes that delegation explicit (Lundfall, 2020; Vogelsteller & Buterin, 2015); it conveys the position of a graph node and is neither a security audit nor a guarantee against malicious contracts.
- Liquidity providers and market makers. To control inventory and spreads, these participants need to find transfer hubs and see how flow is structured. Transfer counts and AWP are the suitable tools for that goal (T. Do et al., 2023), and their benefit is lower screening cost, not higher profit and loss.
- Airdrop and governance programs. Sybil clusters routinely abuse token distributions and governance eligibility (Douceur, 2002; Luo et al., 2025). EndorseRank as defined here offers little protection below the top of its ranking: a closed farm collects about three times its

population share and can concentrate it on one target, so about 25 addresses lift a contract into the cohort's top tenth, although reaching the top would take about 47,000 (Section 5.6); the bound behind these figures holds for C-PR as well. Programs can require diverse incoming endorsements from owners with independent histories, discount mass that circulates inside closed clusters, or send teleportation and dangling mass only to a vetted seed set; none of these stops an attacker who buys endorsements from honest owners.

- Analytics and compliance providers. Exchanges, funds and compliance teams need counterparty rankings that outsiders can recompute. C-PR, EndorseRank and AWP can be published for defined windows and parameters and audited, unlike proprietary machine learning models (Packin & Lev-Aretz, 2024), provided that their Sybil exposure is disclosed and they are not presented as profit predictions.

In none of these scenarios does reputation replace collateral.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties is useful only if each of them can recompute it and see how it might be manipulated. C-PR, EndorseRank and AWP can be recomputed from public logs and a public daily price series with the open pipeline (Appendix 8.1), and Sections 5.6 and 5.7 document how they can be manipulated.

5.5 Societal and practical impact

In the DeFi market, vetting a counterparty is expensive, and the capacity to do it is unevenly spread. Wealthy actors may be able to pay for proprietary risk tools, while retail participants work from anecdotal evidence or skip vetting altogether. A score built from the permissions users have already granted, and cheap to recompute (Table 4.1), lowers the barrier to vetting the contracts that ask for those permissions. Such a score shows how many owners would be exposed if a contract misused its allowances, which transfer records do not show; an approval granted earlier is one route by

which tokens are drained (Section 2.17). It is not shown to anticipate drains: the drain heuristic rose with every score, the transfer-based ones included (Section 4.6).

The results support fitting a score to each decision over recommending a single one, which also lowers the risk of using a flow score to decide authorization. The degree contrasts cited below were computed after the labels were known.

Where the decentralized credit scoring systems discussed by Packin and Lev-Aretz (2024) are opaque, a score built from public events and public prices with versioned parameters supplies the explanatory information that regulators suggest should be required when DeFi protocols are analyzed (Zetzsche et al., 2020). It also brings a duty, described in Section 5.7, to ensure that an ordinal, graph-based statistic is not mistaken for an identity, a credit history, or a security audit.

Any reputation metric that affects how value is allocated is open to exploitation; the next section models what it costs to manipulate a walk over allowance edges, EndorseRank and C-PR among them.

5.6 Sybil attack cost model on allowance graphs

A Sybil attack exploits cheap pseudonyms to create a false impression of trustworthiness (Douceur, 2002). On a transfer graph, a malicious actor can inflate in-degree and in-flow artificially by moving assets among accounts it controls, paying for this in transaction fees and possible inventory loss (Cong et al., 2023). Allowance graphs expose a different attack surface, since each edge records an authorization, not an actual transfer. In this section we formulate a simple cost model for inflating a walk over allowance edges with fake allowances, computed on EndorseRank's graph; its bound holds for C-PR as well (Section 5.6.3). It builds on the optimal link farms of Gyöngyi and Garcia-Molina (2005) and on the result of Cheng and Friedman (2005) that symmetric functions such as PageRank are not Sybil-proof (Section 2.6), and adds the cost of an allowance edge and the size of the gain on this study's graph.

5.6.1 Attack setup

Let the attacker control a set of addresses $\mathcal{S}$, and write $\mathcal{H} = V \setminus \mathcal{S}$ for the honest addresses in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) of weight $w_{uv} > 0$ records that u has authorized v to spend its tokens. The weight is that of Equation (3.7). With damping parameter $d \in (0, 1)$, EndorseRank finds the PageRank vector $\mathbf{r}$ on G_{approve} as the solution of (Page et al., 1999):

$$\mathbf{r} = d\, P^{\top}\mathbf{r} + \frac{1-d}{|V|}\mathbf{1}, \tag{5.2}$$

where P is the transition matrix of Equation (2.1). A node with out-edges splits its row in proportion to its outgoing allowance weights. A dangling node, one without out-edges, has every entry of its row equal to $1/|V|$, so its mass is spread uniformly over all nodes (Langville & Meyer, 2006). The fixed point is the same as that of the iteration in Equation (3.1), which the implementation runs and which feeds dangling mass back through the uniform teleportation vector. The attacker's objective is to maximize the mean EndorseRank of a target set $\mathcal{G} \subseteq \mathcal{S}$,

$$J(\mathcal{G}) = \frac{1}{|\mathcal{G}|}\sum_{i \in \mathcal{G}} r_i, \tag{5.3}$$

subject to a budget constraint on the cost of the attack, which is defined next.

5.6.2 Cost components

The attacker creates a set of edges $E_{\mathcal{S}}$, all with sources in $\mathcal{S}$, and may also buy a set $E_{\mathcal{H}\to\mathcal{S}}$ of edges from honest owners into $\mathcal{S}$. Costs are counted over q score windows, the number of score refreshes for which the targets are meant to hold their rank. Gas is the only cost that every edge incurs.

1. Gas cost. Each `approve` or `permit` call consumes gas; write c_{gas} for the expected fee of one allowance update. EndorseRank reads the `Approval` events of the twelve months before each anchor, so an approval enters only the graphs whose window holds it; inside a closed farm its time weight does not matter, since row normalization removes it (item 2). A closed

farm therefore has to emit its approvals again at least once every twelve months, and over score refreshes that span y such periods $C_{\text{gas}} = y\,|E_{\mathcal{S}}|\,c_{\text{gas}}$.

2. Exposure cost. An approval locks no tokens, and the owner need not hold any balance of the token when it approves. The amount costs nothing either: an unlimited allowance takes the full value weight V without any balance behind it, and after row normalization a single out-edge gives $P_{uv} = 1$ whatever its weight. The list of five tokens does not raise this cost, since approving a listed token needs no balance either. An edge inside $\mathcal{S}$ therefore exposes nothing and costs nothing beyond gas. Exposure arises only when an attacker address that holds tokens approves a spender $v \notin \mathcal{S}$, for instance to look like an ordinary user of known contracts, since v can then move those tokens. Such an edge also lets score mass leave $\mathcal{S}$. With ℓ_{uv} the expected cost to the attacker of one such exposure over one window, $C_{\text{exp}} = q \sum_{(u,v)\in E_{\mathcal{S}},\, v\notin\mathcal{S}} \ell_{uv}$, which is zero for a closed farm.
3. Revocation and detection cost. Legitimate users revoke old allowances from time to time, and a farm of fresh addresses that approve only one another might be identifiable. Let c_{rev} be the expected cost per window of detection and of the revocations the attacker issues to avoid it; over q windows, $C_{\text{rev}} = q\,c_{\text{rev}}$.
4. Bought honest edges. The attacker can also pay honest owners to approve its addresses, for instance through an app that asks users for permission. A bought edge (h, a), with $h \in \mathcal{H}$ and $a \in \mathcal{S}$, stays in the graph for twelve months after h last approves it, and while it is in force, and it loses share of h's row as it ages, so keeping it effective costs, per window,

$$c_{\text{in}}(h, a) = c_{\text{gas}} + p_h + \rho_h, \tag{5.4}$$

where p_h is the payment to the honest owner and ρ_h is a risk premium the owner asks against the allowance being spent by the attacker.

Over the q windows, the total cost of the attack is

$$C = C_{\text{gas}} + C_{\text{exp}} + C_{\text{rev}} + q \sum_{(h,a)\in E_{\mathcal{H}\to\mathcal{S}}} c_{\text{in}}(h, a). \tag{5.5}$$

For a closed farm, $C_{\text{exp}} = 0$ and $E_{\mathcal{H}\to\mathcal{S}}$ is empty, so $C = y\,|E_{\mathcal{S}}|\,c_{\text{gas}} + q\,c_{\text{rev}}$.

5.6.3 Influence and a cost-effectiveness bound

By the usual sensitivity results for PageRank, an edge moves **r** in proportion to the stationary mass at its source node and to its normalized outgoing weight (Langville & Meyer, 2006). Consider first a closed farm: every node of $\mathcal{S}$ has at least one out-edge, all of these edges end in $\mathcal{S}$, and no honest owner approves an address in $\mathcal{S}$. Let D be the set of dangling nodes, all of which then lie in $\mathcal{H}$, and let $r_D = \sum_{j\in D} r_j$ be their stationary mass. By Equation (5.2), a node $i \in \mathcal{S}$ receives $d\sum_{j\in\mathcal{S}} r_j P_{ji}$ from the farm, $(1-d)/|V|$ from teleportation and $d\,r_D/|V|$ from the dangling rows. The rows of the farm's nodes sum to one inside $\mathcal{S}$, so summing over $\mathcal{S}$ gives $\sum_{i\in\mathcal{S}} r_i = d\sum_{i\in\mathcal{S}} r_i + |\mathcal{S}|\,(1-d+d\,r_D)/|V|$, that is,

$$\sum_{i\in\mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|}\cdot\frac{1-d+d\,r_D}{1-d}. \tag{5.6}$$

The first factor is the farm's population share, and the second says how many times that share the farm collects. The second factor equals one only when $r_D = 0$, that is, on a graph without dangling nodes; otherwise the dangling rows pass part of the honest mass to the farm. Since the dangling nodes lie outside $\mathcal{S}$, $r_D \le 1 - \sum_{i\in\mathcal{S}} r_i$, and substituting this into Equation (5.6) gives a bound that holds on any graph:

$$\sum_{i\in\mathcal{S}} r_i \le \frac{|\mathcal{S}|}{(1-d)\,|V| + d\,|\mathcal{S}|}. \tag{5.7}$$

For a farm that is small against $|V|$, the bound is close to $1/(1-d)$ times the population share. On the EndorseRank graph of the matched cohort at T_{obs}, with $2{,}513{,}982$ nodes, the pipeline's own PageRank code gives $r_D = 0.365$, 0.394 and 0.419 at $d = 0.75$, 0.85 and 0.95, so a closed farm

on this graph collects 2.10, 3.23 and 8.96 times its population share. A direct check with the same code agrees: a ring of 20 addresses, each approving the next, added to the graph collects 3.23 times its population share at $d = 0.85$.

Equation (5.7) limits the farm's total mass, not the score of one target, and the attacker can concentrate that mass. Take a star in which f farm addresses each approve the target t and t approves each of them back. Every farm address a has t as its only out-neighbor, so $P_{at} = 1$, and the row of t sums to one over the farm. Write $c = (1 - d + d\, r_D)/|V|$ for the mass that each node receives from teleportation and the dangling rows. The balance equations $r_a = c + d\, r_t P_{ta}$ and $r_t = c + d \sum_a r_a$ then give

$$r_t = \frac{c\,(1 + df)}{1 - d^2}.$$

At $d = 0.85$, $c = 1.93 \times 10^{-7}$, which is the common score of every address that receives no allowance. With $f = 10$ (eleven addresses and twenty approvals), the target reaches $r_t = 6.60 \times 10^{-6}$ on the graph with the star added, above the median EndorseRank of the cohort contracts but only 2.4×10^{-4} of the highest (0.0278). The score grows linearly in f: each further farm address adds $d\, c/(1 - d^2)$ to r_t and costs two approvals. Solved for f, the formula needs about 25 farm addresses to reach the cohort's 90th percentile, about 2,292 its 99th and about 47,073 its highest score. Looking ordinary therefore takes a few dozen approvals and reaching the top about 94,000, two per farm address, since the highest-ranked cohort contract holds about 3% of the graph's stationary mass. Because the graph keeps only the approvals of the twelve months before each freeze, the farm has to emit these approvals again at least once a year to keep the target's rank (Section 5.6.2). This study did not measure c_{gas}, so these counts are not turned into a price.

Honest in-edges are therefore not needed to reach the upper part of the ranking, but bought ones add to the gain. When the farm has no out-edges to honest addresses, an edge (h, a) bought from an honest owner h adds, to first order,

$$\Delta \sum_{i \in \mathcal{S}} r_i \approx \frac{d\, r_h\, P_{ha}}{1 - d}$$

to the farm's mass, where P_{ha} is the share of h's row that the new edge takes. A fresh large or unlimited approval bought from h, with $V(z)$ close to one, takes about $\sigma(0)/(o(h) + \sigma(0))$ of h's row; for the median owner in the graph that share is 0.683. The edge an attacker would buy is therefore a large or unlimited approval from an owner whose own approvals are old, small or few, preferably one with a high r_h. Such an approval puts the owner's balance of the token at the attacker's disposal, so the risk premium ρ_h in Equation (5.4) grows with what the owner holds and is close to zero only for an owner that holds none of the token. In the capped allowance layer of C-PR such an approval takes a large share of a row only if it is large against the owner's other allowances (Section 3.6.3). The attack's cost-effectiveness is

$$\eta = \frac{\Delta J(\mathcal{G})}{C}, \tag{5.8}$$

with $\Delta J(\mathcal{G})$ the gain in the objective of Equation (5.3) that the attack edges produce, held over the q windows that C in Equation (5.5) pays for.

Equations (5.6)–(5.8) show that EndorseRank, as defined here, is open to cheap inflation by closed farms. Under uniform teleportation and uniform redistribution of dangling mass, a closed cluster is not held to its population share. The coupled walk is open to the same farm. The derivation of Equation (5.7) uses only the uniform teleportation and the uniform spread of dangling mass, which C-PR shares; a farm address whose only out-edges are approvals follows them with weight one whatever λ is (Equation (3.9)), and a single out-edge takes the whole row whatever its amount, so the cap U does not protect C-PR either. The farm factors and star sizes above were computed on the allowance graph only; on the union graph of C-PR the dangling mass, and with it c, differ and were not computed. The control that does bound closed clusters changes the walk itself: if teleportation and dangling redistribution go only to a vetted seed set, as in TrustRank (Gyöngyi et al., 2004), a cluster outside that set gains mass only through edges from honest owners, which the attacker has to buy at the price of Equation (5.4). AWP's restart rule gives the transfer side no protection either: a farm whose addresses send small amounts to one another raises their activeness and so

draws restart mass to itself, and EndorseRank with activity restarts inherits the same exposure. The Sybil-adjusted stability proxies (Section 3.9.3) flag repeated transfers from a few addresses but not a farm of fresh addresses, so they cannot replace an adversarial assessment, which we leave to future work.

5.7 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Earlier legal work has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain (Zetzsche et al., 2020). Packin and Lev-Aretz (2024) add a warning of their own: decentralized credit scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. C-PR, EndorseRank and AWP are all computed deterministically from on-chain events and a public daily price series with open-source algorithms, so anyone can recalculate them using the open-source software (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because contracts with little activity, new ones among them, will score lower whatever their code.

Ethically, these scores should never be used as security ratings, as creditworthiness measures in the traditional sense, or as evidence that the owners who approve a contract are distinct individuals (Douceur, 2002). They are rank-based indicators of a contract's position in the defined graph topology. If a deployment grants financial services on the strength of score thresholds, it should clearly document the graph definitions behind the scores, the timeframe, and the limitations, above all the exposure to Sybil attacks. A regulatory framework centered on interpretability would prefer this kind of ranking to proprietary machine learning models like those Kotb (2023) reviews, as long as providers make no exaggerated claims about how well they can predict loan repayment rates (Packin & Lev-Aretz, 2024).

Privacy on an open blockchain is limited by the ledger's very nature (Nakamoto, 2008), and com-

puting the reputation scores draws on nothing beyond public data. Further attempts to uncover the identity behind each address are discouraged, and the scores should not be treated as permanent, fingerprint-like identifiers. This study scores contract addresses alone and makes no attempt to link any address to a real-world individual (Section 3.16).

5.8 Threats to validity

The findings reported here should be read against the limitations below, which are grouped under construct, internal, external and adversarial validity.

5.8.1 Construct validity

The threats below concern whether the scores and checks measure what they are read as measuring.

- These ratings are not credit ratings: no default labels from uncollateralized lending enter the analysis.
- Same-window checks rest on local allowance and transfer statistics.
- Holdout construct. A future new approval is still an endorsement event and a future new sender a flow event: both lie outside the score window, yet neither is an independent credit or trading construct. Being counts of arrivals, they also favor any score that tracks the current count of counterparties, and in both windows the raw counts rank them best (Section 5.1.3).
- Unit of analysis. The walks over allowances score the receiving side of an approval, and every scored address here is a contract (Section 4.1.1). Claims about the reputation of traders rest on the W0 trader run and the comparison of Section 5.1.6, both on small samples.
- Own-construct agreement is partly mechanical. EndorseRank smooths spender in-approve degree over the whole graph, and AWP does the same for incoming transfers weighted by their value, so each score is expected by construction to agree with its own family. The within-score contrasts of Table 8.6 were computed for this reason; with few ties they compare like with like, but they remain checks of the instruments.

- Value weights. EndorseRank and AWP pass each US-dollar amount through $V(z)$ with a slope set so that the median amount weighs one half (Section 3.5.1), so the weights record the value of the approvals and transfers an edge carries as well as their number and age. The slopes are choices of the revision, checked only in the same window (Table 4.6). The layers of C-PR add dollar amounts without the bound of V, capping each allowance at U (Section 3.6.3). On both spender holdouts EndorseRank ranks both labels slightly better than the allowance layer walked alone (Tables 4.9 and 4.13), without a paired contrast, so how amounts are weighted may matter for this construct, and neither weighting is shown to be the right one. Such weighting choices propagate into **r** and into the rank correlations (Langville & Meyer, 2006).

5.8.2 Internal validity

The threats below concern whether the comparisons are free of choices that could have produced their results.

- Shared score windows. Every method and proxy is computed over the twelve months before its freeze, 1 July 2025 to 30 June 2026 in the same window and W1 and 1 April 2025 to 31 March 2026 in W0. Market regimes peculiar to those months (volatility, incentive programs, protocol changes) may push alignment up or down. An allowance last set before the window began is not an edge even while it is in force, so a contract whose owners approved it long ago and have not approved it again loses those edges. The checks on damping and value slopes measure sensitivity to these parameters, but none of them varies the calendar window.
- Cohort selection. The matched cohort is the 6,483 contracts on Arbitrum One that received non-zero approvals of one of the five tokens from at least three owners in the twelve months before T_{obs}, and results may not carry over to contracts with fewer approvals or with approvals of other tokens only. Nor do the graphs see every cohort contract (Section 4.1.1), and the sample-definition sensitivity shows how strongly the coefficients react when the evaluation population changes (Table 4.7).

- Baseline fidelity. AWP follows the published method in its edge weights and its restarts (Section 3.6.1). The paper gives no parameter values, so k, t_0 and b are this study's choices (see value weights above). The paper scored Ethereum transactions; here AWP scores transfers of five ERC-20 tokens on Arbitrum One, valued in US dollars, on the cohort's ego network, which limits the activeness of a sender outside the cohort to its transfers to cohort contracts.
- Order of analysis. The cohorts, scores, labels, statistics and rule A were fixed in the analysis plan of 9 October 2026, before any log of the study was extracted. Rule B with its margin, the two sensitivity analyses and the comparison on trader outcomes were registered later that day, after the W0 results were known and before the study extracted the July–September logs (Section 3.7.2). Every other comparison, the degree contrasts among them, was computed after the relevant labels were known and is reported with intervals; none of them decides a question.
- Comparators of the rules. Rules A and B compare C-PR with its own layers walked alone, the only contrast that changes the coupling and nothing else. C-PR also differs from EndorseRank and AWP in its amount weights and its uniform restarts, so its contrasts with them (Table 4.18) mix the coupling with those choices; the registered sensitivity analysis with AWP as the comparator is the only rule-bound contrast with a published walk.
- Dependence and leakage. Contracts that share owners or a protocol are resampled as independent, and the post hoc contrasts are not adjusted for multiplicity (Section 3.10). The cohort of each freeze is drawn from the approvals of its own score window, so no label window chooses the contracts around which its data are drawn (Section 3.7); the five tokens, however, were chosen by their rows in the score window of T_{obs}, which holds the label window of W0.
- Post hoc definition of the main analysis. The amounts, the token list and the score windows of the revised analysis were defined after the registered results of all three windows were known (Section 3.5.1). The revised analysis, the main analysis of this thesis, is therefore post hoc, and its verdicts under rules A, B and R1–R3 are a re-analysis, not registered tests. Its slopes

and cap come from quantiles of the W0 score window, which ends at t_1, and the registered analysis reached the same verdicts on rules A and B and on R2 and R3; R1 holds on the revised scores and did not in the registered analysis (Appendix 8.6). Some contrasts differ between the two, among them F3 and the sign of AWP minus the transfer in-degree on new senders in both windows, which lay above zero in the registered analysis (Table 8.12), and the signs of Q in W0 and of S-PR against the transfer walk on new senders in W0.

- Timing environment. All runtimes are wall-clock times from one machine (Appendix 8.1). The standard deviations in Table 4.1 cover only repeated solves of one configuration, and in the damping sweep each graph–damping pair is timed once. Variation between machines is not measured, so the comparison is stated in ratios instead of seconds.

5.8.3 External validity

The threats below concern how far the results carry beyond the data studied.

- Single chain and window. The evidence comes from Arbitrum One alone, from contracts that at least three owners approved on one of five tokens, over the twelve months of scored activity before each freeze and two three-month label windows. Extending the findings to other rollups, Layer-1 chains or lending-centered venues would need cross-protocol validation, since DeFi venues differ in protocol design (Werner et al., 2022) and in market structure (Schär, 2021).
- Sample scaling versus alignment scaling. The subsample stages bear only on runtime on subgraphs of the matched-cohort graph (Section 4.2.1); they do not show that the τ profile holds at other population sizes.
- Ego-network depth. Each graph keeps only the edges incident to the cohort contracts, so the findings about propagation do not carry over to graphs with longer endorsement paths (Section 5.3.2).

5.8.4 Adversarial validity

The threats below concern what an attacker could do to the scores.

- Sybil and wash trading. Rankings may be manipulated through adversarial allowance or transfer patterns (Douceur, 2002); the Sybil-adjusted proxies only partly address this, and the cost model of Section 5.6 is analytical, not an empirical red-team evaluation.
- Strategic approval behavior. Once a score built on allowances is live, rational actors may grant allowances strategically to farm rank. An observational τ estimated before deployment may overstate how well the ranking resists such behavior.

5.9 Limitations

On this sample, the strengths of C-PR are a ranking of both spender labels above EndorseRank in both windows, agreement with both families that lies between its two layers, and an order among the contracts that the transfer walk leaves tied. Its weaknesses are the failure of both rules, a ranking of later senders below AWP and the transfer in-degree, a ranking of later approvals below the in-approve degree in both windows and, in W1, below EndorseRank with activity restarts, a liquidation-free close rate ranked below the in-approve degree, a Sybil-stability coefficient below AWP's and the highest cost of the three walks.

The coupled operator was tried at three fixed values of λ, with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by dollar amounts, with each allowance capped at U. Layers weighted as in EndorseRank and AWP, node-dependent or learned weights, mixtures of the stationary vectors instead of the transition rows, and other Markov chains are not evaluated in this thesis, so the answers to RQ4 and RQ5 apply only to that family of methods on this dataset. Deployment remains bound by the considerations of Section 5.7.

EndorseRank, the allowance reference, agrees with the allowance checks, ranks differently from AWP and costs little because its graph is sparse. It carries little transfer information, ranks later approvals below the raw in-approve degree in both windows (computed after the labels were known)

and, like C-PR, is exposed to closed farms. None of the scores says anything about contracts that few owners approve or about externally owned accounts, which are not scored (Section 3.2).

The data add four limits. The graphs hold five tokens only (Table 8.4), so a contract that owners approve only on other tokens is outside the cohort. Every price comes from one public service, the daily series of DefiLlama (DefiLlama, n.d.), so an error in it on some date would pass into every amount of that day (Appendix 8.4.2). The account type was read once for each address, at blocks of 9 October 2026 after both label windows (Table 8.1), so a contract that destroyed itself before EIP-6780 (Ballet et al., 2023) is left out as an externally owned account (Section 4.1.1). And the trader cohorts are small (Section 4.1.1), so every trader result has wide intervals.

None of the scores draws on off-chain evidence about who controls an address; their only off-chain inputs are the token list and the daily price series. Unlike attestation tooling such as Human Passport (Human Passport, n.d.), they give no guarantee that two highly ranked addresses are controlled by different actors, so they should be read as orderings of an already admitted population and not as a Sybil defense. Counting only the approvals of owners that pass an attestation gate before computing C-PR or EndorseRank is a plausible deployment, but the gate's own weaknesses would carry over to it (Bordeianu & Popescu, 2026), and evaluating the combination needs labeled Sybil data that this study did not collect.

These limitations narrow what can be concluded. For this cohort of contracts and time frame, C-PR is a walk that carries authorization into a transfer-based reputation score at a loss on inflow, and EndorseRank a cheap, allowance-specific score for spender contracts; neither is a credit scoring method, nor a Sybil-resistant identity mechanism, nor a substitute for protocol-level collateralization (Gudgeon, Perez, et al., 2020). The allowance edge they rest on is worth adding to on-chain reputation, and the approval count is the cheapest way to use it. Chapter 6 restates what has been shown, what that knowledge is worth and what it would take to extend these bounds.

6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives.

6.1 Answers to the research questions

The thesis built a coupled PageRank (C-PR), one walk over ERC-20 allowance and transfer edges mixed at each address with a fixed weight λ, and evaluated it beside its single-layer references, EndorseRank (PageRank with the edge weights of Adaptive Weighted PageRank, AWP, on the latest allowances) and AWP itself (T. Do et al., 2023), its ablation S-PR and the raw degrees, on Arbitrum One contracts scored on the twelve months before each freeze, with holdout windows W0 (April to June 2026) and W1 (July to September 2026). The amounts in US dollars on five tokens, the cohorts and the score windows were defined after the registered results were known; the rule verdicts below are a re-analysis that matches the registered ones except on R1 (Appendix 8.6).

RQ1 asked whether EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, rank the contracts of one sample differently, and how closely each score agrees with allowance and transfer statistics in the same window. They rank the contracts differently, and H1 is supported, largely by construction (Section 4.4). Each score follows the edges it walks on (Table 4.3), which supports H2 only narrowly, because the proxies come from the same events as the graphs. The divergence is the premise of the coupled walk.

RQ2 asked what each score costs to compute, and how much walking both edge sets adds. The cost grows with the size of each graph and with the iterations its walk needs, so EndorseRank, on the sparsest graph, is the cheapest score, and the coupled walks cost more than AWP (Table 4.1); H3 is supported, largely by construction, and on subsamples EndorseRank stays the cheaper at every stage, though the time ratio follows the edge ratio at none (Table 4.2).

RQ3 asked whether scores frozen at t_1 predict the new approval pairs and new transfer senders that spender contracts gain afterwards, and whether any PageRank, the coupled one included, predicts them better than the raw degree of the same edge type. The first part holds in both windows, and over all spenders the edge type decides which label a score anticipates (Tables 4.9 and 4.15). The answer to the second is no for every walk, in contrasts computed after the labels were known: AWP ranks later new senders below the transfer in-degree and EndorseRank later approval pairs below the in-approve degree in both windows (Tables 4.10 and 4.13), EndorseRank with AWP's activity restarts falls short of the degree by less, and C-PR ranks neither label above the degree of the same edge type (Table 4.18). The restart rule changes how far a walk falls short of its count, not whether it does.

RQ4 asked whether C-PR, the method of the thesis, ranks contracts better than either of its layers walked alone. For C-PR, the coupling the plan fixed, the answer is no, on the revised scores as in the registered analysis (Table 8.12). Under rule A, C-PR meets the allowance condition of the primary part but fails its transfer condition and the secondary part (Section 4.7.1; Table 4.10). The seeded ablation ranks new senders slightly above the transfer walk and new approval pairs far below the allowance walk: in this form, allowance edges contribute mainly when the walk moves along them, and little as a seed.

RQ5 asked whether adding the allowance layer improves a walk over the transfers alone. The answer is again no. Under rule B, the exploratory gain over the transfer walk on new approval pairs in W0 recurred in W1, possibly in part through the spenders that walk scores zero, but the loss on new senders exceeded the margin, and on the traders' liquidation-free close rate a loss beyond it could not be ruled out; against AWP it ranked new approval pairs higher and new senders lower, with that close rate unresolved (Table 4.16). Beside the rules, in contrasts computed after the labels were known, C-PR ranks both spender labels above EndorseRank in both windows, its lead over AWP on later approvals comes mainly from the contracts that AWP scores zero, with a small lead among the others in W1, and it ranks later approvals below the in-approve degree in both windows and below EndorseRank with activity restarts in W1 (Table 4.18).

In a comparison registered with rule B that adds no research question (Section 5.1.6), the approval count at the freeze led the transfer count on later liquidation avoidance, the outcome built from neither edge type that bears on counterparty risk, in W1, the primary window ($P = +0.141$, 97.5% interval $[0.026, 0.256]$; Section 4.8), and was positive but not resolved in W0. The like-for-like PageRank pair is not resolved, the signal concerns forced-liquidation avoidance, not trading success, and the samples are small.

Claims C1–C6 (Section 4.9) state these findings formally.

The main research question asked what one walk over both edge types adds to ranking these contracts. In contrasts computed after the labels were known, against the walk over allowances alone the coupled walk adds the transfer construct and ranks every later spender label higher, and against the walk over transfers alone it adds delegated authority, which ranks later approval pairs higher, mainly through the contracts that walk cannot rank, at a loss on later inflow. Under the rules fixed in advance it beats neither layer, and in contrasts computed afterwards no walk of the study, the coupled one included, beat the raw degree of the same edge type out of window.

6.2 Original contribution to knowledge

Read against the literature of Chapter 2, the five contributions of Section 1.9 establish the following. Before this work, graph reputation on public ledgers ranked wallets by their transfers (H.-D. Do & Do, 2023; T. Do et al., 2023), by how risk spreads through transaction networks (D. Lin et al., 2025), or by explicit ratings (Almasoud et al., 2020; Bellini et al., 2020), and the reputation tooling deployed in DeFi scored the holder of an address from credentials (Human Passport, n.d.). None of it used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power (Vogelsteller & Buterin, 2015), as the edge of a reputation ranking, coupled it with payments in one walk, or ranked contract accounts by the approvals they receive; approvals had been used to cluster the addresses of one entity (Victor, 2020) and studied as a security risk (Wang et al., 2022). None evaluated on-chain reputation scores on one cohort against the checks of each construct and against

the raw degrees they smooth, with confidence intervals, difference tests and temporal holdouts under decision rules fixed in advance. This thesis supplies both. PageRank and AWP's weighting are taken from the literature (T. Do et al., 2023; Page et al., 1999); the coupled walk changes the transition rule, and EndorseRank, its allowance reference, changes only the edge.

The new knowledge has five parts. First, what an on-chain reputation score of a contract measures depends on the edges it uses, and this can be checked on one cohort with quantified uncertainty. Second, a walk that couples the allowance edge with transfers, a step that the multilayer literature suggested, beats neither of its layers under rules fixed in advance; beside the rules, it ranks every later spender label above the walk over allowances alone and adds to the walk over transfers mainly an order among the contracts that walk leaves tied, at a loss on later inflow. Third, the allowance edge alone yields a score that is cheaper to update, differs from the transfer-based score, ranks later endorsements of spenders in two windows and orders the spender contracts without transfer history, which transfer walks score zero; its post hoc lead over AWP on later endorsements rests on them (Table 4.11). On few traders, the comparison registered with rule B found in the re-analysis, though not in the registered analysis, a count-level lead of allowance over transfer on later avoidance of forced liquidation in the primary window. Fourth, in contrasts computed after the labels were known, no PageRank beat the raw degree it smooths out of window: on the allowance side neither with uniform nor with activity restarts, and on the transfer side AWP fell below the count, with the transfer walk under uniform restarts lower still by point estimate; the restart rule changed only how far a walk fell short of the count of its edges. Fifth, at depth one a walk with uniform restarts is, up to a constant, a linear function of the transition mass into a spender, which explains much of the allowance-side result (Section 5.3.2). The construct-validity framing (Cronbach & Meehl, 1955; Messick, 1995) that makes these claims testable, and the open pipeline whose published summaries let others check every table, are part of the contribution, since whether decentralized systems miscalculate credit scores is a matter of public interest (Packin & Lev-Aretz, 2024).

The scope is deliberately narrow. The thesis does not show that on-chain reputation can replace collateral (Gudgeon, Perez, et al., 2020; Schär, 2021), that the results apply beyond contracts ap-

proved on the five tokens on Arbitrum One in the window studied, or that any score measures creditworthiness, default risk or trading skill, and it does not claim that C-PR or EndorseRank resists Sybils: under uniform teleportation a closed farm of attacker addresses can inflate either cheaply (Section 5.6).

6.3 Future work

Each suggestion below answers a limitation of Chapter 5. Any new score or weighting it leads to should be fixed before it is scored, replicated on registered new data and checked out of window against degree baselines.

- A registered test of the coupled walk. A registration committed before the logs of a later window, such as October to December 2026, are extracted would fix the revised C-PR with its US-dollar amounts, five tokens, value slopes, allowance cap and twelve-month score windows, together with a coupled walk whose layers carry the edges and weights of AWP and EndorseRank and whose restarts mix sending and approving activity (C-AWP, whose ends are AWP and EndorseRank with activity restarts), with the cohort rule, applied only to events up to its freeze, rules A and B and the comparison on traders, which needs more contract traders than W0 and W1 held. C-PR differs from AWP in its coupling, its weights and its restarts at once; C-AWP would set the coupling alone against the published walks. The value-slope check did not cover the holdout labels (Table 4.6), so the slopes and the cap should not be tuned on the new window. Node-dependent or learned weights, per-layer damping and mixtures of the two stationary vectors are further couplings that this thesis does not evaluate; any of them should keep the auditability that sets graph propagation apart from opaque learned scores (Kotb, 2023; Packin & Lev-Aretz, 2024).
- Propagation on a deeper sample. On the ego networks studied, EndorseRank reduces to a reweighted approval count for every depth-one spender (Section 5.3.2); a sample extracted two or more hops out, with EndorseRank and its variant with AWP's activity restarts tested

against the in-approve degree on new data separately for depth-one and deeper spenders, would show whether propagation adds to the count where the walk is more than a reweighted count.

- Validate the results in other contexts. Running the same analysis on other rollups, on Layer-1 networks and over later windows would show whether the results extend beyond Arbitrum One and the months studied; because DeFi is heterogeneous (Schär, 2021; Werner et al., 2022), this is the main threat to external validity.
- Adversarial evaluation. The Sybil cost model is analytical, and no attack was carried out on chain (Section 5.6). Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove the uniform inflow that closed farms exploit, and red-teaming with wash approvals, coordinated spender farms and strategic revocation would measure the realized cost of attacks and test whether the Sybil-adjusted proxies remain informative.
- Composition with attestation tooling. Filtering the approving owners through a Passport-style gate (Human Passport, n.d.) before C-PR or EndorseRank orders the contracts is a plausible deployment; it should be evaluated on labeled Sybil data, with the gate's own weaknesses (Bordeianu & Popescu, 2026; Siddarth et al., 2020) in view.
- Deployment studies. Integrating the scores with risk oracles and governance tooling would test whether their runtime profiles can sustain production refresh cadences. Controlled experiments on fee tiers or monitoring cadence would let evidence replace the adoption hypotheses of Section 5.4.4.

6.4 Closing reflection

In adversarial, pseudonymous markets on-chain reputation does not remove the need for collateral (Gudgeon, Perez, et al., 2020), but it can make behavioral structure readable, and which structure it reads depends on the edges used. A transfer records economic flow and an allowance records

delegated authority. PageRank will rank either graph, and one walk can follow both. On these data the coupled walk ranked every later spender label above the walk over allowances alone, but it beat neither layer under the rules fixed in advance and gained on the walk over transfers mainly among the contracts that walk cannot rank. For a protocol or researcher who needs a fast, auditable signal of which contracts owners have authorized, the allowance edge is the one to use on data like those studied here, above all for spenders without a transfer edge. The approval count is its cheapest form and, post hoc, ranked later approvals better than every PageRank of the study in both windows; EndorseRank is the propagated form, exposed to cheap inflation by closed farms (Section 5.6). For describing transfer hubs and anticipating their inflow, the transfer edge is the one to use, and there too, also post hoc, the count ranked later inflow better than AWP. For screening counterparties against forced liquidation, the approval count led the transfer count among contract traders in the primary window, though not resolvably in W0, and more traders are needed to settle it.

7 References

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4 and behind the registered analysis of Appendix 8.6.

8.1.1 Pipeline stages

The evaluation source is in the folder `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments` of the public repository https://github.com/abitocodes/phd_works, and the paths below are relative to it. The evaluation has five stages. The first three run on Arbitrum One over the window of scan A, 1 October 2023 to 30 June 2026 (UTC, inclusive at both ends); the fourth adds July to September 2026, and the fifth reads the tables that the first four left in BigQuery, together with the DefiLlama daily prices and the Chainlink feeds, and cuts from them the twelve months before each freeze.

1. Plan, cohort and graphs: Commit the analysis plan (`docs/analysis_plan.md`) and the configuration before any log of the study is extracted (commit `43d8045`, 13:41 UTC on 9 October 2026). Copy the logs of the window into a project table (scan A), list the spenders that at least three owners approved (`sql/02_spender_candidates.sql`), read their account types (`scripts/check_account_code.py`) and keep the contracts as the plan's cohort (`scripts/build_cohort.py`). Extract the ego events of the cohort (`sql/03_ego_events.sql`) and aggregate them into the pair tables of both graphs at T_{obs} and at t_1 (`sql/04_anchor_pairs.sql`).
2. Checks and alignment: Score every method at T_{obs} with the solver settings of Section 3.1, compute the checks of Section 3.9 and report Spearman ρ and Kendall τ with 95% paired-

bootstrap intervals (400 resamples, seed 42) and the contrasts of Section 3.10 (`scripts/run_same_window.py`).

3. Temporal holdout W0: Score every method at t_1 and evaluate the spender and trader cohorts against the labels of April–June 2026, with the contrasts of rule A (`scripts/run_holdout.py`).
4. Registered window W1: Commit the registration (`docs/registration_w1.md`) with the status in the configuration set to registered (commit `0d3a057`, 16:49 UTC on 9 October 2026). `scripts/extract_w1.py`, which will not run before that commit, records the hashes of both files in an extraction manifest and runs scan B and the ego extraction of July–September; `scripts/run_registered.py` evaluates rule B and the comparison on outcomes built from neither edge type.
5. Revised analysis (Section 3.5.1): `scripts/usd_inputs.py` builds the token list and the daily prices, `scripts/chainlink_check.py` compares them with the Chainlink feeds, and `scripts/run_usd_pipeline.py` loads both and keeps the events of the five tokens from the ego tables of stages 1 and 4 (`sql/19`, `sql/12`). `scripts/run_usd12m_pipeline.py` then draws the cohort of each freeze from the approvals of its twelve months and computes its pair tables, constants, proxies and labels (`sql/20` to `sql/23`, with `sql/13` to `sql/17` on its own tables), and checks that every allowance and transfer pair it keeps is one of the extracted ones. With `CONTRACT_REP_VARIANT=usd12m` the scripts of stages 2 to 4 then compute the revised analysis, and `scripts/run_robustness.py`, `scripts/sybil_model.py` and `scripts/run_benchmark.py` add the damping, value-slope and sample-definition checks, the Sybil model and the timed runs. `scripts/run_supplementary.py` computes, from the saved scores and with the same resamples, the contrasts that belong to no rule: EndorseRank with activity restarts against the in-approve degree and against EndorseRank, and C-PR against every other score and both raw degrees (Table 4.18).

The order of these steps is recorded in the commit history, in the cost ledger, in the extraction

manifest of W1 and in the run manifest of the revised analysis. The digest of the revised summaries records the code revision `0a94215`, at which they were computed; they were committed in `950a7a4`. The timed runs were made on an otherwise idle machine with an Intel Core Ultra 9 185H processor and 64 GB of memory under Windows 11 (build 26200), with Python 3.11.9, NumPy 2.3.2 and SciPy 1.16.1. `scripts/export_latex.py` wrote the tables to `results/tables/` and the figures to `Figures/generated/` from the summaries.

The experiments folder holds the queries, scripts and tests, the versioned configuration (`config/contract_reputation.yaml`), the plan, the registration and the revision document (`docs/`), the token list and daily prices with their Chainlink comparison (`data/0-usd-token-list-and-prices/`) and the cost ledger. The summaries behind the tables are in `data/3-published-results-for-thesis/revised-usd-12m/`, and those of the registered analysis in `data/3-published-results-for-thesis/`. Each of the two folders holds copies of the configuration, the plan and the registration, a digest `results_digest.md`, which names the summary file and key of every number, and `SHA256SUMS.txt`, so any number in a table can be checked against its summary without re-running the pipeline. The extracted tables and the per-contract scores, stored as parquet parts below 90 MB, are kept in a public Google Drive folder; the committed file `drive-manifest.json` gives its link and lists every part with its checksum.

Synthetic tests check the solver, the edge weights, the paired bootstrap and the US-dollar conversion offline. Real extraction needs Google Cloud credentials and a billing project.

8.1.2 BigQuery extraction design

The public table `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs` is read once for each window. Scan A copies the ERC-20 `Approval` and `Transfer` logs from 1 October 2023 to 30 June 2026, the GMX V2 position closes and the Aave V3 pool events, which no analysis of this thesis uses, into one project table partitioned by month, and scan B does the same for July to September 2026. Every later query reads these project tables, so the chain-wide scan is paid for once per window, and the ego extraction keeps only events with a cohort contract at one

end, which ties the size of the extract to the cohort instead of the whole chain. Appendix 8.2 shows the queries.

Every job is listed in `data/bq_ledger.json` with its query, parameters, estimated and billed bytes and cost. The 56 jobs billed 6.76 TiB, or USD 42.24 at the on-demand rate of USD 6.25 per TiB, against the budget of USD 50 fixed in the configuration. Scan A alone billed 3.29 TiB (USD 20.55) and scan B 0.26 TiB (USD 1.64); the 17 jobs of the registered analysis billed USD 29.71. The revised analysis reads the US-dollar ego tables built by three jobs for USD 1.31, and its own 11 jobs billed USD 2.49. Twelve further jobs on the US-dollar tables billed USD 3.78, three of them run outside `scripts/run_bq.py` and entered in the ledger by hand, and 13 jobs labeled exploratory billed USD 4.95, for analyses and post hoc checks the thesis does not report.

8.1.3 Account-type classification

Every address that could enter a cohort was classified by its code under the rule of Section 3.2. `scripts/check_account_code.py` calls `eth_getCode` for each address at one block, which it records with the time of the read. Table 8.1 lists the four runs, all made on 9 October 2026 for the address sets of the plan, in which approvals of every token counted. The first 20,000 candidates were read from the public endpoint https://arb1.arbitrum.io/rpc, which then limited the request rate, and the other 9,022, at the same block, from https://arbitrum-one.public.blastapi.io, which also served the later runs.

Table 8.1: Account types read with `eth_getCode` on Arbitrum One, one block per run. Contract: non-empty code that is not a delegation designator. No code and EIP-7702 (a delegation designator) are both externally owned accounts. The address sets are those of the plan (approvals of any token); the cohorts of the revised analysis are subsets of them. The W0 run read the spenders that were not cohort candidates, and the W1 run the addresses that no earlier run had classified. Sources: the JSON files beside the outputs in `data/1-raw-blockchain-extracts/cohort/` and `gmx/`.

Address set	Block	Read at (UTC)	Addresses	Contract	No code	EIP-7702
Cohort candidates: non-zero approvals from ≥ 3 owners, Oct 2023 – Jun 2026	513,216,506	9 Oct 2026, 13:55	29,022	27,844	1,121	57
GMX V2 accounts with ≥ 3 closes, Oct 2023 – Jun 2026	513,219,999	9 Oct 2026, 14:10	40,307	7,633	28,075	4,599
W0 spenders with a positive allowance at t_1, not cohort candidates	513,233,527	9 Oct 2026, 15:08	8,224	7,792	413	19
W1 spenders and GMX V2 accounts with ≥ 3 closes in W1, not read before	513,257,476	9 Oct 2026, 16:52	1,153	350	568	235

The matched cohort is part of the plan's cohort and was classified in the first run. The W0 spenders were classified in the first and third runs, since a spender with a positive latest allowance of one of the five tokens at t_1 also held that allowance in the ego data of the plan. The W1 spenders and the trader cohorts take their account types from all four runs, so the revision made no new read. Every read refers to the chain on 9 October 2026, and an address is classified by the code it held on that day.

8.1.4 Benchmark run logs

Table 8.2 lists the wall-clock seconds of every timed solver call of the benchmark, in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its five timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.5 and the full-cohort row of Table 4.2 repeat the first block of this table.

8.1.5 Parameter summary

Table 8.3 collects the default parameters of the revised analysis; every value is read from the versioned configuration or, for the tokens, prices, value slopes and cap, from the inputs committed with the revision (Section 3.5.1).

8.2 SQL Templates

This appendix shows the three query patterns on which the graphs of both analyses rest, abridged; the full files are in `sql/` of the experiments folder (Appendix 8.1.1). Placeholders in double underscores, such as `__DEST__`, are filled in by `scripts/run_bq.py` before a query runs, and names that start with `@` are query parameters.

8.2.1 Log materialization pattern

Scan A and scan B run the same query, `sql/01_materialize_logs.sql`, over different windows. It reads the public log table once, tags each log by kind and keeps the four kinds the study needs

Table 8.2: Wall-clock seconds of every timed solver call of the benchmark (matched cohort, $n = 6{,}483$ contracts), in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.5 and the full-cohort row of Table 4.2 are the first block, reused. The stages are the subsamples of Table 4.2.

Method	Run 1	Run 2	Run 3	Run 4	Run 5	Mean	SD
Full graphs, $d = 0.85$							
EndorseRank	2.799	2.591	2.562	2.751	2.582	2.657	0.109
AWP	7.870	7.925	8.795	9.308	8.750	8.530	0.618
C-PR ($\lambda = 0.5$)	11.737	11.304	11.429	11.527	11.821	11.563	0.214
S-PR	11.145	10.580	11.319	10.743	10.628	10.883	0.330
Full graphs, $d = 0.75$							
EndorseRank	2.179	1.881	1.859	1.959	2.054	1.986	0.132
AWP	6.566	5.925	6.196	5.832	5.260	5.956	0.483
Full graphs, $d = 0.95$							
EndorseRank	7.560	7.353	8.068	8.929	8.654	8.113	0.679
AWP	21.964	21.172	20.230	20.452	20.206	20.805	0.757
Stage $n = 1{,}000$							
EndorseRank	0.530	0.563	0.496	0.466	0.435	0.498	0.050
AWP	1.305	1.305	1.298	1.294	1.334	1.307	0.016
Stage $n = 2{,}000$							
EndorseRank	0.308	0.321	0.330	0.319	0.314	0.318	0.008
AWP	2.493	2.353	2.349	2.358	2.361	2.383	0.062
Stage $n = 5{,}000$							
EndorseRank	1.740	1.619	1.596	1.715	1.689	1.672	0.062
AWP	6.711	6.760	6.461	6.558	6.610	6.620	0.119

Table 8.3: Default parameters used in the evaluation.

Parameter	Value	Role
PageRank damping d	0.85; sweep 0.75, 0.95	Teleport mixture
Convergence tolerance ε	10^{-8} in L_1 norm	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer of C-PR)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer of C-PR)
Value transform b	$\ln 3/m$ per dollar, $m_T = 170$, $m_A = 100$	Value weight $V(z)$ of AWP (m_T) and EndorseRank (m_A)
Allowance cap U	19,000 dollars	C-PR allowance layer; in-approve value
Tokens and prices	WETH, USDC, USDT0, ARB, WBTC; DefiLlama daily close	Events and US-dollar amounts (Table 8.4)
Restart distribution	$\propto X_u$; uniform	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 primary; 0.25, 0.75	C-PR allowance-layer weight, fixed in the plan
S-PR restart vector	C-PR ($\lambda = 1$) on the transfer-layer nodes, renormalized	Endorsement-seeded teleportation
Score windows	2025-07-01 – 2026-06-30; 2025-04-01 – 2026-03-31	Same window and W1; W0. UTC inclusive; the freeze anchors the decay
Extraction (scan A)	2023-10-01 – 2026-06-30	Scored in full by the registered analysis
W0 freeze / labels	2026-03-31 / 2026-04-01 – 2026-06-30	Spender and trader cohorts; rule A
W1 freeze / labels	2026-06-30 / 2026-07-01 – 2026-09-30	Spender and trader cohorts; rule B, margin $\delta = 0.01$
Bootstrap resamples / seed	400 / 42; 2,000 / 42	Paired contract bootstrap; trader comparisons
Timed runs / warm-up	5 / 1	Runtime protocol
Scaling seed	`contract-benchmark-v1`	Deterministic SHA256 subsamples
BigQuery budget	USD 50	Checked before every query

in a project table partitioned by month. With the topic hashes and the GMX and Aave branches shortened, it reads:

```
CREATE TABLE `__DEST__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH) CLUSTER BY kind AS
WITH tagged AS (
  SELECT *, CASE
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] = '0x8c5b...b925'
        THEN 'A'                                    -- ERC-20 Approval
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] = '0xddf2...b3ef'
        THEN 'T'                                    -- ERC-20 Transfer
      WHEN ... THEN 'G'                             -- GMX V2 PositionDecrease
      WHEN ... THEN 'V'                             -- Aave V3 pool events
    END AS kind
  FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp < TIMESTAMP(@end_ts))
SELECT block_timestamp, block_number, log_index, kind,
  FROM_HEX(SUBSTR(address, 3)) AS emitter,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(1)], 27)), NULL) AS a,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(2)], 27)), NULL) AS b,
  IF(kind IN ('A', 'T'), SAFE.FROM_HEX(SUBSTR(data, 3)), NULL) AS val -- ...
FROM tagged WHERE kind IS NOT NULL
```

For an approval, `a` is the owner and `b` the spender; for a transfer, `a` is the sender and `b` the recipient. In both, `emitter` is the token contract and `val` the 32-byte amount, and addresses are stored as 20-byte values.

8.2.2 Ego-network extraction pattern

`sql/03_ego_events.sql` reads the materialized table of one window and keeps the approvals and transfers with a contract of the plan's cohort, held in `contract_rep.cohort`, at one end. Abridged, for the approvals:

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.approvals___SUFFIX__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH) AS
SELECT l.block_timestamp, l.block_number, l.log_index,
       l.emitter AS token_address, l.a AS owner, l.b AS spender,
       u256(l.val) AS value
FROM `dissertation-bq.contract_rep.__SRC__` AS l
WHERE l.kind = 'A'
  AND l.block_timestamp >= TIMESTAMP(@start_ts)
  AND l.block_timestamp < TIMESTAMP(@end_ts)
  AND (l.a IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`)
       OR l.b IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`));
-- transfers___SUFFIX__: the same with kind = 'T', from_address = a,
-- to_address = b and zero amounts dropped: AND LTRIM(TO_HEX(l.val), '0') != ''
```

The function `u256`, defined at the top of the file, reads the 32-byte amount as a double in raw base units. Approvals of zero are kept, because an approval of zero sets an allowance to zero, while zero-value transfers are dropped. The revised analysis does not run this query again: `sql/12_usd_ego.sql` keeps from these tables the events of the five tokens that have at one end a contract approved on them by at least three owners in scan A, and values a transfer at its token's price on the UTC day of its block; `sql/21_usd12m_cohorts.sql` then draws the cohort of each freeze from the approvals of its score window.

8.2.3 Latest-allowance and pair-table pattern

In the revised analysis `sql/14_usd_anchor_pairs.sql` turns the ego events up to one anchor into the edge lists of both graphs; `sql/04_anchor_pairs.sql` did the same for the registered analysis in raw base units, with $b = 1$ and no cap. Abridged:

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_allow_triples___TAG__` AS
SELECT r.token_address, r.owner, r.spender, r.amount * p.price_usd AS usd,
```

```
       TIMESTAMP_DIFF(TIMESTAMP(@anchor_ts), r.block_timestamp, MICROSECOND)
         / 8.64e10 AS age_days
FROM (
  SELECT *, ROW_NUMBER() OVER (
           PARTITION BY token_address, owner, spender
           ORDER BY block_number DESC, log_index DESC) AS rn
  FROM `dissertation-bq.contract_rep.u_approvals`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp <= TIMESTAMP(@anchor_ts)
) AS r
JOIN `dissertation-bq.contract_rep.u_prices` AS p
  ON p.token = r.token_address AND p.day = DATE(TIMESTAMP(@anchor_ts))
WHERE r.rn = 1 AND r.value > 0;

CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_allow_pairs___TAG__` AS
SELECT owner, spender,
       SUM(sig(age_days) * vt(usd, @b_allowance)) AS w_er,  -- EndorseRank
       SUM(LEAST(usd, @cap_allowance)) AS w_raw,    -- C-PR allowance layer
       MAX(sig(age_days)) AS max_sig
FROM `dissertation-bq.contract_rep.u_allow_triples___TAG__`
GROUP BY owner, spender;
-- u_transfer_pairs___TAG__: per (from_address, to_address) up to the anchor,
-- w_awp = SUM(sig * vt(usd, @b_transfer)), w_raw = SUM(usd * sig), MAX(sig)
```

The functions `sig` and `vt`, defined at the top of the file, are the logistic decay of Equation (3.3) with $k = 0.01$ and $t_0 = 180$ days and the value transform of Equation (3.4), set to one from $bz = 40$ on, where it equals one in double precision. The window function picks the latest approval of each token, owner and spender by block number and log index; it is kept only when its value is positive, so a revocation removes that token from the pair, and it is valued at the price of the anchor's day. The slopes $b_A = \ln 3/100$ and $b_T = \ln 3/170$ per dollar and the cap $U = 19{,}000$ dollars are passed as parameters from `constants.json`. Each pair row carries the weights of every method, so all scores at one anchor come from the same rows. The same query also writes the EndorseRank and AWP weights with the slope multiplied and divided by ten, from which the value-slope check of Table 4.6 is computed.

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al. (1999) and Langville and Meyer (2006), and the rank correlations Spearman (1904) and Kendall (1938).

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\text{PR}} = dP^{\top} + \frac{1-d}{|V|}\mathbf{1}\mathbf{1}^{\top},$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed, every directed graph has unique PageRank scores (Langville & Meyer, 2006).

Three restart vectors in this thesis are not uniform. AWP's, proportional to activeness, is zero at every address that has sent nothing, that of EndorseRank with activity restarts is zero at every address that has approved nothing, and S-PR's, the normalized score of the allowance walk, is zero at every node of the transfer layer that has no allowance score, so none of the three Google matrices is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d. In general it is not strictly positive, however, because a node with no restart mass that no path from a restart node reaches keeps a score of zero. Uniqueness means that, once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ arises from their edges and restart distributions.

8.3.2 Continuity in edge weights

For fixed $d \in (0, 1)$, PageRank depends continuously on the entries of P. If $\tilde{P}$ is a second row-stochastic matrix on the same nodes, with PageRank vector $\tilde{\mathbf{r}}$, then

$$\|\mathbf{r} - \tilde{\mathbf{r}}\|_1 \leq \frac{d}{1-d} \max_i \left\| P_{i\cdot} - \tilde{P}_{i\cdot} \right\|_1,$$

a bound that does not depend on $|V|$ (Langville & Meyer, 2006).

Continuity in P is not the same as continuity in the edge weights. A row $P_{i\cdot} = w_{i\cdot}/o_i$ moves continuously with the weights only while the out-strength o_i stays positive. When a node gains its first out-edge or loses its last one, its row jumps between the uniform dangling row and a row concentrated on its out-neighbors, however small the weight involved. In the worked example of Section 2.13, adding an edge $D \to A$ of weight 10^{-12} moves $\mathbf{r}$ by 0.37 in the $\|\cdot\|_1$ norm and raises r_A from 0.1375 to 0.3202.

The damping sweep is covered by continuity: it keeps P fixed and changes d smoothly, and for $d < 1$ the PageRank vector varies continuously with d. The value-slope check is covered only in part. Since $V(z) > 0$ for every $z > 0$, a change of the slope b adds or removes no edge in exact arithmetic, but in double precision an edge that carries only tiny amounts can round to zero weight, and a tenfold change of b is too large for the bound above to say much, so the stability of that check rests on the data (Table 4.6). Empirically, the family means of EndorseRank move by at most 0.0031 and those of AWP by at most 0.021 across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.5).

8.3.3 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges and $n = |V|$ nodes, the sparse product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations, each iteration also updates vectors of length n, and building the sparse operator from the edge list is $O(m)$, so power iteration over I iterations is $O(m + (m + n)I)$. Both scores run the same solver. In the measured calls an iteration of AWP takes about 1.95 times as long as one of EndorseRank, less than the edge ratio of 2.21, and AWP needs more iterations (Section 4.2), so

neither m nor I alone sets the time ratio.

8.3.4 Properties of Spearman's ρ

Spearman's ρ lies in $[-1, 1]$. A value of $\rho = 1$ means perfect monotone agreement, and $\rho = -1$ perfect reverse agreement. The evaluation pipeline computes ρ and τ_b with standard tie-aware implementations. When n is large, ρ and τ usually agree in sign, but they differ in magnitude. Across the fourteen pairs of EndorseRank or AWP with a proxy (Tables 8.7 to 8.9), every $|\tau|$ exceeds 0.05, the ratio ρ/τ lies between about 1.2 and 1.5 except for AWP against the inbound-counterparty ratio, where $\tau = 0.055$ and $\rho = 0.163$, and $|\rho|$ is always the larger of the two; no pair has coefficients of opposite sign.

8.3.5 Properties of Kendall's τ

Among the scores of this thesis ties are rare: on the matched cohort no score ties more than 3.4% of the contract pairs, and Section 4.4 gives the shares of EndorseRank and AWP. The denominator of τ_b in Equation (2.6) differs from n_0 only through ties, so for EndorseRank against AWP it differs from n_0 by less than two percent, and τ_a, computed from the pair counts of Section 4.4, lies within two percent of the reported τ_b.

Ties in a proxy cap τ_b as well: the integer proxies, such as the in-degree and the in-approve degree, tie many pairs, and the largest τ_b that any score can reach against such a proxy falls with the share of tied pairs. Because the PageRank scores tie few pairs, this cap differs little between two of them compared on one proxy, so such a contrast is not distorted by it; a contrast between a PageRank and a raw degree, or between two proxies, can be, to the extent that their ties differ (Section 3.10).

8.4 Data Description

This appendix describes the extracted data behind Chapter 4: the volume of the ego extraction and of the two graphs, the five tokens with their prices and allowance amounts, and the activity of GMX V2 contract accounts by quarter.

8.4.1 Extracted events and graphs

The ego extraction of scan A, 1 October 2023 to 30 June 2026, was made for the plan's cohort of 27,844 contracts. It holds 389,866,882 approval logs, zero-value ones included, and 1,538,404,067 transfer logs of non-zero value of every ERC-20 token, each with a contract of that cohort at one end. The revised analysis keeps, for each freeze, the approvals and transfers of the five tokens in its score window that have a contract of the freeze's cohort at one end (Section 4.1 gives their number); of the July–September logs of W1 it keeps 12,200,682 approvals and 58,625,425 transfers. At T_{obs} the pair tables built from the kept events hold 3,810,253 allowance pairs and 8,432,635 transfer pairs, and at t_1, 4,117,103 and 9,198,965. The 1,304 and 6,511 pairs at T_{obs} whose value weight is zero in double precision, and the addresses that only they touch (Section 4.1), are not part of the solver graphs of EndorseRank and AWP (Table 4.1).

8.4.2 Tokens, prices and allowance amounts

Section 3.5.1 gives the rule by which the five tokens were chosen. In the score window of T_{obs} they hold 79.40% of the rows of all tokens in the ego tables of the plan's cohort (transfers in the window and positive latest allowances at T_{obs} set in it). Table 8.4 gives for each of them the address, decimals and feed, the rows held and kept, the US-dollar volume of the kept transfers and the data behind the daily prices; its caption counts the rows left out, those of the other tokens and those of the five tokens in which no contract of the revised cohort takes part.

DefiLlama's series under the Arbitrum One address begins in August 2024 for WETH and WBTC and in January 2025 for USDT0, before the first score window opens, so every one of the 548 days that the windows cover is priced under the token's own address. No day of the five tokens is filled from the previous day or replaced as a data error, and no kept transfer precedes its token's first price. Against the Chainlink feeds read on the chain on 20 of these days per token, the DefiLlama prices differ by a median of 0.010% to 0.056% per token and by at most 0.24%, for WBTC.

Unlimited allowances are common. Of the 4,185,287 positive latest allowances at T_{obs}, 2,084,669

Table 8.4: The five tokens of the revised analysis, with their Chainlink feed (market-risk category), the rows of the score window of T_{obs} (1 July 2025 to 30 June 2026) that the ego tables of the plan's cohort hold for them (approvals and transfers) and the rows that the revised analysis keeps, the US-dollar volume of the kept transfers, the days of the price series filled from the second DefiLlama key, filled from the previous day and replaced as one-day errors (of the 548 days from 1 April 2025 to 30 September 2026 that the windows of the revised analysis cover), and the median and largest absolute difference, in percent, between the DefiLlama price and the Chainlink feed read on the chain on the 20 check dates among those days. The ego tables of the plan's cohort hold 135,799,669 approvals and 580,802,372 transfers in that score window. Of these, 49,138,325 approvals and 117,646,035 transfers belong to other tokens (of the 44,119 other tokens in the ego tables of that score window and W1) and are left out; 674,782 approvals and 20,434,298 transfers of the five tokens are left out because neither end point is a contract of the T_{obs} cohort of the revised analysis; and 0 kept transfers precede their token's first price. At T_{obs}, 2,084,669 of the 4,185,287 positive latest allowances set inside the score window are unlimited and 2,331,420 reach the cap $U = 19{,}000$ dollars of the allowance layer of C-PR.

Token	Address	Dec.	Chainlink feed	Rows, plan	Rows, kept	Transfer USD	2nd key	Filled	Errors	CL diff. (%)
WETH	`0x82af…bab1`	18	ETH / USD (low)	201,126,112	191,667,668	1.9×10^{11}	0	0	0	0.04 / 0.14
USDC	`0xaf88…5831`	6	USDC / USD (low)	186,761,761	180,274,238	4.9×10^{11}	0	0	0	0.01 / 0.02
USDT0	`0xfd08…cbb9`	6	USDT / USD (low)	95,067,280	92,700,928	1.0×10^{11}	0	0	0	0.01 / 0.15
WBTC	`0x2f2a…5b0f`	8	WBTC / USD (low)	41,730,499	40,610,808	7.7×10^{10}	0	0	0	0.05 / 0.24
ARB	`0x912c…6548`	18	ARB / USD (low)	25,132,029	23,454,959	1.0×10^{10}	0	0	0	0.06 / 0.24

(49.8%) are of 1.15×10^{77} base units or more, close to $2^{256} - 1$, and 2,331,420 (55.7%), the unlimited ones included, are worth at least the cap U. The 4,185,287 allowances belong to 2,508,872 owners, of whom 1,389,569 hold at least one unlimited allowance and 256,803 hold both an unlimited and a finite one. In the registered analysis, which kept every token in raw base units, 99.52% of the ego transfers and 99.78% of the 28,069,418 latest allowances at T_{obs} were of ten base units or more, where V with $b = 1$ is within 10^{-4} of one, and 1,532,301 of the 13,029,306 owners with a positive latest allowance held both an unlimited and a smaller allowance.

8.4.3 GMX V2 contract traders by quarter

The trader cohorts are GMX V2 accounts that hold contract code (Section 4.1.1). Table 8.5 counts their position closes, the `PositionDecrease` events of the protocol's event emitter (GMX, n.d.), by calendar quarter, together with the liquidations among them and the number of contract accounts that closed at least one and at least three positions in the quarter.

Table 8.5: GMX V2 closes of contract accounts by quarter. October 2023 to June 2026: closes by the 7,633 contracts among the 40,307 GMX V2 accounts with at least three closes in these months (Table 8.1). July to September 2026 (the label window of W1): closes by the contract accounts of scan A and of the W1 run. Liquidation: a close with the liquidation order type or emitted through the liquidation handler. Source: the decoded closes in `data/2-processed-tables-and-evaluations/gmx/`.

Quarter	Closes	Liquidations	Accounts with a close	Accounts with ≥ 3 closes
2023 Q4	1,119	104	180	105
2024 Q1	7,956	692	1,247	961
2024 Q2	11,509	589	2,120	1,165
2024 Q3	7,947	456	1,807	724
2024 Q4	7,343	904	1,321	698
2025 Q1	6,929	498	497	273
2025 Q2	12,080	379	368	250
2025 Q3	28,753	897	1,750	1,567
2025 Q4	40,240	1,596	1,915	1,744
2026 Q1	6,026	233	269	171
2026 Q2	3,783	125	169	127
2026 Q3 (W1)	6,805	182	165	131
October 2023 – June 2026	133,685	6,473		

Contract accounts closed the most positions in the second half of 2025 and far fewer in 2026. A trader cohort further requires the account to be a node of the graph at the freeze: of the 127 contract accounts with at least three closes in April–June 2026, 87 form the trader cohort of W0, and of the 131 in July–September, 70 form that of W1.

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, and their paired differences out of window. Apart from the trader rows of Table 8.10 and the grid at the end, whose pairs were registered with W1 and are computed on the revised scores as a re-analysis, none of these comparisons was fixed in advance, and none carries

a claim of the thesis. The two scores walk on different edges and are built for different constructs, so on each check the score ahead is, almost every time, the one whose edges the check counts. Outcomes built from neither edge type are the exception, since neither score was built for them, and are reported in Section 4.8. Their coverage differs as well: no allowance edge touches 191 of the matched contracts, and no transfer edge touches 1,192. Out of window neither score beats the raw degree it smooths: AWP ranks later new senders below the transfer in-degree, and EndorseRank ranks later new approval pairs below the in-approve degree, with uniform and with activity restarts (Sections 4.6 and 4.5.4; computed after the labels were known), so a lead of EndorseRank over AWP does not show that propagation over approvals adds anything to the count of approving owners.

8.5.1 Same window

Table 4.3 gives the family means of the two scores with their intervals, and the last row of Table 4.4 their agreement. The two patterns cross: AWP tracks the transfer family most closely and the allowance family least, while EndorseRank tracks the allowance family most closely. Table 8.6 tests five paired differences on the same contract resamples (Section 3.10), numbered i to v in the order of its rows.

Table 8.6: Paired bootstrap tests of differences between Kendall τ coefficients of EndorseRank and AWP in the same window (matched cohort, $n = 6{,}483$ contracts). Family entries use the family-mean τ; the same contract resamples are used for a and b, so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.560	0.196	+0.363	[0.341, 0.382]	1.000
AWP transfer minus EndorseRank transfer	0.736	0.196	+0.539	[0.518, 0.560]	1.000
EndorseRank allowance minus AWP allowance	0.560	0.226	+0.333	[0.311, 0.354]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.419	0.026	+0.393	[0.373, 0.412]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.650	0.218	+0.432	[0.410, 0.452]	1.000

All five intervals lie above zero. AWP's transfer alignment is higher than EndorseRank's (contrast ii, $+0.539$, $[0.518, 0.560]$), EndorseRank's allowance alignment is higher than AWP's (contrast iii,

$+0.333$, $[0.311, 0.354]$), and AWP leads EndorseRank on the Sybil-stability family (contrast iv, $+0.393$, $[0.373, 0.412]$). The remaining two compare EndorseRank with itself: its allowance alignment is higher than its transfer alignment (contrast i, $+0.363$, $[0.341, 0.382]$), and its in-approve-degree alignment is higher than its in-degree alignment (contrast v, $+0.432$, $[0.410, 0.452]$). EndorseRank ties only 0.20% of the contract pairs, so its own ties cap none of these coefficients, and the ties of the proxies do not explain gaps of this size (Appendix 8.3.5). With the contracts outside the allowance graph kept as isolated nodes instead of being scored zero, none of the family means changes (Table 4.4).

The opposite directions of contrasts ii and iii follow from the constructs. Each score is ahead on the family built from its own edges, and the main proxies of each family include the in-degree that the corresponding PageRank smooths, so the agreement is partly built in and checks the intended construct of each score. Neither difference says which score is the better measure of reputation.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.7 to 8.9 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ, Kendall τ and the 95% interval of τ; the transfer proxies are the ones used in the evaluation of AWP by its authors (T. Do et al., 2023).

8.5.2.1 Transfer family

Table 8.7 gives the coefficients of the two scores with the in-degree and the in-value.

AWP agrees with the in-value almost as closely as with the in-degree (Table 8.7), for the reason given in Section 4.3. EndorseRank's transfer coefficients have intervals that overlap neither zero nor AWP's intervals. They mainly reflect that contracts that many owners approve receive transfers from more senders than the rest of the cohort.

Table 8.7: Domain alignment of EndorseRank and AWP with the transfer proxies (in-degree and in-value of each contract), same window (matched cohort, $n = 6{,}483$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-degree	0.302	0.218	[0.199, 0.238]
In-value (sum inbound)	0.256	0.175	[0.157, 0.194]
AWP			
In-degree	0.883	0.742	[0.732, 0.752]
In-value (sum inbound)	0.887	0.730	[0.720, 0.740]

8.5.2.2 Allowance family

Table 8.8 gives the coefficients of the two scores with the in-approve degree and the in-approve value.

Table 8.8: Domain alignment of EndorseRank and AWP with the allowance proxies (in-approve degree and value of each spender contract), same window (matched cohort, $n = 6{,}483$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-approve degree	0.804	0.650	[0.639, 0.661]
In-approve value (sum)	0.628	0.469	[0.454, 0.484]
AWP			
In-approve degree	0.320	0.242	[0.223, 0.262]
In-approve value (sum)	0.294	0.211	[0.193, 0.229]

EndorseRank follows the in-approve degree more closely than the in-approve value, for the reason given in Section 4.3. AWP's coefficients lie above zero: among the cohort's contracts, those that many owners approve also tend to be central in the transfer graph.

8.5.2.3 Sybil-adjusted stability family

Table 8.9 gives the coefficients of the two scores with the inbound counterparty ratio, the transfer tenure and the active months.

Table 8.9: Domain alignment of EndorseRank and AWP with the Sybil-adjusted stability proxies from transfer activity, same window (matched cohort, $n = 6{,}483$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
Inbound counterparty ratio	-0.290	-0.197	[-0.213, -0.180]
Transfer tenure (days)	0.199	0.133	[0.114, 0.153]
Active months	0.203	0.143	[0.124, 0.164]
AWP			
Inbound counterparty ratio	0.163	0.055	[0.033, 0.079]
Transfer tenure (days)	0.763	0.588	[0.575, 0.600]
Active months	0.768	0.616	[0.603, 0.629]

AWP is ahead of EndorseRank on tenure and active months, as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.6, contrast iv). On the inbound counterparty ratio the two coefficients have opposite signs, negative for EndorseRank and positive for AWP. The ratio falls when the same senders return, so contracts that many owners approve are, on this measure, contracts whose senders return. AWP's positive coefficient comes from coverage: the $1{,}192$ contracts outside the transfer graph have ratio zero and AWP score zero, so they fill the bottom of both orderings. The family describes how much longitudinal transfer information each score carries.

8.5.3 Out of window

Table 8.10 collects the paired differences between the two scores in both holdout windows. In every row a is EndorseRank and b AWP. The spender rows were computed after the labels were known. On the spenders of W0, EndorseRank ranks new approval pairs above AWP, a paired difference of

Table 8.10: Paired bootstrap differences between EndorseRank and AWP out of window (W0 spenders $n = 7{,}914$; W0 traders $n = 87$; W1 spenders $n = 7{,}960$; W1 traders $n = 70$). None of them enters rule A or rule B: the spender rows were computed after the labels were known, and the trader rows are the EndorseRank-minus-AWP pair of the comparison on an outcome built from neither edge type (Tables 4.19 and 8.11). The W1 spender rows were computed from the saved W1 scores after the W1 evaluation, with the resamples of Table 4.15. Within each block a and b share the contract resamples (400 for the spenders and 2,000 for the traders; seed 42).

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
W0, spenders ($n = 7{,}914$)						
EndorseRank minus AWP	new approval pairs	0.303	0.196	+0.107	[0.086, 0.130]	1.000
EndorseRank minus AWP	new transfer senders	0.272	0.409	-0.138	[-0.151, -0.123]	0.000
W0, traders with at least three closes ($n = 87$)						
EndorseRank minus AWP	liquidation-free close rate	0.236	0.254	-0.019	[-0.173, 0.144]	0.401
W1, spenders ($n = 7{,}960$)						
EndorseRank minus AWP	new approval pairs	0.299	0.157	+0.142	[0.119, 0.164]	1.000
EndorseRank minus AWP	new transfer senders	0.259	0.398	-0.139	[-0.154, -0.125]	0.000
W1, traders with at least three closes ($n = 70$)						
EndorseRank minus AWP	liquidation-free close rate	0.297	0.263	+0.034	[-0.067, 0.134]	0.750

$+0.107$ (interval $[0.086, 0.130]$). On new transfer senders the order reverses, with a difference of -0.138 ($[-0.151, -0.123]$). Every label falls after all scored events. The first gap rests on coverage. AWP scores zero for the $1{,}342$ spenders without a transfer edge before the freeze, about one in six, which only allowance edges order, and these gained new approval pairs more often than the rest (26.7% against 20.9%). On the $6{,}489$ spenders that had received a transfer from another address, AWP is ahead on both labels (Table 4.11), without a paired test.

In July–September, where AWP again scores zero the spenders without a transfer edge, the pattern repeats: EndorseRank leads on new approval pairs ($+0.142$, $[0.119, 0.164]$) and AWP on new senders (-0.139, $[-0.154, -0.125]$).

On the W0 traders (Table 4.12), only AWP's intervals lie above zero on the number of profitable closes and on the realized gain, two labels that grow with the number of closes. On the two shares, which do not, both scores are negative, with intervals below zero on the non-loss share only. Neither score ranks these contract traders by the share of successful closes.

8.5.4 All trader labels in both windows

Table 8.11 sets the allowance side against the transfer side on all six trader labels in both windows. Its pairs were registered with W1 before the study extracted the logs of July–September (Section 3.7.3); here they are computed on the revised scores, after the registered results were known, so the table is a re-analysis (Appendix 8.6). P and Q on the liquidation-free close rate enter rules R1 and R2, and P on the two shares enters rule R3 (Claim C6, Section 4.8). In W1 the counts (P) favor the allowance side on both liquidation labels with intervals above zero, while the three PageRank pairs are positive with intervals that include zero. In W0, P is positive on both liquidation labels with intervals that include zero, and the three PageRank pairs lie below zero, their intervals including zero as well. On the four profit labels of W1 every pair favors the transfer side: the intervals of Q, of the C-PR layers and of the EndorseRank pair lie below zero on all four, and those of P on the two shares. In W0 every interval on the profit labels includes zero except that of the C-PR layers on realized gain.

8.6 The Registered Analysis

The registered analysis is the one that the analysis plan and the registration of W1 fixed before the study extracted the data it judges (Appendix 8.1.1): amounts in raw token base units, every ERC-20 token, the value transform with $b = 1$, scores from every event since 1 October 2023, and the plan's cohort of $27{,}844$ contracts. The revised analysis, with amounts in US dollars, five tokens and scores from the twelve months before each freeze, was defined after its results were known (Section 3.5.1), and the chapters report it. Table 8.12 sets the decision contrasts and verdicts of the registered analysis beside the same contrasts on the revised scores; its last four rows, degree contrasts computed after the labels were known, carry no verdict. The two columns reach the same verdicts except on R1, which holds on the revised scores and not in the registered analysis; only the verdicts of the registered column are outcomes of tests fixed before the study extracted the data they judge. The complete registered results, with every coefficient, interval and verdict, are published

Table 8.11: Allowance-side minus transfer-side scores, $\Delta\tau$ with 95% paired bootstrap interval, on all six trader labels of the window W1 ($n = 70$ GMX V2 contract traders) and of W0 ($n = 87$). P: in-approve degree minus transfer in-degree. Q: EndorseRank with activity restarts minus AWP (same edge weights and restart rule). C-PR: $\lambda = 1$ minus $\lambda = 0$ (same USD build). ER: EndorseRank minus AWP. Profitable closes and realized gain grow with the number of closes; the other four labels do not. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Label	P (counts)	Q (PageRanks)	C-PR layers	ER minus AWP
Window W1, July–September 2026 ($n = 70$ traders)				
Liquidation-free close rate	0.141 [0.037, 0.240]	0.023 [-0.062, 0.113]	0.019 [-0.093, 0.128]	0.034 [-0.067, 0.134]
No liquidation	0.141 [0.044, 0.232]	0.027 [-0.066, 0.129]	0.021 [-0.087, 0.132]	0.038 [-0.069, 0.149]
Profitable closes	-0.115 [-0.257, 0.025]	-0.215 [-0.399, -0.042]	-0.213 [-0.402, -0.028]	-0.200 [-0.390, -0.023]
Realized gain	-0.109 [-0.247, 0.032]	-0.299 [-0.457, -0.146]	-0.294 [-0.476, -0.116]	-0.292 [-0.460, -0.134]
Profitable share	-0.162 [-0.295, -0.025]	-0.262 [-0.436, -0.100]	-0.266 [-0.461, -0.070]	-0.246 [-0.427, -0.073]
Non-loss share	-0.222 [-0.328, -0.106]	-0.324 [-0.471, -0.188]	-0.316 [-0.481, -0.152]	-0.315 [-0.468, -0.176]
Window W0, April–June 2026 ($n = 87$ traders)				
Liquidation-free close rate	0.095 [-0.027, 0.234]	-0.012 [-0.153, 0.149]	-0.057 [-0.209, 0.092]	-0.019 [-0.173, 0.144]
No liquidation	0.109 [-0.022, 0.257]	-0.005 [-0.162, 0.165]	-0.056 [-0.211, 0.098]	-0.007 [-0.170, 0.165]
Profitable closes	-0.002 [-0.141, 0.129]	-0.027 [-0.170, 0.113]	-0.089 [-0.244, 0.070]	-0.046 [-0.205, 0.113]
Realized gain	-0.026 [-0.176, 0.112]	-0.110 [-0.270, 0.040]	-0.257 [-0.447, -0.071]	-0.139 [-0.311, 0.022]
Profitable share	-0.043 [-0.182, 0.081]	0.044 [-0.084, 0.183]	0.012 [-0.125, 0.150]	0.074 [-0.066, 0.223]
Non-loss share	-0.074 [-0.208, 0.052]	0.078 [-0.055, 0.217]	0.047 [-0.090, 0.191]	0.108 [-0.034, 0.256]

unchanged in `data/3-published-results-for-thesis/` with their digest, `results_digest.md`.

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate

Reference: [Unisa SoC/CAES ethics clearance ref: TBD]

Date of issue: [TBD]

(Scanned certificate to be inserted here.)

Table 8.12: Decision contrasts and verdicts of the registered analysis (amounts in raw base units, every ERC-20 token, $b = 1$, scores from every event since 1 October 2023), exactly as obtained under the analysis plan and the W1 registration, beside the same contrasts on the revised scores (amounts in USD, the five listed tokens, scores from the twelve months before each freeze). The revised analysis, its score windows included, was defined after the registered results were known; its column applies the registered rules and thresholds to the revised scores and is not a registered test. The last block holds degree contrasts that no rule of either analysis decides; they are reported in both. $\Delta\tau$ with 95% paired bootstrap interval (400 and 2,000 resamples, seed 42); the rules and their thresholds are in Sections 3.7 and 3.7.2.

Quantity	Registered analysis	Revised analysis
Matched cohort (contracts)	27,844	6,483
W0 spender cohort	33,101	7,914
W1 spender cohort	34,587	7,960
W1 traders	70	70
EndorseRank vs. AWP, same window (τ)	0.217 [0.209, 0.225]	0.199 [0.181, 0.221]
A, W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	$+0.008$ [0.003, 0.013]	$+0.031$ [0.021, 0.042]
A, W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.030 [-0.034, -0.026]	-0.030 [-0.040, -0.020]
Rule A, primary part	not met	not met
Rule A, secondary part	not met	not met
B, F1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.101$ [0.092, 0.111]	$+0.170$ [0.149, 0.189]
B, F2: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.025 [-0.029, -0.021]	-0.025 [-0.038, -0.015]
B, F3: C-PR minus C-PR ($\lambda = 0$), liquidation-free close rate	-0.044 [-0.100, -0.002]	$+0.000$ [-0.010, 0.010]
Rule B (F1, F2 and F3)	not met	not met
Rule B, sensitivity 1 (AWP as comparator)	not met	not met
Rule B, sensitivity 2 (isolated nodes)	not met	not met
F6 (rule A, primary part, on W1)	fails	fails
P, W1: in-approve degree minus transfer in-degree	$+0.063$ [-0.026, 0.148]	$+0.141$ [0.037, 0.240]
Q, W1: EndorseRank (activity restarts) minus AWP	$+0.051$ [-0.085, 0.213]	$+0.023$ [-0.062, 0.113]
R1 (count-level lead)	not met	met
R2 (PageRank-level lead)	not met	not met
R3 (specificity)	applies	applies
W0: EndorseRank minus in-approve degree, new pairs	-0.011 [-0.016, -0.006]	-0.090 [-0.100, -0.079]
W0: AWP minus transfer in-degree, new senders	$+0.038$ [0.035, 0.042]	-0.038 [-0.043, -0.032]
W1: EndorseRank minus in-approve degree, new pairs	-0.029 [-0.032, -0.025]	-0.082 [-0.090, -0.073]
W1: AWP minus transfer in-degree, new senders	$+0.020$ [0.016, 0.023]	-0.041 [-0.047, -0.034]